Li-ion Cell Manufacturing: an Engineer's Course

- Module 0 — Roadmap and Line Overview
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- Module 6 — Cell Assembly: Winding, Stacking, Welding, Housing (Stage 7)
- Module 7 — Electrolyte Filling and Wetting (Stage 8)
- Module 8 — Formation (Stage 9)
- Module 9 — Degassing, Final Sealing, Aging, Grading and Final Test (Stages 10–12)
- Module 10 — Equipment List and Line Integration
- Module 11 — Factory Layout, Dry Rooms and Safety Zoning
- Module 12 — Parameter Handbook, Quality Engineering and Process FMEA
- Module 13 — Defect and Troubleshooting Guide, Failure Analysis
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Module 0 — Roadmap and Line Overview

Li-ion Cell Manufacturing: an Engineer's Course · as of 27 Oct 2026

Goal of this module: be able to walk a visitor through an entire cell line in ten minutes, name the bottleneck, and say where cost, energy and quality risk sit.

How the course works

Every process module follows the same 11 parts: simple concept → engineering principle → industrial process → process flow → equipment → parameter table → defects → factory example → interview questions → quiz → hands-on exercise.

You already know APQP, PFMEA, SPC and line commissioning from automotive, so the course maps battery practice onto those tools instead of re-teaching them. Pack, BMS and BESS basics are skipped.

Evidence labels (used in every table)

Label	Meaning	How to treat it
[V] Verified	Published by a manufacturer, government, peer-reviewed paper or equipment OEM, with a link	Quote with the source
[T] Typical practice	Range reported across several papers, textbooks or industry guides	A starting window for DOE, not a spec
[E] Estimate	Engineering estimate made for this course, assumptions stated	Use for sizing and learning only
[P] Proprietary	Real values exist but cell makers keep them confidential	Get these from your plant's process specification

A cell plant's recipe (slurry formulation, coating speed, formation protocol) is its core IP. Public numbers give you the window; your line's control plan gives you the target.

Learning roadmap

#	Module	File	You should be able to…
0	Roadmap and line overview	this file	Walk a visitor through the line in 10 minutes
Phase A	**Electrode making**		
1	Materials and incoming quality	`01-materials-and-incoming-quality.md`	Write incoming specs for NMC, LFP, graphite, foils, separator, electrolyte
2	Slurry preparation	`02-slurry-preparation.md`	Set a mixing sequence and read a viscosity curve
3	Coating and drying	`03-coating-and-drying.md`	Size a coater oven; diagnose streaks, cracks, edge build-up
4	Calendering	`04-calendering.md`	Pick target density/porosity and set roll gap and force
5	Slitting, notching, cutting	`05-slitting-notching-cutting.md`	Specify burr limits; choose laser vs die notching
Phase B	**Assembly and finishing**		
6	Assembly: winding and stacking	`06-cell-assembly.md`	Compare cylindrical, prismatic and pouch assembly
7	Electrolyte filling and wetting	`07-electrolyte-filling.md`	Set fill mass, vacuum steps and wetting time
8	Formation	`08-formation.md`	Explain SEI and design a first-charge protocol
9	Degassing, aging, grading, final test	`09-degassing-aging-grading-final-test.md`	Use OCV/K-value data to screen micro-shorts
Phase C	**Factory and quality**		
10	Equipment and line integration	`10-equipment-and-line-integration.md`	Build a full equipment list and balance the line
11	Factory layout and dry rooms	`11-factory-layout-and-dry-rooms.md`	Zone a plant by dew point; plan material flow
12	Parameter handbook, QC and PFMEA	`12-parameters-quality-pfmea.md`	Write a control plan and PFMEA for a cell line
13	Defects and failure analysis	`13-defects-and-failure-analysis.md`	Run an 8D on a self-discharge escape
Phase D	**Business and industry**		
14	Economics and factory operations	`14-economics-and-operations.md`	Build a cost-per-kWh model for 1, 5, 10 GWh plants
15	Global and Indian ecosystem	`15-global-and-indian-ecosystem.md`	Separate announced from operating capacity; map suppliers
16	Capstone: interview guide, glossary, resources	`16-capstone-interview-glossary-resources.md`	Pass a technical interview; design a 5 GWh line on paper

Why this order: electrode quality sets most of a cell's capacity, resistance and safety, and defects made there cannot be fixed downstream. Learning in line order also matches how you walk a line during commissioning.

Suggested pace: Phase A weeks 1–4 · Phase B weeks 5–7 · Phase C weeks 8–10 · Phase D and capstone weeks 11–12 (6–8 hours a week).

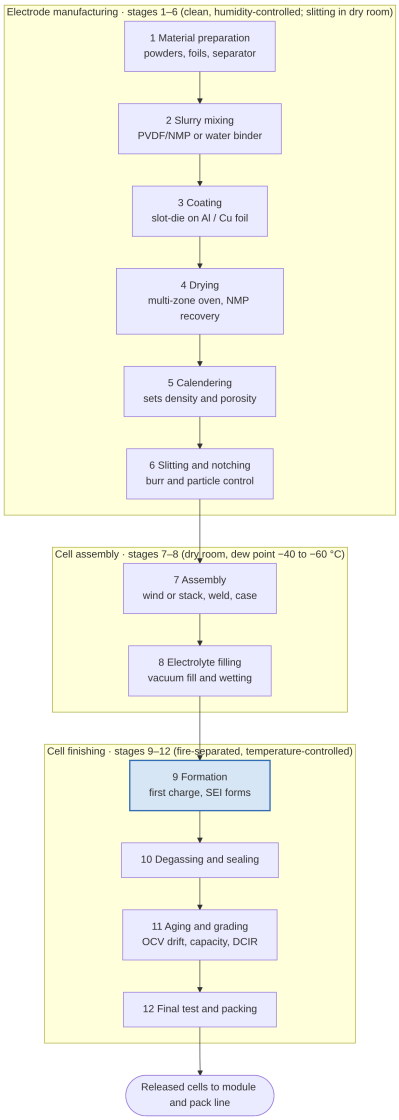
Where each final deliverable lives

#	Deliverable you asked for	Module
1	Learning roadmap	0
2	Process flowchart	0 (whole line), 2–9 (per stage)
3	Equipment list	10

#	Deliverable you asked for	Module
4	Parameter handbook	12 (consolidated), 2–9 (per stage)
5	Factory layout explanation	11
6	Defect and troubleshooting guide	13
7	Process FMEA	12
8	Cost model	14
9	Global and Indian ecosystem map	15
10	Interview preparation guide	16
11	Glossary	16
12	Books, papers, videos, courses	16

The line in 12 stages

A cell line has three blocks: **electrode manufacturing** (stages 1–6), **cell assembly** (7–8) and **cell finishing** (9–12). Think of it as a paint shop feeding a body shop feeding an EOL test area: continuous web processes first, then discrete parts, then long electrochemical dwell times.



#	Stage	What it does	Main equipment	Critical-to-quality parameters	Typical defects	Environment
1	Material preparation	Qualify and condition powders, foils, separator, electrolyte	Silos, vacuum conveying, dryers, gravimetric feeders	Moisture, particle size (D50), purity, magnetic impurities	Metallic contamination, moisture pickup	Controlled, dust-tight
2	Slurry mixing	Disperse active material, carbon and binder in solvent	Planetary or twin-screw mixer, degassing, filters	Solid content, viscosity, mixing energy, temperature	Agglomerates, gelation, bubbles	Controlled; NMP area solvent-rated
3	Coating	Lay slurry on Al (cathode) or Cu (anode) foil	Slot-die coater, unwinder, tension control	Wet thickness, areal loading (mg/cm²), edge profile	Streaks, pinholes, edge build-up	Clean, dry
4	Drying	Evaporate solvent, fix electrode structure	Multi-zone air-flotation oven, NMP recovery	Zone temperatures, air velocity, residual solvent	Cracking, binder migration, delamination	Clean, dry
5	Calendering	Compress coating to target density and porosity	Two-roll press, gap control, heated rolls	Line force, roll gap, density, thickness	Wrinkles, foil breakage, density variation	Clean, dry
6	Slitting and notching	Cut mother rolls into lanes, form tabs	Rotary knife slitter, laser or die notcher	Burr height, width tolerance, dust	Burrs, particles, heat-affected zone	Dry room
7	Assembly	Build stack or jelly roll, weld tabs, case it	Winder or stacker, ultrasonic and laser welders	Alignment, anode overhang, weld strength	Misalignment, separator damage, weak welds	Dry room
8	Electrolyte filling	Dose electrolyte, wet the pores	Vacuum filling station, soak chambers	Fill mass, vacuum profile, wetting time	Under-fill, splash, moisture ingress	Dry room, tightest dew point
9	Formation	First charge: build the SEI on the anode	Formation cyclers, pressure fixtures	C-rate, voltage cut-offs, temperature, pressure	Lithium plating, gas, low capacity	Fire-separated, temperature-controlled
10	Degassing and final seal	Remove formation gas, close the cell for good	Vacuum degas and reseal (pouch), seal pin weld (prismatic)	Vacuum level, seal temperature and time	Leaks, seal wrinkles	Dry room
11	Aging and grading	Hold cells, measure OCV drift, sort	Aging racks, OCV/IR testers, capacity testers	OCV drop (K-value), capacity, DCIR	Micro-shorts, high self-discharge	Fire-separated, temperature-controlled
12	Final test and packaging	EOL inspection, labelling, shipment	Vision, insulation tester, leak tester, sorting robots	Voltage window, insulation resistance, dimensions, ID	Escapes, labelling errors	Warehouse with fire protection

How chemistry changes the process

The line architecture is the same for every chemistry; what changes is powder handling, slurry recipe, moisture tolerance, electrode density and formation. High-nickel NMC/NCA is the most process-sensitive; LFP is the most forgiving but the hardest to disperse.

Chemistry	Cathode / anode	Process consequences	Moisture sensitivity	Typical formats
LFP	$LiFePO_4$ (carbon-coated, fine particles) / graphite	High surface area needs more carbon and binder and lower slurry solids; dispersion is the main mixing challenge; lower calendered density than NMC **[T]**	Moderate	Large prismatic, blade, increasingly 46-series cylindrical
NMC (532 → 811)	$LiNi_xMn_yCo_zO_2$ / graphite (± silicon)	Dense electrodes, high calendering force; residual lithium on high-Ni powder can gel PVDF/NMP slurry, so powder stays dry and slurry pot life is short **[T]**	High, rising with Ni	Cylindrical, pouch, prismatic
NCA	$LiNiCoAlO_2$ / graphite-silicon	Behaves like high-Ni NMC: strict powder dryness, tight dew point	Very high	Mainly cylindrical
LTO	NMC or LMO / $Li_4Ti_5O_{12}$	Anode works above the SEI potential, so formation is short and gentle; anode can sit on Al foil; low energy density	Low–moderate	Niche prismatic and cylindrical

Chemistry	Cathode / anode	Process consequences	Moisture sensitivity	Typical formats
Sodium-ion	Layered oxide or Prussian-blue analogue / hard carbon	Largely drop-in on Li-ion lines; Al foil on both electrodes cuts cost and allows shipping at 0 V; Prussian-blue cathodes need strict water removal **[T]**	Oxide cathodes high	Cylindrical, prismatic

For your PFMEA: chemistry mostly moves the severity and occurrence scores at mixing, drying and dry-room stages. A high-Ni line needs tighter powder-moisture and dew-point controls than an LFP line of the same output.

How cell format changes the line

Format changes only stages 7–10. Electrode making is nearly identical, except for electrode width and notching pattern.

	Cylindrical	Prismatic	Pouch
Electrode build	Continuous winding into a jelly roll	Flat winding or stacking	Stacking (Z-fold or lamination-stack)
Line character	Very high speed, many small cells **[T]**	Fewer, large cells; stacking/winding is the assembly bottleneck	Like prismatic, plus pouch forming
Tab joining	Tabbed or tabless (full-edge); laser welding to can and cap	Ultrasonic pre-weld of foil bundles, laser weld to terminals	Ultrasonic weld of foil bundles to tab leads
Housing	Deep-drawn steel or Al can; crimped or welded cap	Al can, laser-welded lid	Al-laminate pouch, heat-sealed
Filling	Before closing	Through fill port; often two fills	Before sealing, with a gas pocket
Degassing	None after closing	Through fill port, then seal pin welded	Pierce gas pocket, vacuum, reseal, trim
Main risks	Winding telescoping, weld spatter, core burrs	Stack alignment, lid weld porosity	Seal integrity, swelling, pouch damage

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Where cost, energy, time and yield sit

Lever	What the sources say	Label	Source
Energy: coating and drying	Coating and drying including NMP recovery about 46% of production energy and about 11.5% of manufacturing cost	[V] research estimate	EEPower on ORNL study
Energy: whole plant	2 GWh prismatic NMC333 plant model: ~44.6 Wh of energy per Wh of cell capacity, dominated by drying and dry room	[V] model	SINTEF, Batteries 2021
Time: formation	PEM RWTH: forming is one of the most time-consuming and energy-intensive steps	[V]	Battery-News
Time: wetting + formation	Industry protocol cited by ORNL: total ~3–7 days excluding aging	[V]	Wood, NAATBatt 2019
Time: incl. aging	Industry sources cite ~1.5 to 3 weeks from first charge to release	[T]	Oxmaint
Yield	New lines ramp over 12–24 months from low yields; real numbers rarely published	[E]/[P]	Module 14

Engineer's takeaways:

1. Scrap at grading costs the full material plus 1–3 weeks of rack time. Catch defects at coating and calendering, where scrap is cheapest.
2. Dry-room air volume and leakage are an OPEX line you can engineer down.
3. Formation-rack count, not the coater, often caps plant output.

Factory environment primer

Zone	Typical dew point	Why	Label
Mixing and coating	Humidity-controlled; drier for high-Ni	Powders and slurry pick up water; NMP is a solvent hazard	[T]
Electrode handling, slitting, notching	about −35 to −45 °C	Absorbed water reacts with electrolyte later	[T]
Assembly and filling	−40 °C common, down to −60 °C for sensitive chemistries	$LiPF_6 + H_2O \rightarrow HF$, which attacks SEI and cathode	[T]
Formation and aging	Normal humidity; strict temperature and fire separation	Cells are sealed; risk is now thermal	[T]

Sources: AFRY (−35 to −45 °C during assembly), igus whitepaper (1% RH ≈ −40 °C dew point), Cleanroom Technology / ABN (−40 °C standard, −60 °C option).

Automotive parallel: treat dew point like booth temperature and humidity in a paint shop — a process parameter with an SPC chart, alarm limit and reaction plan, not a facilities metric.

Module 0 check

Interview questions

1. Walk me through a cell line from powder to shipped cell, and name the bottleneck.
2. Why is cathode slurry usually PVDF in NMP while anode slurry is water-based?
3. Why do assembly and filling need a lower dew point than coating?
4. What is formation, and why can't it be skipped or rushed?
5. Why is a defect found at grading much more expensive than one found at coating?
6. How does moving from NMC to LFP change your PFMEA?

Quiz

1. Which foil carries the anode, and why can sodium-ion cells use aluminium on both sides?
2. Name the two steps that dominate plant energy use.
3. In which stage does the SEI form?
4. Which format needs degassing through a gas pocket?
5. What does a high OCV drop during aging usually indicate?
6. What does the K-value measure?
7. Which build method is continuous and fastest: winding or stacking?
8. Under ACC PLI, how much capacity was commissioned on time by October 2025?

Hands-on exercise: on one page, sketch a process flow for a 5 GWh LFP prismatic plant. For each of the 12 stages write (a) the one parameter you would put on an SPC chart first, (b) the defect you most fear, (c) the room class. Keep it — it becomes your control-plan skeleton in Module 12.

Quiz answers

1. Copper. Lithium alloys with aluminium at anode potentials; sodium does not, so Al works for both electrodes.
2. Electrode drying (with NMP recovery) and dry-room air conditioning.
3. Formation (stage 9).
4. Pouch.
5. A self-discharge problem, often a micro-short from a particle or burr.
6. The rate of OCV drop over time during aging, usually in mV/day.
7. Winding.
8. 1.4 GWh (2.8% of 50 GWh), all by Ola Electric (IEEFA).

Next: Module 1 — Materials and incoming quality

Module 1 — Materials and Incoming Quality (Stage 1)

Li-ion Cell Manufacturing: an Engineer's Course · Phase A · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: roughly three-quarters of a cell's cost is material, and most field failures trace back to something that came in on a truck — a metallic particle, a wet powder, a thin spot in the separator. Incoming quality in a cell plant is closer to semiconductor practice (ppb-level contamination) than to automotive sheet-metal practice.

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1. The concept in simple language

A cell is a sandwich: a **cathode** (active powder glued to aluminium foil), a **separator** (porous plastic film), an **anode** (graphite glued to copper foil), all soaked in **electrolyte** (a lithium salt in organic solvents) and sealed in a **can or pouch**. Every layer must be pure, dry and dimensionally consistent, because the layers are only tens of microns thick and a single 50 µm metal particle can bridge them.

2. Engineering principle

Three properties govern almost every incoming specification:

1. **Electrochemical purity.** Foreign metals (Fe, Cu, Zn, Cr, Ni as free metal) dissolve at the cathode potential, migrate and re-deposit on the anode as dendrites → micro-short → self-discharge or thermal event. Hence **magnetic-impurity limits at ppb level**.
2. **Moisture.** Water + $LiPF_6$ → HF. HF attacks the cathode, dissolves transition metals and damages the SEI. Every hygroscopic material (cathode powder, separator, binder, electrolyte) carries a water spec.
3. **Particle and surface morphology.** Particle size (D10/D50/D90), specific surface area (BET) and tap density decide how much binder and carbon the slurry needs, how dense the electrode can be calendered, and how much lithium is lost forming the SEI.

3. Bill of materials of a typical cell

Indicative mass shares for an EV-type cell **[T]**: cathode active material is the largest single item (roughly 30–45% of cell mass for NMC; LFP's lower energy density means more of it per kWh), anode active material ~15–20%, electrolyte ~10–15%, foils ~10–15%, separator ~2–4%, housing and terminals the rest. Use your own cell's BOM for anything real.

4. Material-by-material specification

4.1 Cathode active materials (CAM)

Item	LFP	NMC (532/622/811)	NCA
Composition	$LiFePO_4$, carbon-coated (1–3% C) **[T]**	LiNi_xMn_yCo_zO₂	$LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$ (typical)
Function	Li source / host on charge-discharge	same	same
Key properties	Fine primary particles (sub-µm) aggregated to a few µm; high BET; ~3.2–3.3 V plateau	D50 ~ 3–15 µm secondary spheres or single-crystal; 3.6–3.7 V nominal	Similar to high-Ni NMC
Critical specs	D10/D50/D90, BET, tap density, carbon content, Fe^{2+}/Fe^{3+} phase purity, moisture, magnetic impurities	Same plus **residual lithium (LiOH + Li_2CO_3)**, cation mixing, pH	Same as NMC; residual Li critical
Moisture sensitivity	Moderate	High; rises with Ni	Very high

Item	LFP	NMC (532/622/811)	NCA
Storage	Sealed bags, dry store	Sealed, nitrogen or dry-air, FIFO, limited exposure time after opening **[T]**	As high-Ni
Common defects	Off-spec carbon coating, Fe impurities, agglomerates	Residual Li too high (gels PVDF slurry), surface carbonate, cracked secondaries, metallic Fe from processing	Same
Global suppliers	Hunan Yuneng, Shenzhen Dynanonic, Wanrun, Lopal **[T]**	Umicore, BASF, EcoPro BM, L&F, POSCO Future M, Ronbay, Easpring **[T]**	Sumitomo Metal Mining, EcoPro BM **[T]**
India	Altmin (Telangana, ₹750 cr plant under construction); Himadri (Odisha, 40,000 t/yr targeted by late 2027); Gujarat Fluorochemicals (small LFP line commissioned) **[V]**	No commercial-scale Indian NMC CAM yet reported **[T]**	—

India sources: Swarajya, 2026.

4.2 Anode active materials

Item	Graphite (natural / synthetic)	Silicon additives	Hard carbon (Na-ion)	LTO
Composition	C, spheroidised natural or synthetic (from needle coke)	SiOx or Si-C composite, a few % blended into graphite **[T]**	Disordered carbon from biomass/resin	$Li_4Ti_5O_{12}$
Key properties	D50 ~10–20 µm, BET low (~1–4 m^2/g), high first-cycle efficiency **[T]**	Large volume change on lithiation; low first-cycle efficiency	High irreversible capacity	Zero-strain, 1.55 V vs Li
Critical specs	D50, BET, tap density, ash content, magnetic impurities, first-cycle efficiency	Si content, particle size	Same as graphite	Same
Defects	High BET → more SEI loss; metallic impurities; off-spec coating	Swelling, cracking	High ICL	—
Global suppliers	BTR, Shanshan, Putailai, Shangtai, POSCO Future M **[T]**	Shin-Etsu, BTR, Sila, Group14 **[T]**	Kuraray, Chinese makers **[T]**	Toshiba (cells) **[T]**
India	Epsilon Advanced Materials (Karnataka; anode qualification plant; ₹145 cr incentive); HEG and Graphite India investing in anode plants **[V]**	—	—	—

4.3 Conductive additives

Item	Carbon black	CNT (carbon nanotubes)	Graphene / VGCF
Function	Electronic bridges between active particles	Long conductive network at low loading	Niche
Typical loading	1–3% of electrode dry mass **[T]**	0.5–1% (often as pre-dispersed paste) **[T]**	—
Key specs	BET, oil absorption, ash, metals	Purity, dispersion quality, residual catalyst metals (Fe, Co)	—
Suppliers	Imerys (Super P / C-NERGY), Cabot, Orion, Denka **[T]**	Cnano, LG Chem, OCSiAl **[T]**	—
Watch-out	Hard to wet; agglomerates	Catalyst metal residues are a self-discharge risk	—

4.4 Binders and solvents

Item	PVDF + NMP (cathode)	CMC + SBR + water (anode)	Alternatives
Composition	Polyvinylidene fluoride dissolved in N-methyl-2-pyrrolidone	Carboxymethyl cellulose (thickener/dispersant) + styrene-butadiene rubber latex (binder)	PAA, water-based cathode binders, PTFE (dry process)
Typical loading	1–4% PVDF (cathode dry mass) **[T]**	~1–2% CMC + ~1–2% SBR **[T]**	—

Item	PVDF + NMP (cathode)	CMC + SBR + water (anode)	Alternatives
Key specs	PVDF molecular weight, moisture; NMP purity ≥99.5–99.9%, water <~200 ppm **[T]**	CMC degree of substitution and viscosity; SBR solids, particle size; water conductivity (DI water)	—
Why	PVDF is stable at cathode potential; NMP dissolves it	Cheap, low-tox, stable at anode potential	Remove NMP cost and hazard
Hazards	NMP is reprotoxic; needs recovery and exposure control	Low	—
Suppliers	PVDF: Arkema, Syensqo (Solvay), Kureha, Dongyue · NMP: BASF, Mitsubishi Chemical, Chinese makers **[T]**	CMC: Ashland, Nippon Paper, Daicel · SBR: Zeon, JSR (now ENEOS Materials), LG Chem **[T]**	—
India	Gujarat Fluorochemicals produces fluoropolymers incl. battery-grade PVDF plans **[T — verify]**	Domestic CMC makers exist for other industries **[T]**	—

4.5 Electrolyte

Item	Detail
Composition	~1.0–1.2 M $LiPF_6$ in carbonate blend (EC + EMC/DMC/DEC) plus 1–5% additives (VC, FEC, LiDFOB, PS, etc.) **[T]**
Function	Ionic path between electrodes; additives shape SEI/CEI
Key specs	**Water < ~20 ppm, free acid (HF) < ~50 ppm [T]**, colour (APHA), conductivity, density, metal ions
Storage	Stainless drums/ISO tanks under dry N_2 or Ar; temperature-controlled; FIFO; shelf-life tracked
Defects	Water pickup → HF; discolouration ($LiPF_6$ decomposition); wrong additive dosing
Global suppliers	Tinci, Capchem, Guotai-Huarong, Enchem, Soulbrain, Mitsubishi Chemical **[T]**
India	Gujarat Fluorochemicals produces $LiPF_6$ commercially at Dahej; Neogen Chemicals makes electrolyte under licence (MU Ionic Solutions) and is building $LiPF_6$ with Morita **[V]**

4.6 Separator

Item	Detail
Composition	PE (wet process) or PP / PP-PE-PP (dry process) microporous film, usually ceramic-coated (Al_2O_3 or boehmite) and/or PVDF/aramid-coated **[T]**
Function	Electronic insulator, ionic conductor, thermal shutdown
Key specs	Thickness 9–25 µm (EV often 9–16 µm + coating), porosity ~35–50%, Gurley (air permeability), puncture strength, MD/TD shrinkage at 90–130 °C, shutdown temperature (~130 °C for PE), moisture **[T]**
Defects	Pinholes, gels, thin spots, coating flakes, wrinkles, static charge
Storage	Dry room or sealed with desiccant; temperature-limited to avoid shrinkage
Global suppliers	SEMCORP (Yunnan Energy New Material), Asahi Kasei, SK IE Technology, Toray, Senior, Sinoma **[T]**
India	TCPL Packaging announced entry; no Indian Li-ion separator plant yet in commercial production **[V]**

4.7 Current collector foils

Item	Aluminium foil (cathode)	Copper foil (anode)
Thickness	~10–20 µm (12–15 µm common) **[T]**	~4.5–10 µm (6–8 µm common) **[T]**
Process	Rolled (1xxx series alloys); some carbon-primed	Electrodeposited (ED) mostly; some rolled
Key specs	Thickness tolerance, tensile strength and elongation, pinholes, surface cleanliness (oil residue → poor adhesion), wettability (dyne)	Same plus roughness (Rz) both sides, oxidation/anti-tarnish layer
Defects	Pinholes, oil stains, edge waviness, wrinkles	Oxidation spots, pinholes, low elongation (cracks in winding)
Global suppliers	Dingsheng, Wanshun, UACJ, Showa Denko, Hindalco (Al) **[T]**	Nuode, Jiayuan, Lingbao Wason, SK Nexilis, Lotte Energy Materials, Solus **[T]**

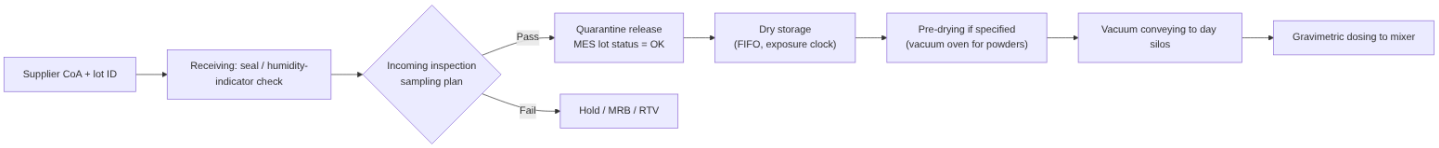
Item	Aluminium foil (cathode)	Copper foil (anode)
India	Hindalco battery-grade aluminium foil (Odisha) and an announced 11,500 t copper-foil plant at Waghodia, Gujarat **[V]**	

4.8 Housings and terminals

Item	Detail
Cylindrical	Deep-drawn Ni-plated steel can (or Al), cap assembly with CID, PTC and vent **[T]**
Prismatic	Deep-drawn or impact-extruded aluminium can (3003 alloy common), laser-welded lid with terminals, vent, fill port **[T]**
Pouch	Al-laminate film: nylon / aluminium / CPP (polypropylene) ~ 110–160 µm **[T]**; tab leads Al and Ni-plated Cu with sealant film
Key specs	Can wall thickness, flatness, burrs, cleanliness, vent burst pressure, laminate peel strength and pinholes
Suppliers	Cans and caps: Kedali, Shenzhen Everwin **[T]**; laminate film: DNP, Resonac (formerly Showa Denko), Zijiang **[T]**
India	Few dedicated battery-can suppliers yet; Indian deep-drawing and stamping base is a localisation opportunity **[E]**

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5. Process flow: receipt to line-side



6. Equipment for material handling

Equipment	Principle	Key parameters	Watch-outs
Big-bag / drum unloading stations	Contained discharge under extraction	Dust containment, operator exposure	Cross-contamination between cathode and anode materials — physically separate lines
Vacuum or dense-phase pneumatic conveying	Powder moved by gas pressure difference	Gas dryness (dew point), velocity (attrition of secondary particles), wear	Conveying NMC too fast cracks secondary particles **[T]**
Magnetic separators / demagnetisers	Permanent or electromagnet traps ferrous particles	Field strength (gauss), cleaning frequency	Must be validated with test pieces; record catches
Sieves (ultrasonic)	Removes oversize agglomerates/foreign matter	Mesh size	Screen wear sheds metal — inspect
Vacuum dryers	Remove adsorbed moisture	Temperature, vacuum, time	Over-drying binder/CMC can degrade
Gravimetric loss-in-weight feeders	Dose by weight loss rate	Accuracy, refill strategy	Calibrate under MSA

Material contact parts are typically ceramic-lined, tungsten-carbide or polymer-coated, and copper/zinc/brass is banned from cathode material contact to avoid metal contamination **[T]**.

7. Incoming inspection methods

Test	Applies to	Method / instrument	Why
Particle size distribution	CAM, AAM, carbon	Laser diffraction (D10/D50/D90)	Packing, rate capability, calendering
Specific surface area	CAM, AAM, carbon	BET N_2 adsorption	Binder demand, SEI loss
Tap density	CAM, AAM	Tap tester	Achievable electrode density
Moisture	All powders, separator, binder	Karl Fischer coulometric (oven method)	HF formation
Residual lithium / pH	High-Ni CAM	Acid–base titration	Slurry gelation
Magnetic impurities	CAM, AAM, carbon	Magnetic extraction + ICP-OES/MS	Micro-shorts

Test	Applies to	Method / instrument	Why
Elemental composition	CAM, electrolyte	ICP-OES	Stoichiometry, contaminants
Phase purity	CAM	XRD	Correct crystal structure
Morphology	CAM, AAM	SEM	Cracks, fines, coating
Electrolyte water / HF	Electrolyte	Karl Fischer; acid titration	HF control
Separator	Separator	Thickness gauge, Gurley, puncture, shrinkage oven, pinhole optical scan	Short-circuit protection
Foil	Al, Cu	Micrometer, tensile tester, dyne pens, pinhole light table	Adhesion, winding cracks
Electrochemical check	CAM, AAM	Coin-cell test (capacity, first-cycle efficiency)	Final functional gate

8. Common material defects and their effects

Defect	Detection	Effect in cell	Containment / prevention
Metallic particle (Fe, Cu, Zn) in powder	Magnetic extraction + ICP; later as OCV drop	Micro-short, high self-discharge, possible thermal event	Supplier magnet/sieve controls, incoming test, line magnets
High residual Li on NMC	Titration	Slurry gelation, gassing in formation	Dry storage, limited exposure time, supplier washing/coating spec
Wet powder	Karl Fischer	HF, capacity fade, gas	Sealed packaging, dew-point-controlled storage, pre-drying
Separator pinhole / thin spot	Optical inline scan at supplier; hi-pot after assembly	Hard short	Supplier inspection data per roll; incoming audit
Foil oil residue	Dyne test	Coating delamination	Supplier cleaning spec
Electrolyte with high HF	Titration	Fade, gas	Supplier spec, storage temperature control

9. Factory example

A cathode line in a new plant shows rising OCV-drop rejects at aging three weeks after a CAM lot change. The 8D finds the supplier changed a conveying hose to a cheaper grade that shed fine stainless particles. Incoming magnetic-impurity testing had used a sampling frequency of one sample per 10 lots. **Fix:** test every lot for magnetic impurities for 3 months after any supplier change notification (PCN), and add a supplier PCN clause to the PPAP. This mirrors automotive PPAP re-submission on process changes.

10. Interview questions

1. Why are magnetic-impurity limits for cathode material expressed in ppb and not ppm?
2. What is residual lithium, why does it matter for PVDF slurries, and how do you measure it?
3. Which incoming tests would you require for a new separator supplier?
4. How does BET surface area of graphite affect first-cycle efficiency?
5. Why is copper foil electrodeposited while aluminium foil is rolled?
6. How would you write a PPAP for a cathode material supplier?

11. Quiz

1. Name the three components of a typical Li-ion electrolyte.
2. What is the typical water limit for electrolyte?
3. Why is brass banned from cathode-powder contact parts?
4. What do CMC and SBR each do in the anode slurry?
5. Which separator property gives thermal shutdown?
6. Which Indian company produces $LiPF_6$ commercially?

Answers

1. Lithium salt ($LiPF_6$), carbonate solvents (EC + linear carbonates), additives (e.g. VC, FEC).
2. On the order of 20 ppm **[T]**.
3. Copper and zinc dissolve at cathode potential and re-plate on the anode, causing shorts.
4. CMC thickens and disperses; SBR binds.
5. PE melting (~130 °C) closes the pores.
6. Gujarat Fluorochemicals (Dahej).

12. Hands-on exercise

Draft an **incoming inspection plan** for an LFP cathode powder: list each characteristic, specification (use [T] windows from this module and mark them "to be agreed with supplier"), test method, sample size and frequency, reaction plan, and whether it is a special characteristic (SC/CC). Then do the same for a ceramic-coated separator. Keep both; they feed the control plan in Module 12.

Sources

- Swarajya — Indian cathode, anode, separator, foil, electrolyte makers (2026)
- PEM RWTH & VDMA — Production Process of a Lithium-Ion Battery Cell, 5th ed.
- Liu et al., iScience 2021
- Supplier names marked [T] are widely reported industry players listed from general knowledge; confirm current status before quoting.

Next: Module 2 — Slurry preparation

Module 2 — Slurry Preparation (Stage 2)

Li-ion Cell Manufacturing: an Engineer's Course · Phase A · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: the slurry fixes the electrode's microstructure before it ever touches foil. If carbon is not broken down and spread around the active particles, no amount of coating or calendering skill recovers the lost conductivity. Treat the mixer like a paint-shop mixing room: batch records, viscosity windows, pot life and filtration are all control-plan items.

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1. The concept in simple language

You are making a very thick, very clean "paint": active powder (the pigment), conductive carbon (makes it conduct), binder (makes it stick) and solvent (makes it flow). The aim is a homogeneous, bubble-free, agglomerate-free paste with the right viscosity to coat, which stays stable until it is coated.

2. Engineering principle

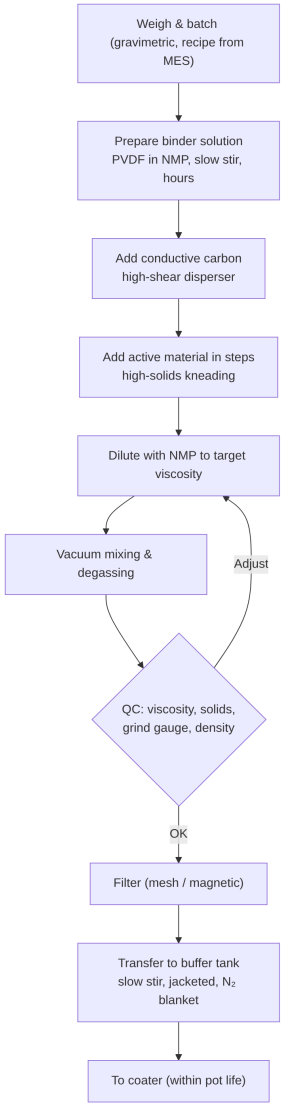
- **Dispersion = breaking agglomerates + stabilising the pieces.** Carbon black arrives as agglomerates of nano-particles; shear energy breaks them; binder adsorbs on the surface and stops them re-agglomerating.
- **Order of addition matters.** What touches the binder solution first gets coated first. Carbon is usually dispersed into the binder solution before the large active particles go in, so carbon ends up as a conductive network around them **[T]**.
- **Mixing intensity has an optimum.** Too little → agglomerates and poor conductivity. Too much → NMC secondary particles crack, carbon chains are shredded to over-fine particles that soak up binder, SBR latex is destabilised, and temperature rises **[T]**.
- **Rheology is shear-thinning.** The slurry must be viscous at rest (no settling) and thin under the shear of the coating die. You control viscosity at a defined shear rate, not one number.
- **Solids content is an economic and drying lever.** Higher solids = less solvent to evaporate in Stage 4 = lower energy and less cracking, but harder to process.

3. Cathode vs anode slurry

	Cathode (NMC, LFP)	Anode (graphite)
Binder system	PVDF dissolved in NMP	CMC (dispersant/thickener) + SBR latex (binder) in deionised water
Typical solids **[T]**	NMC ~65–75 wt%; LFP lower, ~45–60 wt% (high surface area)	~40–55 wt%
Typical dry formulation **[T]**	NMC: ~94–97% CAM, 1.5–3% carbon, 1.5–3% PVDF · LFP: ~90–95% CAM, 2–5% carbon, 2–5% PVDF	~95–97% graphite, 0–1% carbon, 1–2% CMC, 1–2% SBR
Key sensitivities	Moisture and residual Li (high-Ni) → **gelation** (PVDF cross-links under base)	SBR is shear- and temperature-sensitive; pH; foaming
Solvent handling	NMP: reprotoxic, recovered and recycled	DI water
Alternatives	Water-based cathode binders (LFP easier), dry electrode (PTFE fibrillation)	Same water route; PAA binders for Si

4. The industrial process

4.1 Batch process (planetary mixer) — cathode example



Steps in practice:

1. **Weighing and batching.** Gravimetric feeders, recipe pulled from MES, every weight recorded against batch ID. Powders arrive from day silos by vacuum conveying.
2. **Binder solution.** PVDF dissolved in NMP at ~5–10 wt% **[T]** for hours under slow agitation; must be clear and gel-free. Anode: CMC dissolved in DI water (~1–2 wt% solution) **[T]**.
3. **Conductive additive dispersion.** Carbon black or CNT paste into binder solution under high shear (disperser blade or rotor-stator).
4. **Active material addition.** Added in portions. Many recipes use a **high-solids kneading** phase (paste-like) to put high shear energy into the mix, then dilute **[T]**.
5. **Dilution** to target viscosity with solvent.
6. **Vacuum mixing and degassing** at low speed to remove entrained air (bubbles become pinholes on the coater).
7. **QC sampling:** viscosity, solids, grind gauge, density.
8. **Filtration** through mesh (and often magnetic) filters to remove agglomerates and metallic debris.
9. **Transfer and storage** in jacketed buffer tanks with slow stirring; pot life tracked.

Anode-specific: SBR is added **last, at low shear**, because high shear breaks the latex emulsion **[T]**.

4.2 Continuous process (twin-screw extruder)

Powders and liquids are fed continuously into a co-rotating twin-screw extruder whose screw sections convey, knead and disperse; slurry exits continuously. Benefits: smaller footprint, consistent energy input, less waiting

time, lower inventory; challenges: precise powder feeding, start-up scrap, cleaning. Bühler and Coperion are established suppliers; Dürr lists semi-continuous systems of 1–960 L/h and continuous systems of 20–2,000 kg/h **[V]** (Dürr brochure).

4.3 Dry-mixing variants

- **Dry pre-mix:** active material + carbon mixed dry (intensive mixer) so carbon coats particles before solvent is added. Shortens wet mixing.
- **Dry electrode process:** no solvent at all; PTFE binder fibrillated under shear, powder calendered into a free-standing film and laminated onto foil. Removes the coater oven and NMP recovery (the biggest energy users). Still being industrialised; Module 3 returns to it.

5. Equipment

Equipment	Working principle	Main components	Key parameters	Common failures	Suppliers [T]
Double-planetary mixer (+ high-speed disperser)	Two blades orbit and spin; disperser shaft adds high shear	Bowl (jacketed), planetary blades, disperser, vacuum pump, scraper	Planetary rpm, disperser tip speed, time per step, temperature, vacuum	Seal leakage, blade wear (metal shedding), temperature control drift	Inoue, PRIMIX, Ross, Dürr; Chinese makers
Twin-screw extruder	Continuous kneading in screw elements	Barrel zones, screws, gravimetric feeders, liquid pumps	Screw speed, throughput, specific energy, barrel temperature	Feeder inaccuracy, screw wear	Bühler, Coperion
Intensive dry mixer	Rotating pan and rotor	Pan, rotor	Tip speed, time	Wear	Eirich, Hosokawa
Bead mill (for CNT pastes)	Grinding media	Chamber, beads	Bead size, energy	Media wear contamination	Netzsch, Bühler
Binder dissolving tank	Slow agitation	Jacketed tank	Time, temperature	Undissolved gel	Many
Degassing / buffer tanks	Vacuum and slow stirring	Jacketed vessel, anchor stirrer	Vacuum, stir speed, temperature	Settling	Many
Filters	Mesh + magnetic	Bag/cartridge, magnet bars	Mesh size, ΔP	Clogging, bypass	Many
Inline viscometer / density meter	Rotational or vibrational	Sensor	—	Fouling	Anton Paar, Brookfield (AMETEK)

Wear parts that touch slurry are often ceramic-coated or tungsten-carbide to prevent metallic contamination **[T]**.

6. Parameter table

Parameter	Typical range	Purpose	Effect if off	Measurement	Control	Label
Solids content (NMC cathode)	~65–75 wt%	Coating weight, drying load	Low → cracking, more energy; high → poor flow, streaks	Moisture balance (gravimetric drying)	Solvent addition	[T]
Solids content (LFP cathode)	~45–60 wt%	As above	As above	As above	As above	[T]
Solids content (graphite anode)	~40–55 wt%	As above	As above	As above	As above	[T]
Viscosity at defined shear (e.g. 10–50 s^{-1})	Roughly 1–10 Pa·s, recipe-specific	Coatability, no settling	Low → edge flow, settling; high → streaks, bubbles trapped	Rotational viscometer, defined temperature	Solvent dilution	[T] / [P] for exact window
Grind gauge (fineness)	Max particle typically below ~30–50 µm	No agglomerates bigger than coating thickness	Agglomerates → streaks, protrusions, separator puncture	Hegman gauge	Dispersion energy, filter mesh	[T]

Parameter	Typical range	Purpose	Effect if off	Measurement	Control	Label
Slurry temperature	Kept roughly ≤ 40 °C	Avoid binder/solvent issues, gelation, SBR damage	Viscosity drift, gel	Jacket temperature sensor	Cooling jacket	[T]
Vacuum during degassing	Near full vacuum (e.g. −0.08 to −0.095 MPa gauge)	Remove air	Bubbles → pinholes	Gauge	Pump, time	[T]
Mixing time/speed per step	Hours (batch) vs minutes residence (continuous)	Dispersion	Under/over-mixing	Recipe log	MES recipe lock	[P]
Pot life	Hours to a day; shorter for high-Ni	Avoid gelation and settling	Viscosity rise, gel particles	Viscosity re-check	FIFO, max hold time	[T] / [P]
Water content in NMP slurry	Low hundreds of ppm	Prevent gelation & HF	Gelation	Karl Fischer	Dry materials, closed system	[T]

7. Common slurry defects

Defect	Root causes	Detection	Effect downstream	Prevention
Agglomerates	Insufficient shear, carbon added too fast, wrong sequence	Grind gauge, filter ΔP rise	Streaks, protrusions, separator damage, local resistance	Sequence, dispersion energy, filtration
Gelation (NMP slurries)	Residual Li/base on high-Ni CAM, moisture, long hold, heat	Viscosity rising with time, gel particles	Unusable batch, coating defects	Dry powders, low-residual-Li CAM, temperature and pot-life control, acid additives **[T]**
Sedimentation	Viscosity too low at rest, long storage without stirring	Solids difference top/bottom	Loading variation along roll	Rheology design, slow stirring
Bubbles	Air entrainment, insufficient degassing	Visual, density	Pinholes, craters	Vacuum degassing, closed transfer
SBR demulsification	High shear or temperature after SBR addition	Grit, viscosity change	Poor adhesion, agglomerates	Add SBR last at low shear
Metallic contamination	Wear of blades, seals, pumps	Magnetic filter catches, ICP	Micro-shorts	Ceramic/WC wear parts, magnets, PM checks
Viscosity drift	Temperature, solvent evaporation, ageing	Inline viscometer	Coating weight drift	Closed tanks, jacket control, re-check before use

8. Factory example

An NMC811 cathode line sees viscosity rising 30–40% within four hours of discharge, then gel lumps blocking the coater filter. Investigation: a new CAM lot met spec but was opened in a corridor at ambient humidity for a shift change; residual LiOH rose and moisture entered the slurry. **Actions:** limit opened-bag exposure time (alarm in MES), keep CAM in the dry zone end-to-end, add a viscosity-versus-time check at 2 h, reduce pot life for high-Ni recipes. In automotive language: a material-handling special characteristic was missing from the PFMEA.

9. Interview questions

1. Why is carbon black usually dispersed into the binder solution before the active material is added?
2. What causes PVDF/NMP slurry gelation with high-nickel cathodes, and how do you prevent it?
3. Why is SBR added last in an anode slurry?
4. Batch planetary vs continuous twin-screw mixing: when would you choose each?
5. Which slurry properties would you put on SPC charts, and why viscosity at a defined shear rate rather than a single value?
6. What does a rising filter differential pressure tell you?

10. Quiz

1. Typical solids content of an NMC cathode slurry?
2. What does a Hegman gauge measure?
3. Which binder system is used for graphite anodes?
4. Name two consequences of over-mixing.
5. Why degas under vacuum?

Answers

1. Roughly 65–75 wt% **[T]**.
2. Fineness of grind — largest particles/agglomerates in the slurry.
3. CMC + SBR in deionised water.
4. Cracked NMC secondaries, shredded carbon soaking up binder, SBR damage, temperature rise.
5. Entrained air becomes pinholes and craters in the coating.

11. Hands-on exercise

Write a **mixing work instruction and control plan extract** for a 1,000 L batch of LFP cathode slurry: steps, addition order, which parameters are recipe-locked in MES, in-process checks (what, when, spec, reaction plan), pot life and rework rules. Then list three PFMEA failure modes for this step with S/O/D ratings and your reasoning.

Sources

- Dürr — Advanced battery electrode systems brochure
- Kwade et al., Nature Energy 2018 (process–structure links; not opened this session)
- PEM RWTH & VDMA guide, 5th ed.

Next: Module 3 — Coating and drying

Module 3 — Coating and Drying (Stages 3 and 4)

Li-ion Cell Manufacturing: an Engineer's Course · Phase A · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: coat weight (mg/cm²) is the single most important electrode characteristic — it sets capacity and the anode-to-cathode balance. The oven behind the coater is the largest energy user in the plant, and how fast you dry decides whether the binder stays where it is needed. Coating and drying are one machine and must be engineered together.

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1. The concept in simple language

The slurry is pumped through a precision slot onto a moving foil, forming a wet film a few hundred microns thick. The foil then floats through a long oven where hot air evaporates the solvent, leaving a porous dry layer. Both sides of the foil are coated, either one after the other or at the same time.

2. Engineering principles

Coat weight and balance.

- Areal loading m (mg/cm²) × specific capacity (mAh/g) = areal capacity (mAh/cm²).
- The anode must carry slightly more capacity than the cathode it faces: the **N/P ratio** is usually ~1.05–1.20 **[T]**. Too low → lithium plating at the anode → safety risk. Too high → wasted anode and lower energy density.
- So coat-weight tolerance on *both* electrodes is a safety-relevant special characteristic, not just a capacity one.

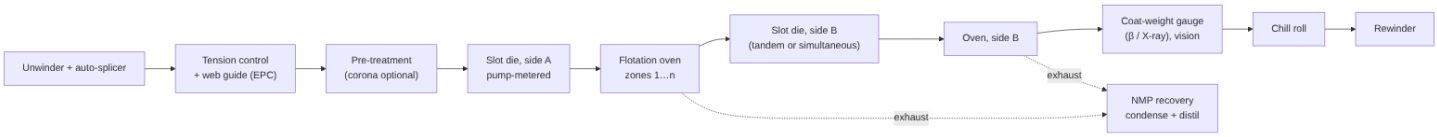
Slot-die physics.

- A metering pump delivers a fixed volumetric flow Q. Wet thickness ≈ Q / (web speed × coating width). The die does not "meter"; the pump does.
- The **coating bead** (liquid bridge between die lips and web) must be stable. The *coating window* in speed–flow–gap space is bounded by air entrainment, ribbing, rivulets/streaks and bead break-up. Vacuum boxes behind the die extend the window at high speed.
- KIT demonstrated slot-die primer coating stable up to 500 m/min with a vacuum box and gaps of 105–250 µm — a lab result for thin primers, not a production speed for active layers **[V]** (KIT, J. Coat. Technol. Res. 2024).

Drying physics.

- Two phases: (1) **film shrinkage** at constant rate while solvent leaves the surface and particles pack together; (2) **pore emptying**, where capillary forces pull liquid (and dissolved/suspended binder) through pores.
- High drying rates in the shrinkage phase drag binder to the surface → **binder migration** → binder-depleted foil interface → poor adhesion and delamination; binder-rich top → higher resistance **[T]**.
- Thick water-based films crack from capillary stress when dried too fast **[T]**.
- Hence the classic **low–high–low** oven profile: gentle start, maximum heat once the film has consolidated, gentle finish **[T]**.

3. The industrial process



3.1 Coating methods

Method	How it works	Pros	Cons	Use
Slot-die	Pre-metered: pump flow through slot	Precise, closed system, high speed, sharp edges, intermittent possible	Die design, setup, sensitive to bead stability	Dominant in gigafactories **[T]**
Comma (knife-over-roll)	Excess slurry metered by gap between comma bar and roll	Simple, tolerant of viscosity	Lower precision and speed, open slurry pool (solvent loss, contamination)	Older lines, lab/pilot, some anodes **[T]**
Gravure / spray	—	Thin layers	Not for thick active layers	Primers, ceramic coatings on separators

3.2 Continuous vs intermittent

- **Continuous:** stripes run the length of the roll with uncoated margins for tabs (used for winding with continuous tabs, and for laser-notched stacking electrodes).
- **Intermittent (skip) coating:** coating stops and starts to leave uncoated gaps along the length (some prismatic and pouch designs). Requires a fast-acting valve or shim actuator; start and stop edges ("head and tail") tend to be thick or thin and are a key quality topic **[T]**.

3.3 Single-side, tandem and simultaneous two-side coating

- **Tandem:** side A coated and dried, web turned, side B coated and dried — two ovens or one double-pass oven.
- **Simultaneous:** both sides coated almost at once and dried in a flotation oven (the web floats on air cushions with nothing touching the wet coating). Dürr lists its GigaCoater at 1,400 mm width, 40 m/min, 40–48 m flotation dryer, ~2 GWh/yr; GigaCoaterXL at 2,200 mm, 60 m/min, 50–70 m dryer, ~5 GWh/yr **[V]** (Dürr).

3.4 Drying and solvent recovery

- **Oven types:** convective air-flotation (most common), roll-support ovens, with infrared or laser assist on newer lines.
- **NMP recovery:** exhaust air is cooled to condense NMP, sometimes followed by adsorption or scrubbing, and the recovered NMP is purified by distillation and reused. Dürr states recovery rates typically above 99% **[V]**.
- **Water-based anode drying:** no recovery needed; still multi-zone for crack control.
- **Energy:** coating + drying including NMP recovery is reported at about 46% of production energy **[V]** (EEPower on ORNL).

3.5 Dry electrode coating (emerging)

Solvent-free powder is fibrillated (PTFE), calendered into a film and laminated to foil. It removes the oven and NMP recovery entirely. Industrial maturity is still limited and process windows are largely proprietary **[P]**.

4. Equipment

Unit	Function	Main components	Key parameters	Quality checkpoints	Common failures	Maintenance
Unwinder + splicer	Feed foil without stopping	Shaft, auto-splice, dancer	Tension, splice timing	Splice position tracking (scrap length)	Wrinkles at splice	Bearing, pneumatic checks
Tension control	Constant web tension	Load cells, dancers, drives	Tension per zone (N)	Tension log	Foil breaks (thin Cu!)	Load-cell calibration
Web guide (EPC)	Lateral position	Edge sensor, steering frame	Edge position	Edge position SPC	Drift → edge misregistration	Sensor cleaning
Slot die	Lay down film	Die body, shims, lips, manifold	Gap, flow, die-to-web distance, vacuum box	Wet weight, edge profile	Lip damage (streaks), dried slurry at lips	Lip polishing, shim inspection
Slurry delivery	Constant flow	Buffer tank, gear or progressive-cavity pump, filters, degasser	Flow (mL/min), pressure, temperature	Pump speed vs coat weight	Pulsation (chatter marks), filter clogging	Pump wear check
Oven	Remove solvent	Zones, nozzles, fans, heaters, exhaust	Zone temperature, air velocity, exhaust flow, LEL monitoring	Residual solvent, cracks, adhesion	Uneven airflow (cross-web variation), nozzle blockage	Nozzle cleaning, balancing
Coat-weight gauge	Measure mg/cm²	β-ray or X-ray scanner, cross-web traverse	Accuracy, scan speed	Closed-loop to pump	Drift	Standard-foil calibration
Vision system	Surface defects	Line-scan cameras, lighting	Resolution	Defect map per meter	False calls	Lens cleaning
NMP recovery	Recover solvent	Condensers, cooling, distillation, scrubber/adsorber	Recovery %, NMP purity, emissions	Water content of recovered NMP	Fouling	Exchanger cleaning
Rewinder	Wind electrode roll	Shaft, lay-on roll	Taper tension	Roll hardness	Telescoping	—

Equipment suppliers [T]: Hirano Tecseed and Toray Engineering (Japan, precision coaters), Dürr (Germany, coaters and NMP recovery), PNT and other Korean firms, Lead Intelligent and Yinghe (China). Korean suppliers hold about 40% of front-end electrode equipment sales among the top-25 suppliers, with PNT ranked second globally **[V]** (Control Engineering).

5. Parameter table

Parameter	Typical range	Purpose	Effect if off	Measurement	Control	Label
Areal loading, EV energy cathode	~15–30 mg/cm² per side (chemistry-specific)	Capacity	Capacity spread; N/P error → plating	β/X-ray gauge, punched-disc weighing	Pump closed loop	[T]
Areal capacity	~2–4+ mAh/cm² (energy cells higher)	Cell design	—	Calculated	Design	[T]
N/P ratio	~1.05–1.20	Prevent plating	Plating / lost energy	Design + both gauges	Coordinated weight targets	[T]
Coat-weight tolerance	Often ±1–2% of target	Uniform capacity	Capacity spread, plating risk	Gauge	Closed loop	[T] / [P]
Wet film thickness	~100–400 µm	—	—	Wet gauge / calculation	Flow	[T]
Line speed (production)	Tens of m/min (Dürr: 20–60 m/min by model)	Throughput	Too fast → bead instability, under-drying	Encoder	Drive	[V] Dürr
Web width	Up to ~1,400–2,200 mm on giga lines	Throughput	—	—	—	[V] Dürr
Oven length	Tens of metres (Dürr: 15–70 m by model)	Residence time	—	—	—	[V] Dürr

Parameter	Typical range	Purpose	Effect if off	Measurement	Control	Label
Zone temperatures (NMP)	Roughly 60–150 °C, low–high–low profile	Drying rate	Binder migration, cracks, residual solvent	Thermocouples, web-temperature pyrometer	Zone PID	[T]
Air velocity at nozzles	Recipe-specific	Heat/mass transfer	Cross-web non-uniformity	Anemometer survey	Fan speed	[P]
Residual NMP after drying	Low hundreds of ppm or below	Prevent side reactions	Gas, capacity loss	GC on samples	Oven profile	[T]
Edge build-up ("ear")	Minimised; spec per design	Calendering & winding	Edge cracks, wrinkles	Profile gauge	Shim design, edge-bead control	[T]
Adhesion (peel strength)	Spec per recipe	Integrity	Delamination	90°/180° peel test	Binder, drying profile	[P]
NMP recovery rate	>99%	Cost, emissions	Cost, permit breach	Mass balance	Recovery system	[V] Dürr

6. Coating and drying defects

Defect	What it looks like	Root causes	Detection	Effect	Corrective / preventive
Streaks	Lines along the web	Agglomerates/particles stuck in die lip, damaged lips, dried slurry, unstable bead (low flow)	Vision, coat-weight cross profile	Local thin/thick, lithium plating at thin anode spots	Filter slurry, lip inspection & cleaning, adjust gap/flow/vacuum
Pinholes / craters	Small uncoated spots	Bubbles, contamination (oil, dust), poor wetting of foil	Vision, light table	Local plating, corrosion	Degassing, foil cleanliness (dyne), clean room discipline
Cracking	Mud-crack pattern	Too-fast drying, thick water-based films, low binder	Vision, microscopy	Particle loss, poor contact	Lower initial drying rate, binder/formulation
Uneven coating (cross-web)	Thickness profile not flat	Die manifold design, temperature differences, oven airflow	Gauge cross profile	Capacity spread	Die adjustment (flex lip), oven balancing
Chatter / barring (along-web)	Periodic thickness bands	Pump pulsation, roll eccentricity, vibration	Gauge frequency analysis	Capacity spread	Pulsation damping, bearing checks
Edge defects	Thick edge bead, wavy or ragged edge	Surface tension at edge, shim design	Profile gauge, vision	Calendering cracks, winding wrinkles, misalignment	Shim design, edge-bead management
Head/tail defects (intermittent)	Thick start, thin end	Valve dynamics	Gauge	Over/under capacity locally	Valve tuning
Bubbles	Blisters	Entrained air, solvent boiling (oven too hot early)	Vision	Pinholes	Degassing, oven profile
Delamination	Coating lifts from foil	Binder migration, foil oil/oxide, under-drying	Peel test, tape test	Capacity loss, particles	Drying profile, foil spec, binder
Residual solvent	—	Insufficient drying	GC	Gas, side reactions	Oven profile, speed
Foil wrinkles / breaks	—	Tension errors, thin Cu, heat	Web break detection	Downtime, scrap	Tension tuning, splice quality

7. Factory example

A new anode coater shows cross-web coat weight 3% higher at the drive-side edge. Gauge data rule out the die. An oven airflow survey finds a nozzle header partly blocked by dried SBR residue on the drive side, so that edge dries slower and shows more binder migration and higher retained water — the gauge reads that as weight. **Fix:** clean the headers, add a quarterly nozzle airflow survey to the PM plan, and add cross-web profile Cpk to the daily quality meeting. Lesson: a gauge measures "mass per area", which includes solvent; always check residual solvent when a profile looks wrong.

8. Interview questions

1. Explain why the slot die does not set the coat weight. What does?
2. What is the N/P ratio and why is coat-weight tolerance on the anode a safety characteristic?
3. Describe binder migration and how oven zoning controls it.
4. How would you diagnose a periodic (along-web) coat-weight variation?
5. Simultaneous vs tandem two-side coating: trade-offs?
6. What would dry electrode coating remove from the plant, and what new risks does it bring?

9. Quiz

1. Name three things that bound the slot-die coating window.
2. Typical NMP recovery rate quoted by Dürr?
3. What profile is used across oven zones, and why?
4. What instrument measures coat weight inline?
5. Which defect is most associated with too-fast drying of thick water-based anodes?

Answers

1. Air entrainment, ribbing/streaks (low-flow limit), bead break-up (high speed); vacuum extends it.
2. Above 99% **[V]**.
3. Low–high–low: avoid binder migration early, maximise drying once consolidated, cool gently.
4. β-ray or X-ray scanning gauge.
5. Cracking.

10. Hands-on exercise

A 5 GWh/yr LFP line needs cathode output of X m² per hour. Using your cell design (choose: 300 Ah prismatic, cathode loading 25 mg/cm² per side, 160 mAh/g, 3.2 V, both sides coated, 90% overall material yield **[E]**), calculate (a) required coated area per year, (b) required line speed for a 1,400 mm web with 85% OEE, 7,500 h/yr, (c) how many coaters you need. Then compare with Dürr's 2–3 GWh per GigaCoater figure and explain any gap.

Sources

- Dürr — Advanced battery electrode brochure (coater sizes, speeds, NMP recovery)
- KIT — High-speed slot-die coating of electrode primers (2024)
- EEPower on ORNL dry-processing study
- Zhang et al., electrode drying review (UCL preprint)
- Control Engineering — battery equipment market shifts

Next: Module 4 — Calendering

Module 4 — Calendering (Stage 5)

Li-ion Cell Manufacturing: an Engineer's Course · Phase A · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: calendering trades porosity for energy density. Press too little and the electrode is thick and poorly connected; press too hard and you close the pores electrolyte needs, crack particles and wrinkle the foil. The target is a density (g/cm³) with a tight cross-web and along-web tolerance — exactly like a thickness tolerance on a rolled sheet, but with an electrochemical penalty for missing it.

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1. The concept in simple language

The dried coating is fluffy (40–50% empty space). Two very stiff, very round rolls squeeze it to a target thickness, packing particles closer together so more active material fits in the same volume and the particles touch each other and the foil.

2. Engineering principle

Porosity from density.

```
\varepsilon = 1 - \frac{\rho_{electrode}}{\rho_{true}}, \qquad
\frac{1}{\rho_{true}} = \sum_i \frac{w_i}{\rho_i}
```

where *w_i* are dry mass fractions and ρ_i the true densities of active material, carbon and binder.

*Worked example **[E]**:* NMC811 cathode, 96% CAM (4.75 g/cm³), 2% carbon (1.9 g/cm³), 2% PVDF (1.78 g/cm³). 1/ρ_true = 0.96/4.75 + 0.02/1.9 + 0.02/1.78 = 0.2021 + 0.0105 + 0.0112 = 0.2238 → ρ_true ≈ 4.47 g/cm³. At an electrode density of 3.4 g/cm³, ε ≈ 1 − 3.4/4.47 ≈ 24%.

Trade-offs governed by porosity:

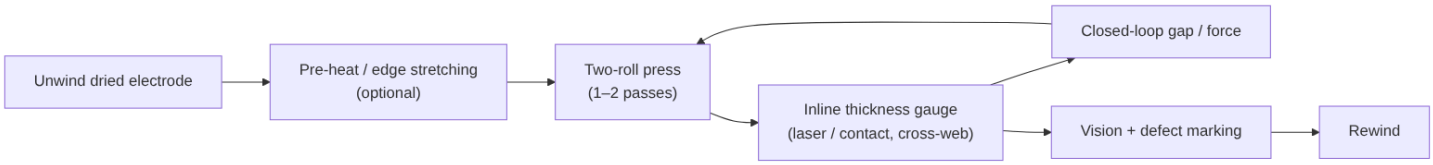
Lower porosity (more compression)	Higher porosity (less compression)
+ Volumetric energy density	+ Ionic transport (rate capability, fast charge)
+ Particle-to-particle and particle-to-foil contact (electronic)	+ Electrolyte wetting, faster filling
− Tortuosity ↑ → ionic resistance ↑ (Bruggeman: effective conductivity ∝ ε^1.5) **[T]**	− Weaker mechanical integrity, particle shedding
− Particle cracking, foil stretching, wetting time ↑	− Lower energy density

So energy cells run denser electrodes, power/fast-charge cells run more open ones.

Mechanics. The coating is compacted only where it is coated; the uncoated foil margin is not. The coated region elongates more than the bare edge, building stress at the boundary → wavy edges, wrinkles, even foil tears. Solutions: pre-stretch rolls for margins, heated rolls, reducing compaction steps, or uncoated-edge pressing **[T]**.

Roll deflection under load makes the gap larger in the middle; rolls are **crowned** or use bending compensation so cross-web thickness stays flat **[T]**.

3. The industrial process



Calenders are often integrated with slitting in one machine ("calendering-slitting integrated") on newer lines **[T]**. Some plants calender twice (rough + finish) for high-density targets.

4. Equipment

Item	Detail
Working principle	Two hardened rolls (forged steel, hardness often >60 HRC), driven, with hydraulic force on chocks; gap or force control **[T]**
Main components	Work rolls (often 500–1,000+ mm diameter on production lines), hydraulic cylinders, gap sensors, roll heating (oil or induction), drives, unwind/rewind, tension control, thickness gauges, roll cleaning brushes and dust extraction **[T]**
Control modes	**Gap control** (constant gap; force varies with coating) or **force control** (constant line load; gap varies). Production usually uses gap control with force monitoring **[T]**
Widths	Dürr lists calendering widths 250–1,300 mm **[V]**
Key parameters	Line load (N/mm of width), roll gap, roll temperature, speed, tension, number of passes
Quality checkpoints	Thickness (cross and along web), density (by punched-disc mass ÷ volume), surface defects, edge waviness
Common failures	Roll surface damage (pits print through), bearing wear (thickness periodicity at roll circumference), hydraulic drift, heating non-uniformity
Maintenance	Roll regrinding schedule, bearing condition monitoring, gauge calibration, roll cleaning
Suppliers **[T]**	Saueressig (Matthews International), PNT and other Korean makers, Chinese makers including Yinghe and Lead Intelligent; Japanese precision press makers

5. Parameter table

Parameter	Typical range	Purpose	Effect if off	Measurement	Control	Label
NMC cathode density	~3.2–3.6 g/cm³ (porosity ~20–30%)	Energy density	Too high → poor rate, cracking; too low → low energy, weak contact	Disc mass ÷ thickness × area	Gap/force	[T]
LFP cathode density	~2.2–2.6 g/cm³	As above	As above	As above	As above	[T]
Graphite anode density	~1.5–1.7 g/cm³ (porosity ~25–35%)	As above; anode porosity also governs plating risk in fast charge	Too dense → plating at high C-rate	As above	As above	[T]
Thickness reduction	~20–40% of dry coating thickness	—	—	Gauge before/after	—	[T]
Thickness tolerance	Often ±2–3 µm per side on EV electrodes	Stack/jelly-roll fit, capacity	Can fit issues, capacity spread	Laser/contact gauge	Closed loop	[T]/[P]
Roll temperature	Ambient to ~120 °C (hot calendering)	Binder softening, easier densification, less springback	Too hot → binder smearing on rolls	RTD in roll / IR	Heating system	[T]
Line load	Hundreds to >1,000 N/mm	Achieve density	Too high → particle/foil damage	Load cells	Hydraulics	[T]
Speed	Similar to or above coating speed; tens to ~100 m/min	Throughput	Too fast → springback, defects	Encoder	Drive	[T]
Springback after calendering	A few % thickness recovery over hours	Final thickness	Over-thick electrode at assembly	Re-measure after rest	Over-compress allowance	[T]

Parameter	Typical range	Purpose	Effect if off	Measurement	Control	Label
Edge wave height	As low as possible; spec per design	Winding/stacking quality	Wrinkles, misalignment	Optical	Edge treatment	[P]

6. Defects

Defect	Root causes	Detection	Effect	Prevention / correction
Foil wrinkles / wavy edges	Differential elongation coated vs uncoated, high line load	Vision, edge-wave sensor	Winding misalignment, tab cracks	Edge pre-stretching, multi-pass, heated rolls, design (narrower margins)
Foil breakage	Excess load, thin Cu foil, edge cracks from coating	Web break detector	Downtime, scrap	Load optimisation, edge quality upstream
Over-densification	Gap too small, coat-weight high spot	Density check	Poor rate, slow wetting, lithium plating (anode)	Closed-loop on thickness, coat-weight control
Under-densification	Gap too large, springback	Density check	Low energy, poor contact, particle shedding	Gap calibration
Particle cracking (NMC secondaries)	High load, low porosity target	SEM cross-section	Fresh surface → gas, fade	Limit density, single-crystal CAM
Periodic thickness variation	Roll eccentricity, bearing wear	FFT of gauge signal vs roll rotation	Capacity spread	Roll regrind, bearing replacement
Cross-web thickness variation	Roll deflection, temperature gradient	Cross-web gauge	Capacity spread, winding telescoping	Roll crown/bending, temperature uniformity
Roll marks / dents	Debris on rolls	Vision	Separator damage	Roll cleaning brushes, dust extraction
Binder smearing / sticking	Roll too hot, binder-rich surface	Visual	Defects, roll contamination	Temperature limits, upstream drying
Delamination	Coating adhesion poor, excessive shear at nip	Peel test	Particles, capacity loss	Adhesion upstream

7. Factory example

After a CAM supplier switch to a single-crystal NMC, the same calender settings give 3.55 g/cm³ instead of 3.45 g/cm³, and cells show slower electrolyte wetting and lower 3C capacity. **Lesson:** density is a function of material as well as press settings; the calendering target belongs in the material change PPAP. Re-run the density–pressure curve (a "compaction curve") for every new material and update the control plan.

8. Interview questions

1. Derive porosity from electrode density and formulation.
2. Why do uncoated foil margins cause wrinkles during calendering, and how do you fix it?
3. Gap control or force control — which would you use in production and why?
4. How does anode porosity relate to fast-charge capability and lithium plating?
5. You see a thickness signal repeating every 2.5 m along the web. Where do you look?
6. What is springback, and how do you allow for it in the target?

9. Quiz

1. Typical density window for an NMC cathode?
2. Effective ionic conductivity in porous electrodes scales roughly as ε to what power (Bruggeman)?
3. Why are calender rolls crowned?
4. Name two benefits of hot calendering.
5. What measurement is used to check density offline?

Answers

1. ~3.2–3.6 g/cm³ **[T]**.
2. ~1.5.
3. To compensate roll deflection so the gap is uniform cross-web.
4. Easier densification at lower load, less springback (binder softens).
5. Punched disc: mass ÷ (area × thickness), minus foil.

10. Hands-on exercise

Using your formulation, build a spreadsheet that converts target porosity (20–35%) into target density and calendered thickness for a given coat weight. Plot thickness vs porosity (column chart of a few points). Then define the control plan line for calendered thickness: spec, gauge, frequency, Cpk target (≥1.67 for an SC), and reaction plan.

Sources

- Dürr — calendering widths
- Kwade et al., Nature Energy 2018 (calendering–structure relationships; not opened this session)
- PEM RWTH & VDMA guide, 5th ed.

Next: Module 5 — Slitting, notching and cutting

Module 5 — Slitting, Notching and Cutting (Stage 6)

Li-ion Cell Manufacturing: an Engineer's Course · Phase A · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: this is where electrodes get their final geometry and where the most dangerous defect — a metal burr or loose particle that can pierce a separator — is created. Width and tab position decide whether assembly can hit its overhang tolerances. Burr height is a safety special characteristic.

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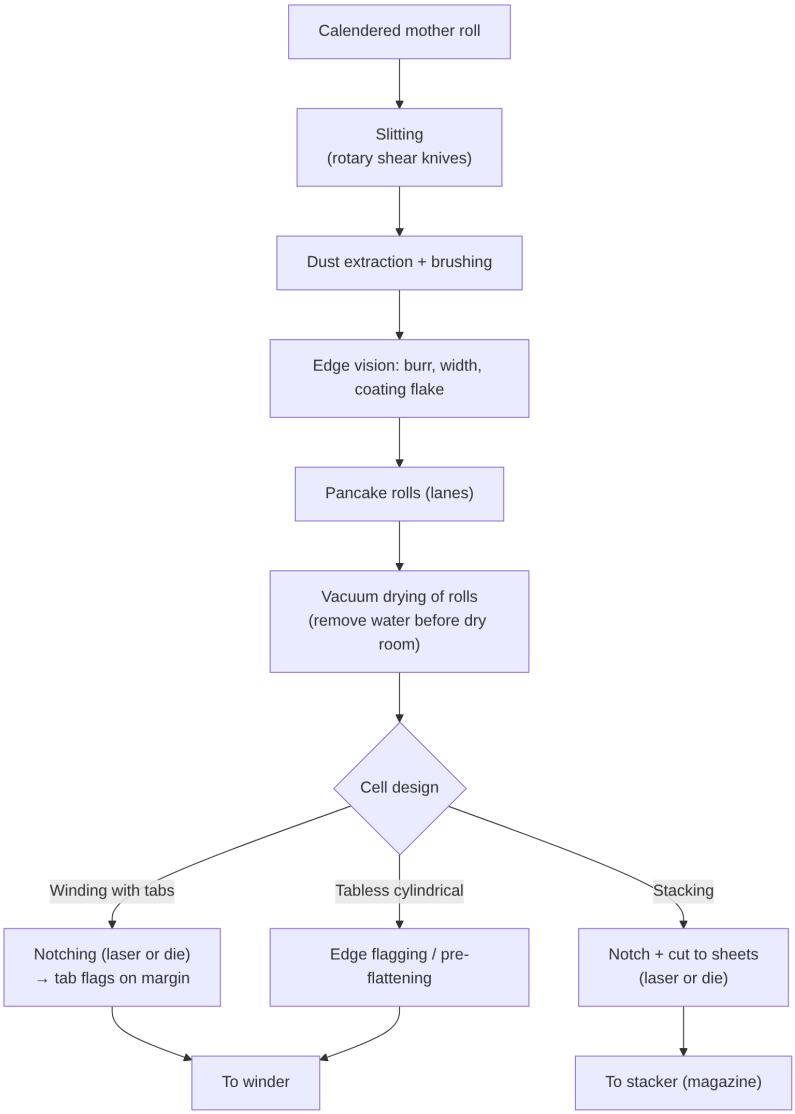
1. The concept in simple language

The wide calendered "mother roll" is cut lengthwise into narrow strips (slitting). For many designs, the uncoated foil margin is then cut into tabs (notching), and for stacked cells each strip is cut into individual sheets (cutting). Every cut must be clean, dimensionally exact and free of loose dust.

2. Engineering principle

- **Shear cutting** of a brittle coating on a ductile foil: the coating fractures, the foil yields then separates. A worn or misaligned blade lets the foil deform before it separates → **burr** (raised edge) and **coating flaking** at the edge.
- **Laser cutting** ablates/melts the material: no tool wear, flexible geometry, but creates a **heat-affected zone (HAZ)**, molten droplets (spatter/"beads"), and coating retraction at the edge.
- **Safety logic:** separators are ~9–25 µm. A burr that is a meaningful fraction of that, pressed by stack pressure and swelling, can cut through → internal short. Burr limits are therefore set well below separator thickness **[T]**.
- **Geometry logic:** the anode must overhang the cathode on all sides (so no lithium plates at uncovered anode edges), and the separator must overhang the anode. Width tolerances and tab position stack up into these overhangs (Module 6).

3. The industrial process



Slitting. Pairs of circular top and bottom knives shear the web (shear slitting); knife overlap, side load, cant angle and sharpness determine edge quality **[T]**. Dürr quotes slitting up to 50 m/min with a lay-on roll **[V]** (Dürr).

Vacuum drying. Electrode rolls (or later, jelly rolls/stacks) are vacuum-dried at elevated temperature for hours to reduce absorbed water before entering the dry room or before filling **[T]**. Exact temperature/time/vacuum per material are recipe-specific **[P]**.

Notching.

- *Laser notching:* pulsed fibre lasers (ns, increasingly ps/fs for less HAZ) cut tab shapes from the moving web at high speed. Flexible (tab shape is software), no tool wear; risks are HAZ, spatter, and fumes **[T]**.
- *Mechanical die notching:* punch and die, very clean edges when sharp, high precision; but die wear (regrinding), changeover cost and speed limits **[T]**.

Cutting to sheets (stacking designs). Die or laser cut into individual electrodes, then delivered to stacker magazines. Some lines combine notching, cutting and stacking in one machine ("cut-and-stack").

4. Equipment

Equipment	Working principle	Key parameters	Quality checkpoints	Common failures	Maintenance	Suppliers [T]
Slitter (shear)	Rotary top/bottom knives	Knife overlap, side force, speed, tension, knife sharpness	Width, burr, edge flake, dust	Blade wear → burrs; knife misalignment	Blade change by length cut, alignment checks	Nishimura, Kampf, Yinghe, Lead Intelligent, Korean makers
Laser notcher	Pulsed laser ablation with scanner	Power, pulse width, frequency, scan speed, focus, assist gas, extraction	Burr, HAZ width, bead/spatter, tab dimension	Lens contamination, focus drift, extraction failure	Optics cleaning, power calibration	IPG, Trumpf, Coherent (sources); Hymson, Han's Laser, Lead Intelligent (machines)
Die notcher / cutter	Punch and die	Clearance, stroke speed, die sharpness	Burr, dimension	Die wear, chipping	Regrind schedule by strokes	Japanese and Korean precision die makers; Chinese makers
Vacuum dryer (rolls)	Heat + vacuum	Temperature, vacuum, time, N_2 backfill dew point	Moisture by KF	Leaks, uneven heating	Leak tests	Many
Dust extraction and brushes	Suction + brushing	Air velocity, brush condition	Particle counts	Clogged filters	Filter change	Many
Inline vision	Line-scan cameras	Resolution (few µm/px)	Burr, flakes, dimensions	False calls	Calibration	Keyence, Cognex, Basler-based integrators

5. Parameter table

Parameter	Typical range / spec	Purpose	Effect if off	Measurement	Control	Label
Burr height	Well below separator thickness; often specified around ≤10 µm	Prevent separator puncture	Internal short	Microscope / laser profilometer on samples; inline vision	Blade/laser maintenance	[T]
Electrode width tolerance	Tenths of a mm or tighter for stacking	Overhang control	Overhang loss, plating, misalignment	Vision, caliper	Knife setup, EPC	[T]
Tab position / pitch	Tight; design-specific	Tab alignment after winding	Tab misalignment, weld failure	Vision	Encoder sync	[P]
HAZ width (laser)	Minimised; tens of µm-level	Edge integrity	Coating retraction, binder damage	Microscopy	Laser parameters	[T]
Spatter / beads	None loose; fused beads within spec	Prevent particles	Shorts	Vision, tape lift test	Extraction, parameters	[T]
Dust / loose particles	Per site particle spec	Prevent shorts	Self-discharge, shorts	Particle counts, tape test	Extraction, brushes	[T]
Knife life	Meters cut per regrind (track)	Burr control	Rising burrs	Burr trend	Preventive change	[P]
Electrode moisture after vacuum drying	Hundreds of ppm or lower	Reduce HF	Gas, fade	Karl Fischer	Drying recipe	[T]/[P]

6. Defects

Defect	Root causes	Detection	Effect	Prevention
Burrs	Worn/misaligned knives, wrong overlap, die wear, laser beads	Vision, microscopy	Separator puncture, short	Tool life control, burr SPC, laser parameter windows
Coating flaking at edge	Poor adhesion, dull blades, HAZ	Vision, tape test	Particles → micro-shorts	Adhesion upstream, sharp tools

Defect	Root causes	Detection	Effect	Prevention
Dimensional error (width, tab)	Web wander, tension, encoder slip	Vision	Overhang loss, tab misalignment	EPC, tension control, encoder checks
Edge waviness / camber	Calendering stresses released	Visual	Winding telescoping	Calendering edge control
Tab cracks / folds	Handling, sharp corners in notch design	Vision	Weld failures, resistance	Tab radius design, guiding
Moisture pickup	Exposure time before drying	KF	HF, gas	Max exposure time, dry-room transfer

7. Factory example

Weekly burr audits show a steady rise from 4 to 9 µm over 5 km of slit length on anode lanes, then a sudden jump to 14 µm. Root cause: a knife regrind interval set by calendar time rather than metres cut; a speed increase on the coater pushed more metres through per week. **Fix:** knife change by cumulative metres in the MES, burr SPC with a reaction limit at 8 µm, and inline vision for flake detection. The same logic as tool-life management on a press line.

8. Interview questions

1. Laser vs die notching: compare quality, cost, flexibility, and maintenance.
2. How would you set the burr specification and justify it?
3. What stacks up into anode overhang, and how does slitting contribute?
4. Why vacuum-dry electrodes before cell assembly, and how do you validate the recipe?
5. How do you control loose particles generated at this stage?

9. Quiz

1. What is a HAZ?
2. Why must the anode overhang the cathode?
3. Name two knife setup variables in shear slitting.
4. Which measurement method verifies moisture after vacuum drying?
5. What failure does a 15 µm burr risk with a 12 µm separator?

Answers

1. Heat-affected zone — material altered by laser heat near the cut.
2. To avoid lithium plating where cathode faces no anode.
3. Overlap, side force, cant angle (also sharpness, speed).
4. Karl Fischer titration.
5. Separator puncture → internal short.

10. Hands-on exercise

For a stacked pouch cell with cathode 200 × 100 mm, define anode and separator sizes and tolerances so that the anode overhangs the cathode by ≥1 mm and the separator overhangs the anode by ≥1 mm on every side **[E]**, including slitting/cutting tolerance, stacking placement tolerance and thermal shrinkage. Do a worst-case and an RSS tolerance stack-up, and decide which tolerances are SCs.

Sources

- Dürr — slitting speeds
- Comau dry-room lab (laser and mechanical notching development)
- PEM RWTH & VDMA guide, 5th ed.

Next: Module 6 — Cell assembly

Module 6 — Cell Assembly: Winding, Stacking, Welding, Housing (Stage 7)

Li-ion Cell Manufacturing: an Engineer's Course · Phase B · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: assembly turns continuous electrodes into discrete cells. Three things decide quality here — **alignment** (overhangs), **cleanliness** (no particles in the stack) and **joints** (tab welds). It is the most "automotive-like" part of the plant: high-speed automation, welding, vision, poka-yoke and takt time.

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1. The concept in simple language

Cathode, separator, anode and separator are layered — either rolled up like a Swiss roll (winding) or laid flat like a deck of cards (stacking). The bare foil tabs of all cathode layers are welded together to the positive terminal, the anode tabs to the negative. The assembly goes into a can or pouch, which is closed except for a fill opening.

2. Engineering principles

Overhang rules (the core design constraint).

- Anode > cathode on every edge, so every cathode area faces anode (otherwise lithium plates on anode edges).
- Separator > anode on every edge, so electrodes never touch.
- Typical anode-over-cathode overhang is around 0.5–2 mm per side and separator-over-anode a similar order **[T]**; exact values are design IP **[P]**.
- Overhang is consumed by width tolerance (Module 5), placement/alignment tolerance, winding telescoping, and separator shrinkage.

Winding vs stacking.

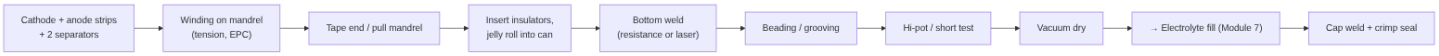
	Winding (jelly roll)	Stacking
Motion	Continuous rotation	Discrete pick-place or Z-fold
Speed	Very high	Lower per layer; improving with multi-station and lamination stacking **[T]**
Space use in prismatic can	Rounded corners waste space; bending stress at fold radius	Flat layers fill the can; better volumetric efficiency
Stress	Inner-turn curvature stress, deformation on swelling	Uniform
Current path	Long electrode, tab positions matter (multi-tab or tabless improves)	Each layer has its own tab → low resistance
Typical	Cylindrical (all), many prismatic	Pouch, many large prismatic (e.g. blade-type)

Welding physics.

- **Ultrasonic welding** (20–40 kHz **[T]**) joins stacks of thin foils (dozens of layers) by high-frequency scrubbing under pressure — solid-state, no melting, ideal for Al and Cu foil bundles.
- **Laser welding** (fibre lasers; green/blue wavelengths absorb better in copper) joins thicker parts: foil bundle to connector, connector to terminal, lid to can. Beam wobble reduces porosity and spatter **[T]**.
- **Resistance welding** joins cylindrical anode tab to can bottom through the jelly-roll core with an electrode rod **[T]**.

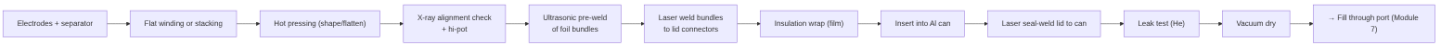
3. Process flows by format

3.1 Cylindrical (e.g. 18650, 21700, 46-series)



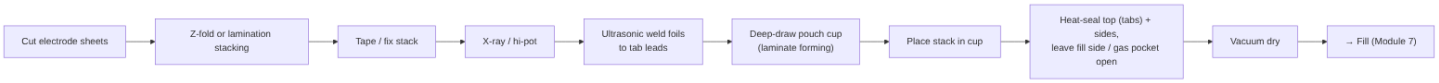
- **Tabbed design:** a single (or few) tab welded to the foil; the current travels along the whole electrode length.
- **Tabless / full-edge design (46-series):** the uncoated foil edge is flattened ("flagged" and pressed) and laser-welded to a current-collector disc at each end — much lower resistance and heat in large-diameter cells **[T]**.

3.2 Prismatic



Large prismatic cells often contain two or more jelly rolls/stacks in parallel in one can **[T]**.

3.3 Pouch



Z-fold stacking: a continuous separator zig-zags while cathode and anode sheets are inserted alternately into the folds. **Lamination stacking:** electrode and separator are heat-laminated into mono-cells or bi-cells first, then stacked — better alignment and handling at speed **[T]**.

4. Equipment

Equipment	Principle	Main components	Key parameters	Quality checkpoints	Common failures	Suppliers [T]
Winder (cylindrical / prismatic)	Electrodes and separators wound on mandrel under tension	Unwinds ×4, tension dancers, EPC, mandrel, cutters, taping, CCD	Tension per material, speed, EPC accuracy, mandrel geometry	Alignment (CCD), overhang, telescoping, roll diameter, hi-pot	Tension spikes → separator neck-down, misalignment, wrinkles	Lead Intelligent, Yinghe, CKD, Koem; Korean makers
Z-fold stacker	Separator zig-zag, electrode pick-place	Magazines, robots/grippers, vision, separator unwind	Cycle per layer, placement accuracy, separator tension	Per-layer vision, X-ray	Mis-picks (double sheets), static, particles	Lead Intelligent, Hymson, Geesun, Korean makers
Lamination stacker	Heat-laminate mono-cells then stack	Laminating rolls, cutters, stacking table	Lamination temp/pressure, accuracy	Peel strength, alignment	Delamination	Korean and Chinese makers
Hot press	Flatten/shape roll or stack	Heated platens	Temperature, pressure, time	Thickness	Separator damage if overheated	Many
Ultrasonic welder	20–40 kHz vibration	Generator, converter, booster, horn, anvil	Energy, amplitude, pressure, time	Peel/pull test, weld-area vision	Horn wear, foil tearing, under-weld	Branson (Emerson), Schunk, Telsonic, Herrmann
Laser welder	Fibre (IR), green or blue laser	Source, scanner/wobble head, fixture, shielding gas	Power, speed, wobble, focus	Penetration (cross-section), porosity, spatter, leak test	Spatter → particles, porosity, cracks	Trumpf, IPG, Coherent; Han's Laser, Hymson, United Winners
Resistance welder	Current through contact	Power supply, electrodes	Current, time, force	Pull test	Electrode wear	Amada, Miyachi

Equipment	Principle	Main components	Key parameters	Quality checkpoints	Common failures	Suppliers [T]
X-ray / CT inspection	Transmission imaging	Source, detector, AI evaluation	Resolution	Anode overhang per layer, tab alignment, foreign particles	Throughput limits	Nikon, Zeiss, Comet Yxlon, Chinese makers
Hi-pot / insulation tester	Apply DC voltage across electrodes before filling	Tester, fixtures	Test voltage (hundreds of V), dwell, leakage limit	Insulation resistance	False passes (fixture contact)	Hioki, Chroma
Helium leak tester	Mass spectrometer detects He	Chamber, vacuum, MS	Leak rate limit	Weld integrity	Contamination	Inficon, Pfeiffer

5. Parameter table

Parameter	Typical range / spec	Purpose	Effect if off	Measurement	Control	Label
Anode-over-cathode overhang	~0.5–2 mm per side	Prevent plating	Plating at edges, safety	X-ray/CT, vision	Width tolerance + alignment	[T]/[P]
Separator-over-anode overhang	~1–3 mm	Prevent contact	Short at edges	X-ray, vision	As above	[T]/[P]
Winding alignment (layer-to-layer)	Tenths of mm	Overhang	Telescoping, plating	CCD per turn	EPC	[T]
Winding tension	Recipe per material	Tight roll without stretch	Neck-down, wrinkles, loose roll	Load cells	Dancers	[P]
Ultrasonic weld energy/amplitude	Recipe per bundle	Joint strength, no foil tear	Weak or torn joints	Weld log, peel test	Generator closed loop	[P]
Laser weld penetration	Design-specific	Mechanical + electrical joint	Leak, high resistance	Cross-section (destructive), leak test	Power/speed	[P]
Hi-pot test voltage	Hundreds of volts DC	Detect shorts/particles before filling	Escapes → failures later	Insulation tester	Recipe	[T]
Insulation resistance	Megaohm range minimum	—	—	—	—	[T]
Can/lid seal leak rate	Very low (He leak test, mbar·L/s scale)	Hermeticity	Moisture ingress, leakage	He leak test	Laser weld control	[T]
Particles in dry room (assembly)	ISO class per site	Prevent shorts	Self-discharge	Particle counters	HVAC, gowning	[T]

6. Defects

Defect	Root cause	Detection	Effect	Prevention
Misalignment / overhang loss	Web wander, width error, stacking placement error, telescoping	X-ray/CT, CCD	Plating, shorts	EPC, vision feedback, X-ray 100% on critical designs
Separator damage (holes, folds)	Burrs, particles, tension spikes, gripper damage	Hi-pot, visual	Short	Clean handling, tension control, gripper design
Foreign particle in stack	Dust, weld spatter, cutting debris	Hi-pot, X-ray, later OCV drop	Micro-short	Extraction at welding, dry-room cleanliness, spatter shields
Weak ultrasonic weld	Horn wear, contamination, wrong energy	Peel test (sample), weld monitoring	High resistance, open circuit	Horn PM, monitoring signatures
Torn foil at weld	Excess energy/amplitude	Vision	Particles, high resistance	Parameter window

Defect	Root cause	Detection	Effect	Prevention
Laser weld spatter / porosity	Contamination, wrong parameters, Cu reflectivity	Vision, leak test, cross-section	Particles, leaks	Wobble, green/blue lasers, cleaning
Lid weld leak	Gap, contamination, power drift	He leak test	Electrolyte leak, moisture ingress	Fixture gap control, monitoring
Pouch forming cracks	Over-deep draw, laminate defect	Visual, pinhole test	Leak	Draw-depth limits, laminate spec
Short (hi-pot fail)	Any of above	Hi-pot	Scrap (cheap now, expensive later)	100% hi-pot before filling

7. Factory example

A prismatic line has a 0.8% hi-pot failure rate, three times the target. Pareto by station and shift shows failures concentrated after the laser station that welds foil bundles to connectors on the night shift. Spatter shields are cleaned once per shift; at night, extraction filter differential pressure runs high (filter changed on day shift only). Particle counts inside the station rise in the last 3 hours of the night shift. **Fix:** extraction ΔP interlock, shield cleaning every 4 hours, and a particle-count trend on the station HMI. Classic process-capability problem with a maintenance root cause.

8. Interview questions

1. Compare winding and stacking for a 300 Ah prismatic LFP cell.
2. Why use ultrasonic welding for foil bundles and laser welding for terminals?
3. What does the hi-pot test detect, and why is it done before filling?
4. Explain tabless cylindrical design and why it matters for large-diameter cells.
5. How would you verify anode overhang in production, and what sampling would you use?
6. What causes telescoping in a jelly roll and how do you prevent it?

9. Quiz

1. What frequency range does ultrasonic welding use?
2. Which laser wavelengths absorb better in copper?
3. In Z-fold stacking, what is continuous: the separator or the electrodes?
4. Why are two or more jelly rolls often placed in one large prismatic can?
5. Name the inspection method used to measure overhang inside an assembled stack.

Answers

1. ~20–40 kHz **[T]**.
2. Green and blue.
3. The separator.
4. To limit roll thickness/size for process and thermal reasons while reaching high capacity.
5. X-ray / CT.

10. Hands-on exercise

Balance an assembly line for a 5 GWh/yr, 280 Ah, 3.2 V prismatic LFP cell (≈ 0.9 kWh per cell). (a) Compute cells per year and required takt time at 7,500 h/yr. (b) If a stacker builds one cell in 60 s **[E]** and a winder in 20 s **[E]**, how many machines do you need for each technology at 85% OEE? (c) Which downstream stations (ultrasonic welding, laser welding, lid sealing) become bottlenecks if their cycle is 15 s **[E]**? Draw the line with parallel stations.

Sources

- Control Engineering — equipment market by process step (Chinese suppliers 72% of top-25 assembly sales)
- RWTH eLab pouch line (separation, deep drawing, Z-folding, ultrasonic welding, heat sealing, filling)
- PEM RWTH & VDMA guide, 5th ed.

Next: Module 7 — Electrolyte filling and wetting

Module 7 — Electrolyte Filling and Wetting (Stage 8)

Li-ion Cell Manufacturing: an Engineer's Course · Phase B · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: the cell is a sealed sponge of micro-pores; electrolyte must reach every pore before the first charge. Too little electrolyte or a dry spot means local lithium plating and short life; any moisture at this step becomes HF inside a sealed can. Filling is a dosing problem (mass accuracy) and a soaking problem (time, vacuum, temperature) in the driest room of the plant.

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1. The concept in simple language

Pull a vacuum on the cell so the pores are empty, inject a precise mass of electrolyte, let pressure push it in, and wait — sometimes with repeated vacuum and pressure cycles — until the electrodes and separator are fully soaked. Then close the fill opening temporarily or permanently.

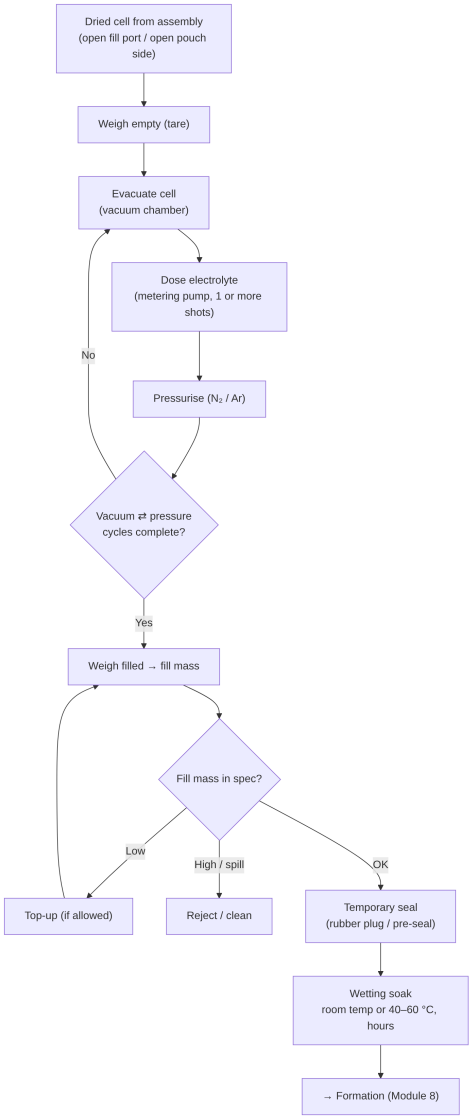
2. Engineering principle

- **Capillary penetration** into the electrode follows a Washburn-type behaviour: penetration depth grows with the square root of time and depends on pore size, contact angle and viscosity. Denser, thicker electrodes (Module 4) wet slower **[T]**.
- **Trapped gas** blocks pores; vacuum before dosing removes it, and pressure after dosing pushes liquid in. Cycling vacuum and pressure ("breathing") speeds wetting **[T]**.
- **Temperature** lowers viscosity and speeds wetting, but raises volatility and side reactions.
- **Fill amount** must fill all pore volume (electrodes + separator) plus a reserve consumed by SEI formation and ageing. It is often expressed as grams of electrolyte per Ah of capacity **[T]**; exact values are design IP **[P]**.
- **Water chemistry:** $LiPF_6 + H_2O \rightarrow LiF + POF_3 + 2\ HF$. This is why filling sits in the strictest dew-point zone and why electrolyte water spec is ~20 ppm **[T]**.

3. Electrolyte preparation and supply

Item	Practice
Composition	~1.0–1.2 M $LiPF_6$ in EC + linear carbonates (EMC/DMC/DEC) with additives (VC, FEC, LiDFOB, PS…) **[T]**
Where made	Usually bought pre-mixed from electrolyte makers; some large cell makers blend in-house **[T]**
Storage	Stainless drums/ISO containers under dry N_2 or Ar, temperature-controlled, shelf life tracked
Supply	Closed, inert-gas pressurised lines from storage to filling machines; filters; no air contact
Checks	Water (KF), free acid (HF titration), colour, conductivity, per lot and at point of use for drift

4. The industrial process



By format

Format	Fill access	Closing after fill	Second fill?
Cylindrical	Open can top before cap	Cap placed and crimped (or welded) — final seal immediately	No
Prismatic	Fill port in lid	Temporary rubber/plug seal; final seal-pin welded after formation and degas	Often yes — top-up after formation **[T]**
Pouch	Open side with gas pocket	Pre-seal; final seal after degassing (Module 9)	Rarely

Wetting time

An industrial protocol cited by ORNL: electrode wetting 6–24 h at room temperature before the first formation cycles, a further 12–24 h after them, and electrolyte wetting 12–24 h at 40–60 °C; wetting plus formation took about 3–7 days in total **[V]** (Wood, ORNL, NAATBatt 2019). Modern lines shorten this with pressure cycling and warm soaks; their exact recipes are proprietary **[P]**.

5. Equipment

Equipment	Principle	Main components	Key parameters	Quality checkpoints	Common failures	Maintenance
Filling machine (multi-station rotary or linear)	Vacuum + metered dosing + pressure	Vacuum chambers, dosing pumps (ceramic piston or gear), nozzles/needles, valves, inert-gas supply, load cells	Vacuum level and hold, shot volume, pressure, cycle count, temperature	Pre/post weights, fill mass Cpk	Nozzle drips/clogs ($LiPF_6$ crystallises), pump wear, vacuum leaks, seal wear	Nozzle cleaning, pump calibration (MSA on mass), leak checks
Checkweighers	Weigh before/after	Precision load cells in dry room	Resolution (0.01 g-level for large cells)	Gauge R&R	Drift with temperature, static	Daily calibration with standard weights
Soak / wetting chambers or racks	Hold at temperature (± pressure)	Heated racks, pressure vessels	Temperature, time, pressure	Time-temperature log per cell	Uneven heating	Temperature mapping
Electrolyte supply system	Closed inert-gas transfer	Drums/tanks, pumps, filters, piping, moisture sensors	Line pressure, water content	KF at point of use	Moisture ingress at connections	Leak tests, purging
Fume/solvent extraction	Local exhaust	Hoods, scrubbers	Air velocity	Area monitoring (HF, solvent)	Saturation	Filter change

Typical suppliers **[T]**: filling machines from Chinese automation makers (Lead Intelligent, Yinghe and others) and Korean/Japanese specialists; pumps from precision-dosing makers.

6. Parameter table

Parameter	Typical range / spec	Purpose	Effect if off	Measurement	Control	Label
Fill mass	Design value (g); often ~1.3–2 g/Ah order depending on chemistry/porosity	Fill pores + reserve	Low → dry spots, fade; high → gas space loss, swelling, cost	Weigh before/after	Pump calibration, closed loop	[T]/[P]
Fill mass tolerance	Tight; often within about ±1%	Uniform cells	Capacity/life spread	Checkweigher, Cpk	—	[T]
Pre-fill vacuum	Deep vacuum (e.g. a few kPa absolute or lower)	Remove gas from pores	Slow/incomplete wetting	Vacuum gauge	Pump, leak tightness	[T]
Pressure step	Up to a few bar inert gas	Push electrolyte into pores	Slow wetting	Gauge	Valves	[T]
Vacuum–pressure cycles	Several	Wetting speed	Dry spots	Recipe log	PLC recipe	[P]
Soak temperature / time	RT or 40–60 °C; 6–24 h stages in a cited protocol	Complete wetting	Plating in formation	Rack logs	Recipe	[V] ORNL slide
Dew point in filling zone	Strictest zone (−40 to −60 °C class)	Prevent HF	Gas, fade	Dew-point sensors	HVAC	[T]
Electrolyte water at point of use	~20 ppm order	HF control	Gas, fade	KF	Closed transfer	[T]
Time from vacuum drying to fill	Limited (hours)	Prevent moisture pickup	HF	MES timestamps	Max-time interlock	[P]

7. Defects

Defect	Root causes	Detection	Effect	Prevention / correction
Under-fill	Pump wear, bubbles in supply, nozzle clog, leak	Post-fill weight	Dry spots, fade, plating	Pump PM, degassed supply, 100% weighing

Defect	Root causes	Detection	Effect	Prevention / correction
Over-fill	Calibration error, double shot	Weight	Swelling, venting, cost	Interlocks
Spill / electrolyte on surface	Nozzle drip, splash during pressurisation	Visual, conductivity swab	Corrosion, poor seal (pouch), poor weld (seal pin)	Nozzle design, wipe/clean station, sealing area protection
Incomplete wetting	Short soak, dense electrodes, low temperature	Formation anomalies (high impedance, plating), EIS	Capacity loss, plating	Soak recipe validation (DOE), EIS check
Moisture contamination	Dew-point excursion, wet electrolyte, long exposure	KF, gas in formation	HF, gas, fade	Dew-point SPC, exposure-time interlocks
Leakage after fill	Bad temporary seal, damaged port	Weight loss, smell, sniff test	Moisture ingress	Seal maintenance
Contamination (particles)	Nozzle wear, dirty fixtures	Later OCV drop	Micro-short	Clean fixtures, filters

8. Factory example

A prismatic line shows bimodal capacity at grading: a sub-population 1.5% low. Correlating MES data shows the low cells all came from filling station 7 of 12. Gauge R&R on that station's checkweigher is fine; but its dosing pump shows a slowly rising pressure ripple — a worn piston seal letting small gas pockets into the shot. Weight still passed because the low limit was set wide. **Fix:** tighten the fill-mass limit to a Cpk-based window, add per-station fill-mass SPC with run rules, and schedule pump seal replacement by cycle count. The data link from cell ID → station → parameters is why traceability (Module 14) matters.

9. Interview questions

1. Why is electrolyte filled under vacuum, and why cycle vacuum and pressure?
2. How do you set the fill mass for a new cell design?
3. Why do prismatic cells often get a second fill after formation?
4. How would you validate the wetting time?
5. What does $LiPF_6$ do with water, and what are the consequences inside the cell?
6. What data do you need to trace a capacity problem back to a filling station?

10. Quiz

1. Name three electrolyte components.
2. What closes a cylindrical cell after filling?
3. Which reaction makes moisture so damaging?
4. In the ORNL-cited industrial protocol, what temperature range is used for the warm electrolyte wetting step?
5. What is the main measurement for fill quantity?

Answers

1. $LiPF_6$ salt, EC + linear carbonates, additives.
2. Cap placement and crimp (or weld).
3. $LiPF_6 + H_2O \rightarrow HF$ (plus LiF, POF_3).
4. 40–60 °C **[V]**.
5. Weighing before and after (gravimetric).

11. Hands-on exercise

For your 280 Ah LFP prismatic cell, estimate the pore volume: cathode and anode porosity (from Module 4), electrode volumes, separator porosity (~40% **[T]**) and volume. Convert to electrolyte mass (density ~1.2 g/cm³

[T]), then add a reserve of 10–20% **[E]**. Compare your result in g/Ah with the [T] range above, and write the control-plan line for fill mass with a Cpk target.

Sources

- Wood, ORNL — formation and wetting (NAATBatt 2019)
- Wood et al., Joule 2019 — formation challenges (not opened this session)
- PEM RWTH & VDMA guide, 5th ed.

Next: Module 8 — Formation

Module 8 — Formation (Stage 9)

Li-ion Cell Manufacturing: an Engineer's Course · Phase B · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: formation is the first, carefully controlled charge that builds the solid electrolyte interphase (SEI) — a nanometre-thin film on the anode that decides the cell's cycle life, self-discharge and safety. It is slow, energy-hungry and occupies huge rack space, so it is both the cell's "birth" and the plant's capacity bottleneck. Every hour cut from formation without hurting the SEI is money.

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1. The concept in simple language

A brand-new cell cannot be charged like a phone battery. The first lithium that moves into the graphite reacts with the electrolyte and forms a protective skin on the graphite. If that skin forms slowly and evenly, the cell lives long. If it forms fast and patchy, the cell loses lithium, makes gas and may plate metallic lithium. So the first charge is slow, warm, often under pressure, and fully monitored.

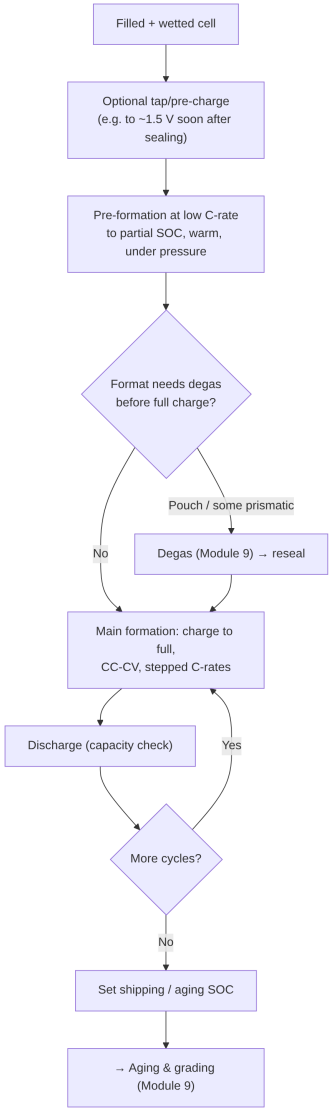
2. Engineering principle

- **SEI formation.** On first charge, the anode potential falls below the stability limit of the electrolyte (EC reduction starts around ~0.8 V vs Li/Li^+ **[T]**). Solvent and additive molecules (VC, FEC) are reduced to an inorganic/organic film (Li_2CO_3, LiF, $ROCO_2Li$, polymers). Once formed, the SEI conducts Li^+ but blocks electrons, stopping further reaction.
- **Irreversible capacity loss.** Lithium consumed in the SEI is lost for ever: first-cycle efficiency for graphite cells is typically ~85–95% **[T]**. That is why cathodes carry excess lithium relative to usable capacity and why silicon (higher loss) is hard.
- **CEI.** A thinner cathode–electrolyte interphase forms on high-voltage cathodes (NMC/NCA), more relevant at high Ni and high cut-off voltage **[T]**.
- **Gas.** SEI reactions release ethylene, CO, CO_2, H_2 **[T]**; pouch and many prismatic designs must remove it (Module 9).
- **Rate–quality trade-off.** Low current gives a dense, uniform SEI and avoids lithium plating while wetting is still incomplete; high current risks plating and a porous SEI. ORNL tests on NMC811 found 10-hour formations caused plating and worse retention, while 26–30-hour protocols performed best and longer gave no further gain **[V]** (ORNL slides).
- **Temperature.** Warm formation (often ~35–45 °C **[T]**) improves kinetics and wetting; too hot accelerates side reactions.
- **Pressure.** Pouch and prismatic cells are often formed between pressure plates so gas does not create voids between layers and the stack stays in contact (hot-press formation) **[T]**.

3. Formation vs normal charging

	Formation	Normal charging
Purpose	Build SEI/CEI, activate cell	Store energy
Current	Low at start (e.g. ~0.02–0.2 C), stepped up **[T]**	0.3–1C+ or fast charge
Voltage window	Staged; may stop at partial SOC for degassing	Full window
Temperature	Controlled, often elevated	Ambient/BMS-managed
Mechanical	Often under fixture pressure	Module/pack compression
Data	Every cell logged at high resolution; used for grading	BMS only
Repeat	Once (1–3 cycles)	Thousands of cycles

4. The industrial process



Notes:

- **Tap charge:** ORNL's NMC811 test matrix applied a 1.5 V tap charge after vacuum sealing **[V]**. Purpose: get the anode potential away from the region where copper can dissolve during long rest periods **[T]**.
- An industrial protocol cited by ORNL: 1–2 formation cycles at 0.1–0.2C, then 1–2 cycles at 0.2–0.5C, with wetting steps in between; ~3–7 days including wetting **[V]**.
- **Faster formation research:** ORNL showed "shallow cycling" between 3.9–4.2 V for NMC532 cut formation time by at least 6× without hurting performance, and a high-SOC C/5 / low-SOC 1C scheme reached ~8.5× faster than baseline **[V]**. Industrial adoption is design-specific **[P]**.

5. Equipment

Item	Detail
Formation cycler (charger/discharger)	Multi-channel bidirectional power electronics; each channel drives one cell (or series strings in some designs) with precise current and voltage. Modern units **recover discharge energy** back to the DC bus or grid **[T]**
Main components	Power modules, channel boards, contact probes/fixtures, temperature sensors per cell, rack chassis, DC bus, controller, data servers
Key specs	Current range, voltage accuracy (typically ~±0.02–0.05% of full scale class **[T]**), current accuracy, channel count, energy recovery efficiency, sampling rate

Item	Detail
Formation racks / hot-press formation	Racks hold trays of cells; for pouch/prismatic, heated pressure plates (servo or pneumatic) apply defined pressure and temperature during formation **[T]**
Temperature control	Climate-controlled chambers or room; rack-level airflow; per-cell thermocouples
Data acquisition	V, I, T, capacity, energy, time per cell; uploaded to MES for grading and traceability; dQ/dV and impedance features for analytics
Safety	Cell-level voltage/temperature limits, smoke and gas detection, fire-rated compartments, rack-level fire suppression or flooding, automatic tray removal of abnormal cells **[T]**
Material handling	Automated stacker cranes and tray conveyors move trays between formation, aging and grading (Module 10)
Suppliers **[T]**	Hangke (Zhejiang Hangke), Lead Intelligent, Chroma ATE, Kataoka, PEC, Digatron, Bitrode; HYNN (cell finishing, partnering PEM RWTH **[V]**)

Chinese suppliers hold about 70% of formation-and-testing equipment sales among the top-25 suppliers **[V]** (Control Engineering).

6. Parameter table

Parameter	Typical range	Purpose	Effect if off	Measurement	Control	Label
Initial C-rate	~0.02–0.2 C	Uniform SEI, avoid plating	Fast → plating, porous SEI	Cycler log	Recipe	[T]
Later C-rate	~0.2–0.5 C	Finish formation faster	—	—	Recipe	[V] ORNL-cited protocol
Formation temperature	~25–45 °C (often warm)	Kinetics, wetting	Hot → side reactions; cold → plating	Cell thermocouple	Chamber control	[T]
Upper cut-off voltage	Chemistry-specific (e.g. 3.65 V LFP, 4.2–4.3 V NMC class)	Full activation	Over-voltage → damage	Cycler	Recipe	[T]
CV termination current	e.g. C/20–C/50	Complete charge	—	Cycler	Recipe	[T]
Fixture pressure (pouch/prismatic)	Design-specific	Layer contact, gas management	Voids, uneven SEI, plating	Load cell	Fixture	[P]
Rest/wetting times between steps	Hours	Wetting complete	Plating	MES timestamps	Recipe	[V] ORNL
Total formation time	Roughly 1–several days (excluding aging)	—	Cost vs quality	MES	Recipe	[V]/[T]
Cell voltage accuracy	High (sub-mV class)	Grading quality	Wrong grading	Calibration	MSA	[T]
First-cycle efficiency	~85–95% (graphite)	Design check	Low → lithium loss, defects	Cycler data	—	[T]

7. Defects and anomalies seen in formation

Anomaly	Likely cause	Detection	Effect	Response
Low first-charge capacity	Under-fill, incomplete wetting, low coat weight	Capacity vs model	Low-grade cell	Trace back to fill/coating data
Voltage plateau anomalies / "voltage dips"	Micro-short, particle	dV/dt, OCV during rest	Self-discharge, safety	Reject; analyse
Lithium plating	Too-high C-rate, cold, poor wetting, N/P too low locally	dQ/dV signatures, teardown (grey/silver deposits)	Capacity loss, dendrites	Lower C-rate early, warm, check wetting
Excessive gas / swelling	Moisture, residual NMP, wrong additive dose, high-Ni residual Li	Thickness, pressure sensors	Swelling, venting	Root cause upstream (KF, drying)

Anomaly	Likely cause	Detection	Effect	Response
High temperature in a cell	Internal short	Per-cell T sensor	Thermal event risk	Isolate tray, safety system
Low first-cycle efficiency	High BET anode, moisture	Charge vs discharge capacity	Lost lithium	Material/process investigation
Contact faults	Dirty probes, misaligned trays	Contact resistance check	False rejects	Probe PM

8. Formation process optimisation

1. **Design of experiments** on C-rate profile, temperature, pressure and SOC windows with cycle-life endpoints (accelerated tests) — the classic DOE you know from automotive, but with a 3–6 month response.
2. **Shallow cycling / high-SOC steps** to grow SEI faster without plating **[V]** (ORNL).
3. **Additive-tailored protocols** (e.g. hold at the reduction potential of VC/FEC).
4. **Energy recovery** cyclers to cut electricity cost.
5. **Use formation data for grading** so the separate grading step can be shortened or merged.
6. **Machine learning** on early formation curves to predict life and screen defects **[T]**.

9. Factory example

A ramping plant needs 20% more output without new racks. The team tests a shortened profile: higher C-rate in the second half of the first charge, 45 °C instead of 35 °C, and a merged capacity check. Accelerated cycle tests at 45 °C after 300 cycles show no retention loss versus control, and dQ/dV shows no plating signature. The change goes through a full PPAP-style change process: customer notification, re-validation, updated control plan. Net capacity gain: ~18% rack throughput **[E]**. The point: formation time is a design-and-validation decision, not a machine setting.

10. Interview questions

1. What is the SEI, how does it form, and why does it matter for life and safety?
2. Why is the first charge slow? What happens if you rush it?
3. Explain first-cycle efficiency and why cathodes are oversized in lithium.
4. Why do pouch cells form under pressure?
5. How would you shorten formation time, and how would you prove it is safe?
6. What data from formation do you feed into grading and traceability?
7. What is the purpose of a tap charge soon after sealing?

11. Quiz

1. Which electrolyte component is reduced first to form much of the SEI?
2. Name two gases released during formation.
3. What did ORNL find about 10-hour formation protocols for NMC811?
4. What share of formation/testing equipment sales do Chinese suppliers hold among the top 25?
5. What feature lets modern cyclers cut energy use?

Answers

1. Ethylene carbonate (EC), with additives such as VC/FEC.
2. Ethylene and CO_2 (also CO, H_2).
3. They caused lithium plating, faster impedance growth and the poorest long-term retention **[V]**.
4. About 70% **[V]**.
5. Energy recovery (regenerative discharge back to the bus/grid).

12. Hands-on exercise

Size the formation area for 5 GWh/yr of 280 Ah LFP prismatic cells (≈0.9 kWh each). Assume total formation occupancy of 24 h per cell **[E]**, channel availability 90% **[E]**, 7,800 h/yr. (a) How many channels do you need? (b) If a rack holds 256 channels, how many racks? (c) Repeat for 12 h and 48 h. (d) Estimate energy per year with 85% recovery efficiency vs no recovery. Plot channels vs formation time as a column chart.

Sources

- Wood, ORNL — formation protocols and fast formation (NAATBatt 2019)
- Battery-News — PEM on formation as a time- and energy-intensive step (HYNN partnership)
- Control Engineering — formation/testing equipment market
- Wood et al., Joule 2019 (not opened this session)

Next: Module 9 — Degassing, aging, grading and final test

Module 9 — Degassing, Final Sealing, Aging, Grading and Final Test (Stages 10–12)

Li-ion Cell Manufacturing: an Engineer's Course · Phase B · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

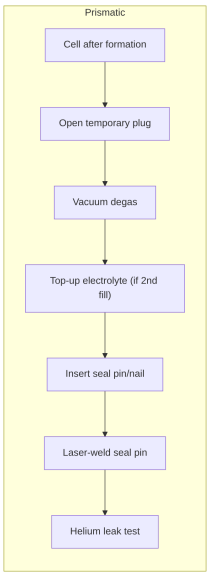
Bottom line: after formation the cell is electrochemically alive but not yet trustworthy. Degassing and final sealing make it hermetic; aging gives hidden defects time to reveal themselves through voltage drift; grading sorts good cells into matched bins; final test and traceability release them to pack lines. This is your end-of-line (EOL) and audit area — the last chance before a defect reaches a customer.

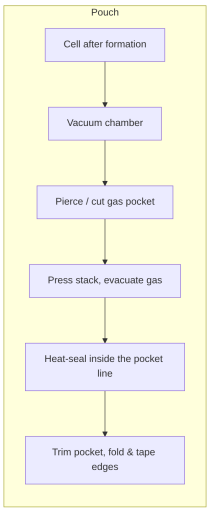
Part A — Degassing and final sealing (Stage 10)

A1. Concept

Formation produced gas. In a pouch, the gas sits in a sacrificial "gas pocket"; in a prismatic can, it sits in the headspace under the fill port. It must be removed and the cell closed permanently. Cylindrical cells are already sealed; their gas stays in the headspace and the vent/CID design accounts for it.

A2. Process flows





A3. Parameters

Parameter	Typical range / spec	Purpose	Effect if off	Measurement	Label
Degas vacuum level and hold time	Deep vacuum, seconds to minutes	Remove gas	Residual gas → swelling, poor contact	Gauge	[T]
Electrolyte lost during degas	Minimised, measured by weight	Keep fill mass	Under-fill	Weight before/after	[T]
Pouch heat-seal temperature	Typically ~170–210 °C class	Melt CPP layer to fuse	Low → weak seal; high → burn-through, thinning	Thermocouple on bars	[T]
Pouch seal time / pressure	Seconds; MPa-level pressure	Seal integrity	Channels, wrinkles	Recipe log	[T]
Seal width	Several mm	Moisture barrier path	Short path → moisture ingress over life	Vision	[P]
Seal-pin laser weld	Power/speed per design	Hermetic close	Leak, spatter	Leak test, vision	[P]
He leak rate (prismatic)	Very low, mbar·L/s order specified per design	Hermeticity	Moisture ingress, electrolyte loss	He mass spectrometer	[T]

A4. Defects

Defect	Cause	Detection	Prevention
Seal channel / leak (pouch)	Electrolyte or particles in seal zone, wrinkles, low temperature	Peel test (sample), burst/vacuum decay test, visual	Keep seal area clean during fill/degas; bar parallelism checks

Defect	Cause	Detection	Prevention
Over-thinned seal (pouch)	Excess temperature/pressure	Cross-section, thickness	Recipe window, bar flatness
Residual gas / swelling	Short degas, more gas after degas	Thickness measurement	Degas recipe; root cause of gas
Seal-pin weld porosity	Electrolyte residue on port	Leak test, vision	Port cleaning before weld
Electrolyte creep on surface	Spill during degas	Visual, swab	Wipe station

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Part B — Aging (Stage 11a)

B1. Concept and principle

Store the cell for days at controlled temperature and watch its open-circuit voltage (OCV). A healthy cell's OCV drifts only slightly as the SEI stabilises. A cell with a **micro-short** (a particle or burr bridging electrodes through the separator, or metal dendrites) leaks current internally and its OCV drops faster.

- **High-temperature aging** (often ~40–60 °C for hours to a few days **[T]**) speeds up SEI stabilisation and makes defects show sooner.
- **Room-temperature aging** (often 7–21 days **[T]**) then gives the measurement period.
- **K-value:** $K = (OCV_1 - OCV_2) / (t_2 - t_1)$, usually in mV/day. Cells above a limit are rejected **[T]**.
- **Statistical limits:** because OCV drift depends on SOC, temperature and lot, many plants use **dynamic limits** — e.g. lot mean + n·σ — instead of a single fixed number **[T]**. This is a direct application of SPC thinking: you are looking for outliers from the lot's own distribution.

B2. Why LFP is harder

LFP's voltage curve is very flat over mid-SOC, so a small charge loss changes OCV very little. Plants therefore age LFP at an SOC where the curve has slope, use longer aging, use higher-resolution voltmeters (µV class), and control temperature tightly **[T]**.

B3. Equipment

Item	Detail
Aging racks / high-bay storage	Automated storage and retrieval (stacker cranes) holding trays for days–weeks; temperature-controlled zones; fire detection and suppression **[T]**
OCV/ACIR testers	Precision DMM + 1 kHz AC-impedance meter with multi-probe fixtures; µV-level resolution for LFP **[T]** (e.g. Hioki, Keysight, Chroma)
Temperature compensation	Measure cell temperature at OCV reading; correct OCV for temperature
Data	Each reading logged to cell ID in MES; K-value computed automatically

B4. Parameters

Parameter	Typical range	Purpose	Label
High-temp aging	~40–60 °C, hours–days	Stabilise SEI, accelerate defect appearance	[T]
RT aging duration	~7–21 days	Measure self-discharge	[T]
SOC during aging	Chemistry-specific (where OCV-SOC slope is useful)	Sensitivity	[T]/[P]
K-value limit	Fixed (mV/day) or dynamic (mean + nσ)	Screen micro-shorts	[T]/[P]
OCV measurement resolution	Sub-mV; µV class for LFP	Detect small drift	[T]

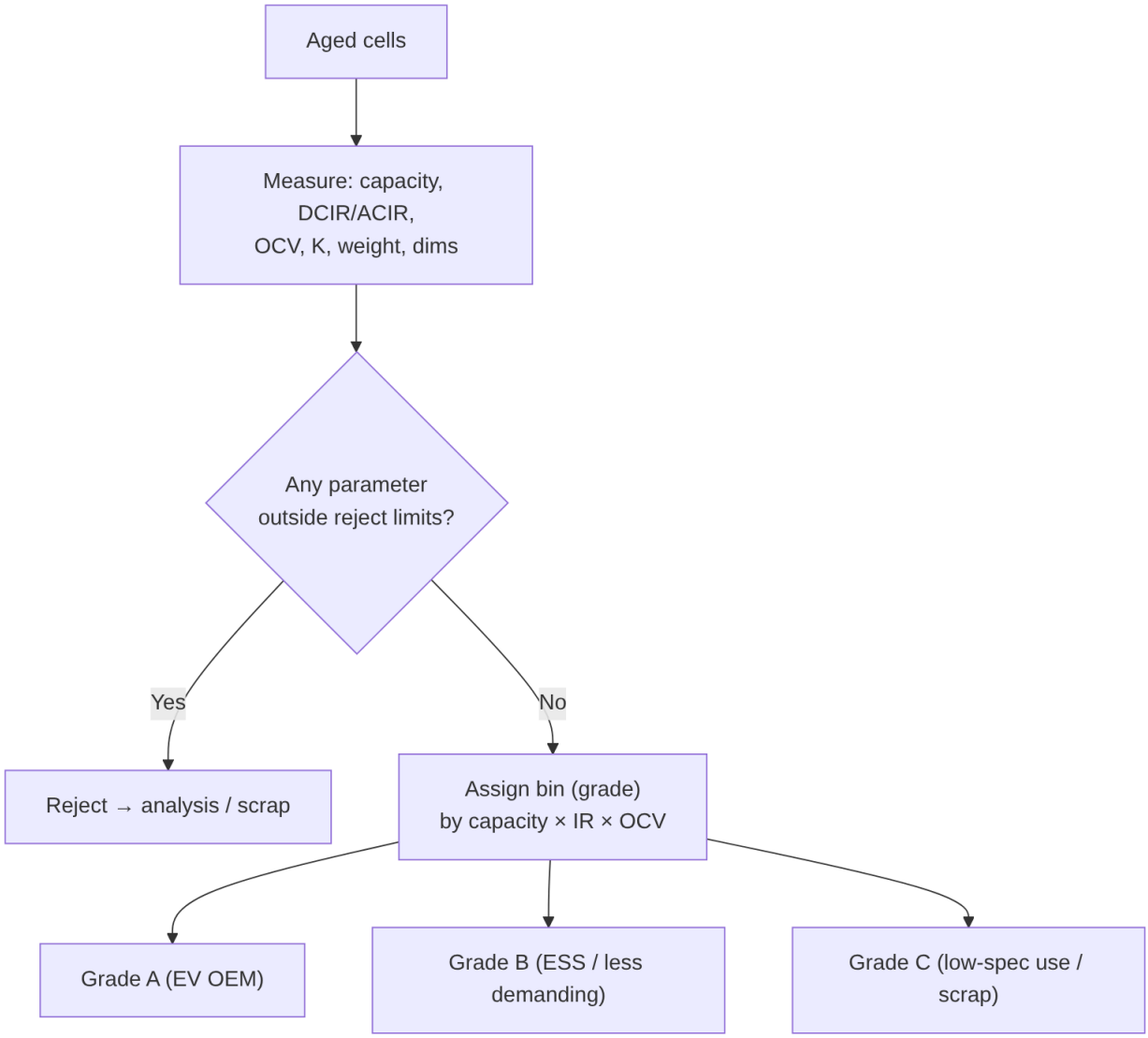
Parameter	Typical range	Purpose	Label
Temperature control during aging	Tight (± 1–2 °C class)	OCV comparability	[T]

Part C — Grading (Stage 11b)

C1. What is measured

Measurement	How	Why
Capacity	Full charge (CC-CV) and discharge at a standard C-rate and temperature on cyclers	Bin by Ah; rated capacity guarantee
DCIR	Current pulse (e.g. seconds-long) at defined SOC and temperature; ΔV/ΔI	Power capability, heat generation
ACIR	1 kHz AC impedance	Fast, non-invasive; ohmic resistance (tabs, welds, electrolyte)
OCV	Precision voltmeter at defined SOC	Matching, self-discharge
Self-discharge (K)	From aging	Defect screen
Weight	Checkweigher	Fill mass / missing parts check
Dimensions / thickness	Laser/contact gauges, flatness	Fits module; swelling check
Insulation (shell-to-terminal)	DC insulation tester	Prismatic shell isolation

C2. How grading works



- **Grades** are commercial categories: e.g. Grade A cells meet full spec with tight bins for OEM packs; lower grades go to less demanding uses at a discount **[T]**. Exact grade rules are customer- and company-specific **[P]**.
- **Bins / cell matching:** within a grade, cells are binned (e.g. narrow capacity and resistance windows) so that series-connected cells in a module age similarly; unmatched cells drift apart and the BMS must balance more **[T]**.
- **Using formation data:** many plants compute capacity from formation discharge to shorten or skip a separate capacity cycle **[T]**.

Part D — Final test, traceability, packaging and release (Stage 12)

Step	Detail
Final electrical	OCV and ACIR re-check (catches anything since grading), insulation resistance (prismatic: terminal-to-can), polarity
Leak test	He leak (prismatic), vacuum decay or sniffing (pouch) on sample or 100% depending on design
Visual	Automated vision: dents, scratches, swelling, electrolyte residue, label/code quality, terminal condition
Dimensions & weight	100% for module-fit characteristics
Traceability	Laser-marked QR/DataMatrix code links to full genealogy: material lots, mixer batch, coater lane/position, calender, slitter, assembly station, fill mass, formation curve, aging data, grade
Labelling	Cell ID, grade/bin, date; customer-specific

Step	Detail
SOC for shipping	Set storage SOC; for air shipment of cells alone (UN3480), IATA requires ≤30% SOC **[T — check current IATA edition]**
Packaging	Insulating trays/blisters, terminal protection, cartons rated for dangerous goods (Class 9)
Storage	Fire-protected finished-goods warehouse, temperature-controlled, FIFO
Release	Lot release with data package; certificate of conformity; type-approval tests done separately (e.g. UN 38.3 transport tests **[T]**, customer validation, Indian BIS requirements where applicable — verify for your product)



Part E — Defects caught at stages 10–12

Defect	Signature	Root cause (typical)	Response
Micro-short	High K-value, OCV drop	Particle, burr, separator damage, metallic contamination	Reject; tear-down sampling; trace back by genealogy
Low capacity	Below bin	Coat weight low, under-fill, plating in formation, lithium loss	Trace via MES to stage
High DCIR/ACIR	Above limit	Weak tab weld, low electrolyte, poor wetting, over-densified electrode	Trace to welding/filling
Leak	Weight loss, He leak fail, smell	Seal/weld defect	Containment by station/lot
Swelling	Thickness out	Gas, moisture	Upstream moisture control
Insulation fail	Low IR shell-terminal	Damaged insulation film, electrolyte residue	Assembly/cleaning

Factory example

A customer reports two cells in a module with OCV 40 mV below neighbours after 3 months in storage. Genealogy shows both from the same slitting lane on the same day; aging data show K-values just inside the fixed limit, but far outside the lot's mean + 4σ. **Actions:** switch to dynamic K limits; quarantine all cells from that lane-day with K above mean + 3σ; tear-down finds edge burrs. **Learning:** fixed reject limits hide outliers within good-looking lots; use SPC logic on EOL data.

Interview questions

1. Explain K-value testing and why dynamic limits are better than fixed ones.
2. Why is self-discharge screening harder for LFP than NMC?
3. ACIR vs DCIR — what does each tell you?
4. Describe pouch degassing and the main sealing failure modes.
5. How do grades and bins differ? Why does a pack maker care about bins?
6. What would you include in a cell's traceability record?

Quiz

1. Which format does not need a separate degassing step?
2. What frequency is used for ACIR?
3. Typical room-temperature aging duration?
4. What does a seal pin close on a prismatic cell?
5. Why measure cell temperature when reading OCV?

Answers

1. Cylindrical.
2. 1 kHz.

3. ~7–21 days **[T]**.
4. The electrolyte fill port.
5. OCV depends on temperature; readings must be compensated to compare cells.

Hands-on exercise

Take 200 simulated cells: generate OCV drops with mean 0.5 mV/day and σ 0.08 mV/day, plus 3 defective cells at 1.1, 0.95 and 0.8 mV/day **[E]**. (a) Apply a fixed limit of 1.0 mV/day: what escapes? (b) Apply mean + 4σ computed on the lot: what is caught? (c) Write the control plan line for K-value screening, including reaction plan for a lot with an abnormal reject rate.

Sources

- PEM RWTH & VDMA guide, 5th ed.
- Battery-News — PEM on cell finishing challenges
- Electrive — PEM model for configuring cell finishing; notes cell-specific equipment, long process times and high energy demand (German)

Next: Module 10 — Equipment and line integration

Module 10 — Equipment List and Line Integration

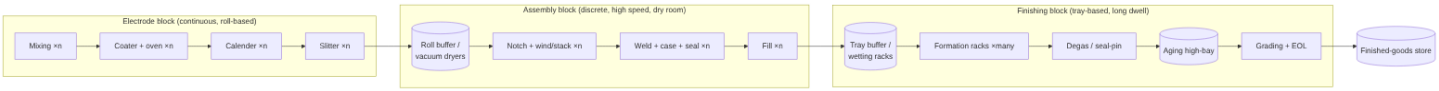
Li-ion Cell Manufacturing: an Engineer's Course · Phase C · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: a cell plant is three different factories joined by buffers: a **continuous web plant** (mixing → slitting), a **discrete high-speed assembly plant** (winding/stacking → filling), and a **warehouse that charges things** (formation → grading). Integration is about sizing each block to the plant capacity, decoupling them with buffers, and wiring every machine into one MES so each cell carries its genealogy.

On suppliers: global names are widely reported players **[T]**; check current product lines before an RFQ. Indian suppliers for core cell-process machines are still scarce; the Indian base today is mainly utilities (dehumidification, HVAC, chillers), material handling, integration and civil/MEP **[T]**.

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1. Line architecture



Rules of thumb **[T]**:

- Electrode block runs in **rolls**; assembly in **cells per minute**; finishing in **trays × hours/days**. Buffers absorb the mismatch and protect the bottleneck.
- Cathode and anode electrode lines are physically separated (avoid cross-contamination — especially copper/graphite into cathode).
- One mixer/coater/calender set is usually shared across several assembly lines; formation and aging scale with cell count × dwell time.

2. Equipment cards

Each card follows the template you asked for. Capacities and cycle times are **[T]/[E]** guides — confirm with OEM quotes.

2.1 Powder handling systems

Parameter	Details
Stage	1
Working principle	Contained big-bag discharge, vacuum or dense-phase pneumatic conveying to day silos, gravimetric loss-in-weight dosing
Main components	Unloading stations, conveying lines, filters, silos, magnets, sieves, feeders, dry-air/N_2 supply
Key parameters	Conveying velocity, gas dew point, dosing accuracy
Production capacity	Tonnes/hour, sized to mixer demand
Cycle time	Continuous / batch refills
Automation	High (recipe-driven from MES)
Quality checkpoints	Dosing weights, magnet catch logs, moisture
Common failures	Blockages, filter blinding, attrition of NMC secondaries, cross-contamination
Maintenance	Filter change, wear-part inspection, magnet cleaning, feeder calibration
Global makers **[T]**	Coperion (K-Tron), Schenck Process, Hosokawa, Chinese integrators
India **[T]**	Indian bulk-handling integrators exist for chemicals/pharma; battery-grade contamination control is the gap

2.2 Mixing systems

Parameter	Details
Stage	2
Working principle	Batch double-planetary + disperser, or continuous twin-screw extrusion
Main components	Jacketed bowl/barrel, blades/screws, vacuum, binder tanks, buffer tanks, filters
Key parameters	Speed, specific energy, temperature, vacuum, sequence
Capacity	Planetary 2–1,500 L batches; continuous 20–2,000 kg/h **[V]** (Dürr)
Cycle time	Hours per batch (planetary) **[T]**
Automation	High; recipe-locked
Quality checkpoints	Viscosity, solids, grind, density
Common failures	Seal leaks, blade wear, temperature drift
Maintenance	Seals, wear coating inspection, cleaning between recipes
Global makers **[T]**	Bühler, Coperion, Dürr, Inoue, PRIMIX, Ross; Chinese makers
India	Imported **[T]**

2.3 Coating machines and 2.4 drying ovens

Parameter	Details
Stage	3–4
Working principle	Pump-metered slot-die coating; air-flotation convective drying; NMP condensation + distillation
Main components	Unwind/splice, tension, EPC, slot dies, pumps, ovens, gauges, vision, chill, rewind, NMP recovery
Key parameters	Coat weight, speed, zone temperatures, air velocity, residual solvent
Capacity	GigaCoater 1,400 mm/40 m/min ≈ 2 GWh/yr; XL 2,200 mm/60 m/min ≈ 5 GWh/yr **[V]** (Dürr)
Cycle time	Continuous
Automation	High; closed-loop coat weight
Quality checkpoints	Coat-weight profile, defects map, adhesion, residual solvent
Common failures	Lip damage, pump pulsation, nozzle blockage, web breaks
Maintenance	Die cleaning, oven nozzle cleaning, gauge calibration, NMP system cleaning
Global makers **[T]**	Hirano Tecseed, Toray Engineering, Dürr, PNT, Lead Intelligent, Yinghe
India	Imported **[T]**

2.5 Calendering machines

Parameter	Details
Stage	5
Working principle	Hydraulic two-roll press with gap/force control, heated rolls
Main components	Rolls, chocks, cylinders, heaters, gauges, unwind/rewind
Key parameters	Line load, gap, temperature, speed
Capacity	Matched to coater output; widths 250–1,300 mm **[V]** (Dürr)
Automation	High
Quality checkpoints	Thickness, density, edge wave
Common failures	Roll damage, bearing wear, hydraulic drift
Maintenance	Roll regrinding, bearing monitoring
Global makers **[T]**	Saueressig (Matthews), PNT, Lead Intelligent, Yinghe
India	Imported **[T]**

2.6 Slitting machines and 2.7 laser cutting / notching

Parameter	Slitter	Laser notcher/cutter
Stage	6	6
Principle	Rotary shear knives	Pulsed laser ablation with galvo scanner
Components	Knife shafts, tension, EPC, extraction, vision	Laser source, scanner, web handling, extraction, vision
Key parameters	Overlap, side force, speed	Power, pulse, speed, focus
Capacity	Up to ~50 m/min slitting **[V]** (Dürr)	High web speeds; machine-specific **[T]**
Quality	Width, burr, flakes	Burr, HAZ, beads, tab dimension
Failures	Blade wear	Optics contamination, focus drift
Maintenance	Blade change by metres	Optics cleaning, power check
Makers **[T]**	Nishimura, Kampf, Lead, Yinghe	Lead, Hymson, Han's Laser, Manz; sources IPG, Trumpf, Coherent
India	Imported **[T]**	Imported **[T]**

2.8 Winding machines and 2.9 stacking machines

Parameter	Winder	Stacker
Stage	7	7
Principle	Continuous rotation on mandrel	Z-fold, cut-and-stack or lamination-stack
Components	4 unwinds, dancers, EPC, mandrel, CCD	Magazines, robots, separator unwind, vision
Key parameters	Tension, speed, alignment	Placement accuracy, time per layer
Capacity	Cylindrical: many cells/min per machine; prismatic: seconds–tens of seconds per roll **[T]**	Tens of seconds to minutes per large cell **[T]**
Quality	CCD alignment, hi-pot	Per-layer vision, X-ray
Failures	Tension spikes, telescoping	Double pick, static, particles
Maintenance	Dancer/bearing checks	Gripper wear, vacuum cups
Makers **[T]**	Lead, Yinghe, CKD, Koem	Lead, Hymson, Geesun, Korean makers
India	Imported **[T]**	Imported **[T]**

2.10 Welding machines

Parameter	Details
Stage	7 (and 10 for seal pins)
Principle	Ultrasonic (foil bundles), laser (terminals, lids, tabless discs, seal pins), resistance (cylindrical bottoms)
Key parameters	Energy/amplitude/pressure (US); power/speed/wobble (laser)
Capacity	Seconds per weld **[T]**
Quality	Weld signatures, pull/peel, leak, cross-section
Failures	Horn wear, spatter, porosity
Maintenance	Horn/anvil change, optics
Makers **[T]**	Branson/Emerson, Schunk, Telsonic, Herrmann (US); Trumpf, IPG, Coherent, Han's, United Winners (laser); Amada (resistance)
India **[T]**	Laser and US welding integrators exist (pack lines); cell-grade sealing lines imported

2.11 Electrolyte filling machines

Parameter	Details
Stage	8
Principle	Vacuum, metered dosing, pressure cycling, weighing

Parameter	Details
Key parameters	Fill mass, vacuum, pressure, cycles
Capacity	Multi-station; cells/min depends on format **[T]**
Quality	Fill mass Cpk, spill detection
Failures	Nozzle crystallisation, pump wear, vacuum leaks
Maintenance	Nozzle cleaning, pump calibration
Makers **[T]**	Lead, Yinghe; Korean/Japanese specialists
India	Imported **[T]**

2.12 Formation equipment and 2.13 aging racks

Parameter	Formation	Aging
Stage	9	11
Principle	Bidirectional power channels with energy recovery; hot-press fixtures	Climate-controlled high-bay storage
Components	Power modules, probes, fixtures, temperature control, fire systems	Racks, stacker cranes, sensors, sprinklers/flooding
Key parameters	Current/voltage accuracy, temperature, pressure	Temperature, time
Capacity	Channel count = cells × hours ÷ available hours	Tray positions = cells × days
Quality	Per-cell curves	OCV/K data
Failures	Probe contact, channel faults	Temperature gradients
Makers **[T]**	Hangke, Lead, Chroma, Kataoka, PEC, Digatron, HYNN	Intralogistics: Daifuku, SSI Schaefer, Chinese makers
India **[T]**	Some Indian power-electronics makers build pack testers; formation-grade cyclers mostly imported	Indian intralogistics integrators available

2.14 Grading equipment

Parameter	Details
Stage	11–12
Principle	Capacity cyclers, OCV/ACIR meters, DCIR pulse, insulation testers, vision, dimension gauges, sorting robots
Capacity	Matched to output; OCV/IR testers seconds per cell **[T]**
Quality	MSA on all gauges (GR&R), temperature compensation
Makers **[T]**	Hioki, Keysight, Chroma, Hangke; sorting by Chinese/Korean integrators
India	Integrators available; instruments imported **[T]**

2.15 Automation, material handling and utilities

System	Role	Notes
AGVs/AMRs	Move electrode rolls between coater, calender, slitter, dryers	Roll weight hundreds of kg to >1 t **[T]**; dry-room-compatible
Stacker cranes / AS/RS	Trays through formation and aging	Fire-zone design critical
Conveyors and tray systems	Cell flow in assembly and finishing	Tray ID (RFID) linked to MES
Dry-room HVAC with desiccant wheels	Dew-point air	Indian makers: Bry-Air (Pahwa Group) and DRI make desiccant dehumidifiers **[T]**
NMP recovery	Solvent recovery	Often from coater OEM or specialist
Chillers, compressed dry air, N_2	Utilities	Indian supply base strong
MES / SCADA	Recipes, genealogy, SPC	Module 14

3. Line balancing — worked example [E]

Plant: 5 GWh/yr LFP prismatic, 280 Ah × 3.2 V ≈ 0.896 kWh/cell.

Step	Calculation	Result
Good cells/year	5,000,000 kWh ÷ 0.896	≈ 5.58 M
Overall yield assumption	92% **[E]**	Starts ≈ 6.07 M cells
Available hours	7,800 h	—
Required output rate	6.07 M ÷ 7,800 h	≈ 778 cells/h ≈ 13 cells/min
Takt time	3,600 ÷ 778	≈ 4.6 s/cell
Stacker at 60 s/cell, 85% OEE	60 ÷ 0.85 ÷ 4.6	≈ 15–16 stackers
Winder at 20 s/cell, 85% OEE	20 ÷ 0.85 ÷ 4.6	≈ 5–6 winders
Formation at 24 h occupancy, 90% channel availability	778 × 24 ÷ 0.9	≈ 20,700 channels
Aging at 10 days	778 × 240 h	≈ 187,000 cells in storage

Takeaways: assembly is balanced by **parallel machines** at a takt of a few seconds; finishing is balanced by **inventory positions**. A one-day cut in aging releases ~18,700 storage positions.

4. Commissioning sequence (what you will actually do)

1. **FAT** at OEM: dry runs, capability on dummy material, safety functions, data interfaces to MES.
2. **Installation & utilities hook-up**; dry-room certification (dew point, particle class, pressure cascade) before any assembly machine runs product.
3. **SAT**: I/O checks, interlocks, emergency stops, fire systems integration.
4. **Process qualification**: material trials, DOE, MSA on gauges, run-at-rate (like PPAP run-at-rate), Cpk on special characteristics.
5. **Ramp-up**: yield learning curve, scrap Pareto, OEE tracking; customer sample (A/B/C samples) and validation.

5. Interview questions

1. Explain why formation and aging are sized by inventory positions, not by machine speed.
2. Where would you place buffers in a cell plant and why?
3. Walk through FAT → SAT → run-at-rate for a stacking machine.
4. How do you separate cathode and anode flows physically and in MES?
5. What makes a dry room "ready" for equipment start-up?

6. Quiz

1. Which block is continuous, which discrete, which tray-based?
2. Takt time for 778 cells/h?
3. Which equipment sizing scales with cell count × hours?
4. Name two Indian makers of desiccant dehumidification systems.
5. What does FAT stand for?

Answers

1. Electrode block continuous; assembly discrete; finishing tray-based.
2. ≈ 4.6 s.
3. Formation channels and aging positions.
4. Bry-Air, DRI **[T]**.
5. Factory acceptance test.

7. Hands-on exercise

Repeat the line-balancing table for a 10 GWh/yr NMC pouch plant with 100 Ah × 3.65 V cells. Choose your own cycle-time assumptions (state them as [E]). Identify the bottleneck, propose buffer sizes in hours between blocks, and draw the line with machine counts.

Sources

- Dürr — electrode equipment specs
- Control Engineering — equipment supplier shares by process step
- PEM RWTH & VDMA guide, 5th ed.

Next: Module 11 — Factory layout and dry rooms

Module 11 — Factory Layout, Dry Rooms and Safety Zoning

Li-ion Cell Manufacturing: an Engineer's Course · Phase C · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

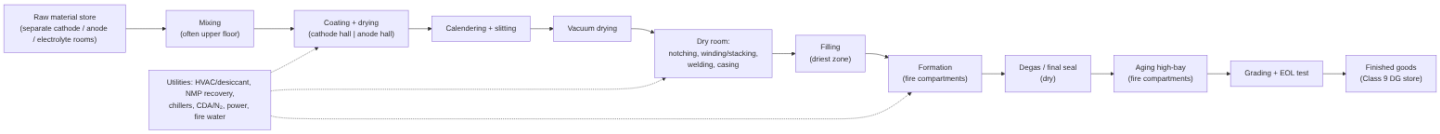
Bottom line: a cell plant layout is driven by three gradients — **moisture** (driest at assembly/filling), **cleanliness** (no metal particles near open electrodes) and **fire risk** (highest where charged cells sit for days). Material should flow in one direction through these zones; people should cross them as little as possible, because every person is a moisture and particle source.

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1. Concept

Picture a long building: powders enter at one end, finished cells leave at the other. Each hall has its own climate: controlled humidity for mixing and coating, ultra-dry air for cutting and assembly, warm and fire-separated for formation and aging, and a dangerous-goods warehouse at the exit.

2. Conceptual layout and material flow



Layout principles [T]:

- **One-way flow** from raw material to shipping; no back-flow of product.
- **Cathode and anode separated** from receipt to slitting (separate rooms, equipment, tools and often gowning colours) to stop copper/graphite reaching cathode and vice versa.
- **Vertical flow** is common: mixing on an upper floor so slurry feeds coaters by gravity/short pipes.
- **Long rectangular coater halls**: coaters with ovens are 50–100+ m long.
- **Dry-room volume minimised**: only processes that truly need it sit inside; use mini-environments (local enclosures) where possible — a big energy lever.
- **Formation and aging in fire compartments** with separation walls and automated removal of abnormal trays.
- **Utilities on the periphery or a mezzanine**, with short duct runs to the dry rooms.
- **Expansion axis**: leave space to add a full line (phase 2) without crossing flows.

3. Area-by-area requirements

Area	Main hazards / sensitivities	Environment	Key design features
Raw material store	Moisture pickup, dust, chemical hazards (electrolyte flammable, $LiPF_6$ water-reactive)	Dry, cool; electrolyte in flammable-liquid store	Segregation, ventilation, spill containment
Mixing	Dust explosion/inhalation, NMP vapour, metal contamination	Humidity-controlled; local extraction	Contained powder handling, solvent-rated electrics where NMP is used, metal-free contact parts
Coating & drying	NMP vapour, heat, LEL in ovens	Clean, humidity-controlled	Oven LEL monitoring, NMP recovery, fire detection
Calender & slitting	Particles, noise	Clean; slitting often already in dry room	Dust extraction, particle monitoring

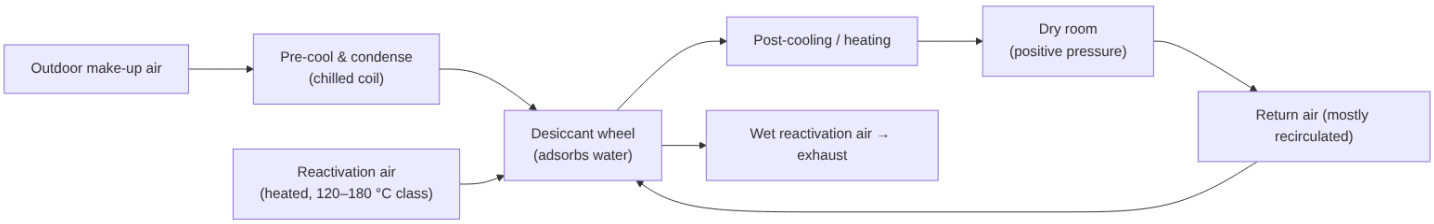
Area	Main hazards / sensitivities	Environment	Key design features
Dry room (assembly)	Moisture, particles, weld fumes	Dew point ~−35 to −45 °C class, down to −60 °C for sensitive steps **[T]**	Airlocks, pressure cascade, gowning, limited headcount
Filling	Moisture, electrolyte vapour, HF	Driest zone	Local exhaust, HF/solvent monitoring
Formation	Charged cells, thermal runaway, gas	Temperature-controlled; normal humidity	Fire compartments, cell-level monitoring, suppression, smoke exhaust
Aging	Large charged inventory	Temperature-controlled	AS/RS with fire zoning, rapid isolation
Grading / EOL	Charged cells handling	Normal	ESD, safety interlocks
FG warehouse	Dangerous goods (Class 9)	Cool, dry	Rack spacing, sprinklers, segregation

4. Dry rooms in depth

4.1 Why dew point, not RH

At these moisture levels relative humidity is near zero and meaningless; **dew point** (the temperature at which water would condense) directly expresses absolute moisture. ~1% RH at room temperature corresponds to roughly −40 °C dew point **[V]** (igus whitepaper). AFRY gives −35 to −45 °C during assembly, equal to ~0.11–0.28 g water per kg dry air **[V]** (AFRY).

4.2 How a dry room works



- Below ~0 °C dew point, cooling coils alone cannot dry the air; **desiccant rotors** are required **[V]** (AFRY).
- Rooms run at **positive pressure** with an airlock cascade so leakage goes out, not in **[V]** (patent example: dew point ≤ −40 °C with overpressure, USPTO 12719139).
- **Moisture loads:** people (breathing, skin), make-up air, door openings, leaks through walls/cable entries, materials brought in, and process equipment **[T]**. People are a major internal source, so headcount and activity in dry rooms are controlled **[T]**.
- **Energy:** dry rooms and electrode drying are the two largest energy users in cell production **[V]** (Linköping SIMS 2021); the reactivation heat for desiccant wheels is the main cost **[T]**.

4.3 Electrode area vs assembly area

	Electrode manufacturing (mixing → calendering)	Cell assembly + filling
Why moisture matters	Powders and slurries absorb water; high-Ni slurries gel; water-based anodes are made with water anyway	Electrodes are dry and about to meet $LiPF_6$; any water becomes HF in a sealed cell
Typical control	Humidity-controlled clean rooms (RH limits; drier for high-Ni) **[T]**	Dew point ~−35 to −45 °C, down to −60 °C for the most sensitive steps **[T]**
Volume	Large halls	Minimise volume; use mini-environments
Cost driver	Ventilation and solvent safety	Desiccant reactivation energy

4.4 Cleanliness

Particles, especially metallic ones, are the main micro-short cause. Typical practices **[T]**: ISO 14644 cleanroom classes (ISO 7–8 class common for assembly; ABN supplies ISO 8 standard with ISO 7 option **[V]** (Cleanroom Technology)), gowning, tacky mats, metal-free tools (ceramic, plastic), copper/brass/zinc bans in cathode zones, particle counters on SPC, and dust extraction at every cutting and welding station.

5. Personnel and material flow

Topic	Practice [T]
Personnel entry	Gowning rooms, air showers, airlocks; separate entries per zone; badge-controlled; headcount limits
Material entry	Pass-through airlocks with purge; wipe-down; packaging removed outside the dry room
Flow separation	Product flow and people flow do not cross where possible; maintenance access from service corridors
Waste	Separate exits for scrap electrodes (contain cobalt/nickel), contaminated solvent, defective cells (charged → special handling)

6. Fire protection and safety zoning

Area	Hazard	Typical measures [T]
NMP coating/recovery	Combustible solvent vapour (NMP flash point ~ 90 °C class **[T]**)	LEL monitoring in ovens, interlocks, rated electrics where required
Electrolyte storage and filling	Highly flammable carbonate solvents (e.g. DMC flash point below room temperature **[T]**), HF formation	Flammable-liquid rooms, spill containment, gas detection, local exhaust
Formation and aging	Thermal runaway of charged cells, toxic and flammable gases	Fire compartments, cell-level voltage/temperature monitoring, early smoke/off-gas detection, sprinklers or water deluge, automated removal of abnormal trays to quarantine, smoke exhaust
FG warehouse	Large charged inventory	Rack spacing, in-rack sprinklers, limited SOC
Whole plant	Electrical, chemical	HAZOP at design, emergency plans, local fire brigade coordination

In India, design follows National Building Code fire provisions, PESO for petroleum/solvent storage where applicable, the Factories Act and state fire-service approvals; international insurers' guidance is often applied too **[T — confirm with your plant's EHS team]**.

7. Factory example

A new plant misses its dew-point target (−45 °C) in the monsoon. Moisture-load analysis shows the design assumed 20 people in the assembly dry room; during ramp-up, 45 engineers and OEM technicians were inside, plus doors propped open for cable pulling. **Fix:** temporary headcount cap with badge counting, door interlocks, sealing of cable penetrations, a temporary dehumidifier for ramp-up. **Lesson:** dry-room design loads must include commissioning-phase occupancy, not just steady-state production.

8. Interview questions

1. How does a desiccant-wheel dehumidifier achieve a −50 °C dew point?
2. Why is the dry room kept at positive pressure, and how do airlocks work in a cascade?
3. What are the main moisture sources in a dry room, and which can you control?
4. How would you reduce dry-room energy consumption?
5. Why separate cathode and anode flows, and how far upstream?
6. Describe fire zoning for a formation hall.

9. Quiz

1. Roughly what dew point corresponds to 1% RH at room temperature?
2. Below what dew point are desiccant dehumidifiers necessary?
3. Which area holds the largest stock of charged cells?
4. Why ban brass tools in cathode areas?
5. What is a mini-environment?

Answers

1. About −40 °C **[V]**.
2. Below about 0 °C dew point **[V]**.
3. Aging (and FG warehouse).
4. Copper/zinc particles dissolve and plate → micro-shorts.
5. A small sealed enclosure around a process with its own dry/clean air, reducing the big room's requirement.

10. Hands-on exercise

Draw a block layout (to rough scale) for a 5 GWh/yr LFP prismatic plant on a 300 × 150 m plot. Show: zones, dew-point classes, airlocks, people and material entry points, fire compartments, utility yard, expansion space for phase 2. Then list the five largest moisture loads for the assembly dry room and how you would control each.

Sources

- AFRY — clean and dry room requirements
- igus — dry cleanrooms whitepaper
- Cleanroom Technology — ABN dry room solution
- Linköping SIMS 2021 — energy of electrode drying and dry room
- USPTO 12719139 — dry room at overpressure, dew point ≤ −40 °C

Next: Module 12 — Parameter handbook, quality and PFMEA

Module 12 — Parameter Handbook, Quality Engineering and Process FMEA

Li-ion Cell Manufacturing: an Engineer's Course · Phase C · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: your automotive quality toolkit transfers almost unchanged. What changes is (1) the **delay** between cause and effect — a defect made at slitting may only show at aging two weeks later — and (2) the **severity scale** — an escaped micro-short is a fire risk. That pushes cell plants towards 100% inline measurement, full genealogy and statistical screening at EOL.

Part A — Manufacturing parameter handbook

Ranges are **windows for learning and DOE starting points**, not specifications. Labels: **[V]** sourced in the stage module, **[T]** typical practice, **[P]** set by your process spec. "Chem" = chemistry-specific; "Equip" = equipment-specific.

Stage	Parameter	Typical range	Purpose	Effect on quality	Measurement	Control	Common deviation	Corrective action	Label
1 Materials	CAM moisture	Low hundreds of ppm or below; tighter for high-Ni	HF prevention	Gas, fade, gelation	Karl Fischer	Dry storage, exposure time	Bag left open	Re-dry or reject; MES exposure clock	[T] Chem
1	Magnetic impurities	ppb-level limits	Micro-short prevention	Self-discharge, safety	Magnetic extraction + ICP	Supplier controls, magnets	Supplier process change	100% lot testing after PCN	[T]
1	Electrolyte water / HF	~20 ppm / ~50 ppm order	HF control	Fade, gas	KF / titration	Closed supply	Drum seal leak	Quarantine lot	[T]
2 Slurry	Solids content	NMC ~65–75%, LFP ~45–60%, anode ~40–55%	Coat weight, drying	Cracks, streaks	Moisture balance	Solvent dosing	Evaporation in tank	Adjust; cover tanks	[T] Chem
2	Viscosity at defined shear	~1–10 Pa·s class	Coatability	Streaks, settling	Rotational viscometer	Dilution	Gelation, temp	Check pot life, temp	[T]/[P]
2	Grind (max particle)	Below ~30–50 µm	No agglomerates	Streaks, separator damage	Hegman	Dispersion, filters	Short mixing	Re-mix, filter	[T]
2	Slurry temperature	≤ ~40 °C	Stability	Gel, SBR damage	RTD	Jacket	Cooling failure	Alarm, hold batch	[T]
3 Coating	Areal loading	~15–30 mg/cm² per side (energy cells)	Capacity, N/P	Capacity spread, plating	β/X-ray gauge	Pump closed loop	Pump drift, density change	Re-zero gauge, check slurry solids	[T] Chem

Stage	Parameter	Typical range	Purpose	Effect on quality	Measurement	Control	Common deviation	Corrective action	Label
3	Coat-weight tolerance	~±1–2%	Uniformity	Capacity spread	Gauge	Closed loop	Cross-web profile	Die adjust, oven balance	[T]/[P]
3	Line speed	Tens of m/min (20–60 m/min by Dürr model)	Throughput	Bead stability, drying	Encoder	Drive	Speed-up without oven change	Re-validate oven	[V] Equip
4 Drying	Zone temperatures	~60–150 °C, low–high–low	Drying rate	Binder migration, cracks	Thermocouples	PID	Heater failure	Alarm, stop line	[T]
4	Residual NMP	Low hundreds ppm or below	Side reactions	Gas	GC	Oven profile	Speed too high	Lower speed	[T]
4	NMP recovery	>99%	Cost, emissions	—	Mass balance	Recovery unit	Condenser fouling	Clean	[V] Equip
5 Calender	Density NMC / LFP / graphite	~3.2–3.6 / ~2.2–2.6 / ~1.5–1.7 g/cm³	Energy vs rate	Rate, wetting, cracking	Disc mass/volume	Gap	Material change	Re-run compaction curve	[T] Chem
5	Thickness tolerance	~±2–3 µm per side	Fit, capacity	Can fit, spread	Laser gauge	Closed loop	Roll wear	Regrind	[T]
5	Roll temperature	Ambient to ~120 °C	Densification	Springback, smearing	RTD	Heater	Uneven heating	Service	[T]
6 Slitting	Burr height	Often ≤ ~10 µm	Separator protection	Shorts	Microscope / vision	Tool life	Knife wear	Change by metres	[T]
6	Width tolerance	Tenths of mm or tighter	Overhang	Plating	Vision	Knife setup, EPC	Web wander	EPC tune	[T]
6	Electrode moisture after vacuum drying	Hundreds of ppm or lower	HF	Gas, fade	KF	Recipe	Short dry	Extend recipe	[T]/[P]
7 Assembly	Anode overhang	~0.5–2 mm per side	Prevent plating	Safety	X-ray/CT	Alignment	Telescoping	EPC, tension	[T]/[P]
7	Hi-pot insulation	Hundreds of V; MΩ minimum	Short screening	Shorts	Insulation tester	Recipe	Particle	Reject; source analysis	[T]
7	Weld quality	Design-specific	Low resistance, integrity	IR, opens	Signature + pull	Generator control	Horn wear	PM by count	[P]
7	Leak rate (lid weld)	Very low (He)	Hermeticity	Moisture ingress	He MS	Laser control	Gap	Fixture	[T]
8 Filling	Fill mass	Design (~1.3–2 g/Ah order)	Wetting + reserve	Fade, plating	Weigh	Pump	Pump wear	Seal change	[T]/[P]
8	Dew point at filling	Driest zone (−40 to −60 °C class)	HF	Gas	Dew-point sensors	HVAC	People load, leaks	Headcount, sealing	[T]

Stage	Parameter	Typical range	Purpose	Effect on quality	Measurement	Control	Common deviation	Corrective action	Label
8	Wetting time/temperature	RT or 40–60 °C, hours	Full wetting	Plating	MES time	Recipe	Shortcut under pressure	Interlock	[V] ORNL slide
9 Formation	Initial C-rate	~0.02–0.2C	SEI quality	Plating	Cycler	Recipe	—	—	[T]
9	Later C-rate	~0.2–0.5C	Throughput	—	Cycler	Recipe	—	—	[V]
9	Temperature	~25–45 °C	Kinetics	Plating / side reactions	Thermocouple	Chamber	Gradient across racks	Mapping	[T]
10 Degas/seal	Pouch seal temperature	~170–210 °C class	Seal	Leak	Thermocouple	Recipe	Bar wear	Replace	[T]
11 Aging	High-temp aging	~40–60 °C, hours–days	SEI stabilisation, defect acceleration	—	Logs	AS/RS	—	—	[T]
11	RT aging duration	~7–21 days	Self-discharge measurement	Escapes if too short	MES	Recipe	Pressure to ship	Do not shorten without validation	[T]
11	K-value limit	Fixed or mean + nσ	Micro-short screen	Escapes	OCV meter	Statistics	Temperature drift	Compensate	[T]/[P]
12 Final	Insulation resistance shell-terminal	MΩ class at test voltage	Safety	Leakage	Tester	Recipe	Film damage	Reject	[T]
12	Shipping SOC (air, UN3480)	≤30%	Transport rules	—	OCV	Recipe	—	—	[T — check IATA]

Part B — Quality engineering approach

B1. Mapping your automotive toolkit

Tool	How it is used in a cell plant
APQP	Same phases. Product design (cell design FMEA, DVP&R), process design (PFD, PFMEA, control plan), validation (run-at-rate, Cpk, cell validation tests), feedback. Cell validation takes months (cycle-life, calendar-life, abuse tests)
PPAP	Customers (pack/vehicle/ESS OEMs) require PPAP-style submissions: PSW, dimensional, material certs, capability, control plan, PFMEA, MSA, lab scope, plus electrochemical test reports
PFMEA	Severity driven by safety (thermal event) and life; AIAG-VDA Action Priority (AP) increasingly replaces RPN
Control plan	Per stage; many characteristics measured 100% inline (coat weight, thickness, fill mass, OCV, IR)
SPC	Continuous inline data → $\bar{X}$-R/I-MR charts, EWMA for slow drifts; cross-web profiles; run rules
Cp/Cpk	Targets like ≥1.33 (normal) and ≥1.67 (special characteristics) as in automotive **[T]**; short-term Cp/Cpk in run-at-rate, long-term Pp/Ppk in production
MSA	Critical: coat-weight gauges, thickness gauges, checkweighers (fill mass), OCV/IR meters (μV/μΩ). GR&R and bias/linearity; temperature effects
Traceability	Cell-level genealogy back to material lots and machine parameters; roll-to-cell mapping (position on the electrode roll → cell ID)
Defect management	Inline vision defect maps on electrode rolls; marking and automatic skipping of defective sections at notching/stacking
RCA / 8D	Same; complicated by delay and need for tear-down analysis (CT, SEM, ICP)

B2. Special characteristics for a cell (example)

Characteristic	Why special	Stage
Coat weight (both electrodes) / N/P	Safety (plating)	3
Burr height	Safety (shorts)	6
Anode overhang	Safety (plating)	7
Hi-pot insulation	Safety	7
Moisture (electrodes, electrolyte, dry room)	Safety & life	1, 6, 8
Fill mass	Life	8
K-value	Safety	11
Weld integrity / leak rate	Safety & life	7, 10

B3. SPC specifics

- **Inline gauges produce huge data**; use subgrouping by roll/lane/position, and watch for **autocorrelation** (successive readings are not independent) — EWMA/CUSUM charts or residual charts are better than plain Shewhart charts for slow drifts **[T]**.
- **Cross-web profile Cpk** (each lane) and **along-web** separately.
- **EOL statistics:** dynamic limits on OCV drop and capacity per lot (outlier detection), plus lot-level trend charts.
- **Linking data:** a defect at EOL is analysed by joining cell ID → roll position → upstream parameter traces (the "digital thread", Module 14).

B4. Control plan extract (example: cathode coating)

Process step	Characteristic	Class	Spec	Method	Sample	Control method	Reaction plan
Coating	Areal loading	SC	Target ± x% **[P]**	β-gauge	100% scan	Closed loop + SPC (EWMA)	Out of limit → auto-mark section; 3 points trend → stop & check slurry solids
Coating	Edge profile	—	≤ y µm above nominal **[P]**	Profile gauge	100%	SPC	Adjust shim; segregate roll
Drying	Residual NMP	—	≤ z ppm **[P]**	GC	1 per roll	Check sheet	Hold roll; adjust oven
Coating	Surface defects	SC (pinholes on anode)	None > size limit	Vision	100%	Defect map to MES	Auto-skip at notching
Drying	Adhesion	—	≥ N/m **[P]**	Peel test	1 per roll start/end	SPC	Hold roll

Part C — Process FMEA (major stages)

Ratings use a 1–10 scale; AP = AIAG-VDA Action Priority (H/M/L) **[E — illustrative ratings]**.

#	Process step	Failure mode	Effect	S	Cause	O	Current prevention	Current detection	D	AP	Recommended action
1	Material receipt	Metallic contamination in CAM	Micro-short, thermal event	10	Supplier process wear	3	Supplier PPAP, magnets at supplier	Lot ICP magnetic test	5	H	100% lot test after PCN; line magnets with catch logging
2	Material storage	High-Ni CAM absorbs moisture	Gelation, gas	7	Exposure after opening	5	SOP	KF on sample	5	H	MES exposure clock, dry storage only
3	Mixing	Agglomerates	Streaks, separator puncture	8	Wrong sequence, short time	4	Recipe lock	Grind gauge per batch	4	M	Inline filter ΔP monitoring
4	Mixing	Gelation	Batch scrap, coating defects	6	Moisture, temp, pot life	4	Jacket cooling, pot-life rule	Viscosity checks	4	M	Viscosity vs time check at 2 h

#	Process step	Failure mode	Effect	S	Cause	O	Current prevention	Current detection	D	AP	Recommended action
5	Mixing	Metal wear debris	Micro-short	10	Blade/seal wear	2	Ceramic/WC wear parts	Magnetic filter	5	H	Wear-part PM by hours; catch trend
6	Coating	Coat weight low (anode)	Lithium plating	9	Pump drift, density change	3	Closed loop	100% β-gauge	2	M	Gauge MSA, redundancy
7	Coating	Pinholes / uncoated spots	Local plating, corrosion	8	Bubbles, foil contamination	4	Degassing, foil spec	Vision 100%	3	M	Auto-skip marked sections
8	Coating	Streaks	Capacity spread, plating	7	Particles at die lips	4	Filtration	Vision, profile	3	M	Lip inspection per roll change
9	Drying	Binder migration / delamination	Particles, capacity loss	7	Drying too fast	4	Oven profile validation	Peel test per roll	5	M	Inline adhesion proxy; airflow survey PM
10	Drying	Residual solvent	Gas, fade	6	Speed too high	3	Speed–oven interlock	GC sample	5	M	Increase sampling during speed change
11	Calendering	Over-densification	Slow wetting, plating	7	Gap error, material change	3	Recipe per material	Thickness gauge, density sample	4	M	Compaction curve in PPAP
12	Calendering	Foil wrinkles	Misalignment, tab cracks	6	Edge stress	5	Edge stretching	Vision	4	M	Design narrower margins
13	Slitting/notching	Burr above limit	Separator puncture → short	10	Knife/die wear, laser beads	4	Tool life by metres	Sample microscopy + vision	4	H	Inline burr vision; tighter tool life
14	Slitting/notching	Loose particles	Micro-short	10	Poor extraction	4	Extraction, brushes	Particle counts	5	H	Extraction ΔP interlock
15	Vacuum drying	Residual moisture	HF, gas, fade	8	Short cycle, leak	3	Recipe, leak test	KF sample	5	H	Moisture sensor in chamber; exposure-time interlock
16	Winding/stacking	Misalignment / overhang loss	Plating, short	9	Web wander, placement error	3	EPC, vision	CCD + X-ray (sample or 100%)	3	M	100% X-ray for new design ramp
17	Winding/stacking	Separator damage	Short	10	Tension spike, gripper	3	Tension control	Hi-pot 100%	3	M	Gripper PM, tension alarms
18	Tab welding (US)	Weak weld	High IR, open	7	Horn wear	4	PM by count	Signature monitoring, sample peel	4	M	Signature-based 100% reject
19	Laser welding	Spatter particles in cell	Micro-short	10	Contamination, parameters	4	Shields, extraction	Vision; later OCV	6	H	Wobble/green laser; spatter shield cleaning interval
20	Lid seal weld	Leak	Moisture ingress, leak	8	Gap, contamination	3	Fixture design	He leak 100%	2	L	—
21	Filling	Under-fill	Fade, plating	8	Pump wear, bubbles	3	Pump PM	100% weighing	2	M	Per-station SPC
22	Filling	Moisture ingress	HF, gas	8	Dew-point excursion	3	HVAC, headcount	Dew-point monitoring	4	M	Interlock filling on dew point
23	Formation	Lithium plating	Capacity loss, dendrites	9	C-rate too high, poor wetting, cold	3	Validated recipe	dQ/dV analytics, capacity	5	H	Wetting check before formation; temperature mapping

#	Process step	Failure mode	Effect	S	Cause	O	Current prevention	Current detection	D	AP	Recommended action
24	Formation	Thermal event in rack	Fire	10	Internal short	2	Upstream controls	Per-cell T/V monitoring	3	M	Automated isolation; suppression testing
25	Degas / pouch seal	Seal channel leak	Moisture ingress over life	8	Electrolyte in seal area	4	Wipe station	Vacuum decay test	4	M	Seal-area protection during fill
26	Aging	Micro-short not detected	Field failure	10	Aging too short, fixed limits	3	Recipe	K-value	5	H	Dynamic limits, LFP SOC selection, µV meters
27	Grading	Wrong bin	Pack imbalance	6	Gauge drift	3	Calibration	MSA	4	L	—
28	Final test	Label/ID mismatch	Traceability loss	7	Handling error	2	Auto marking	Code verification	2	L	—

How to use: pick the H items first; most are **detection** problems with long delays. The cheapest actions are usually **prevention at the source** (tool life, extraction, moisture interlocks) rather than more EOL testing.

Part D — Root cause analysis and 8D in a cell plant

1. **D1–D3 Team and containment:** quarantine by genealogy (lot, roll, lane, station, date range) — this is where traceability pays for itself.
2. **D4 Root cause:** combine data analytics (which upstream variable separates good and bad cells?) with **tear-down**: CT scans for alignment/particles, SEM/EDS on particles (identify Fe/Cu/stainless), ICP for metal content, separator inspection.
3. **D5–D6 Corrective actions:** prevention at source; verify with Cpk and, for life-related issues, accelerated tests.
4. **D7 Prevent recurrence:** update PFMEA, control plan, lessons-learned database across lines and plants.
5. **D8 Closure.**

Interview questions

1. Which characteristics would you classify as special characteristics for an EV cell and why?
2. How do you do MSA on a coat-weight gauge?
3. Why does autocorrelation matter for SPC on inline gauges?
4. Walk me through containment for a customer complaint of high self-discharge.
5. Why are many High-AP items in a cell PFMEA detection problems?

Quiz

1. Typical Cpk target for special characteristics?
2. What replaces RPN in AIAG-VDA FMEA?
3. What tool identifies particle composition during tear-down?
4. Which chart type is good for slow drifts?
5. Name three gauges that need careful MSA in a cell plant.

Answers

1. ≥1.67 **[T]**.
2. Action Priority (AP).
3. SEM with EDS.
4. EWMA (or CUSUM).
5. Coat-weight gauge, thickness gauge, checkweigher, OCV/IR meter (any three).

Hands-on exercise

Extend the PFMEA above with five failure modes for your own chosen format (cylindrical, prismatic or pouch) that are not yet listed. Rate S/O/D with your reasoning, set AP, and add actions. Then write a full control plan for Stage 8 (filling) using the template in B4.

Next: Module 13 — Defects and failure analysis

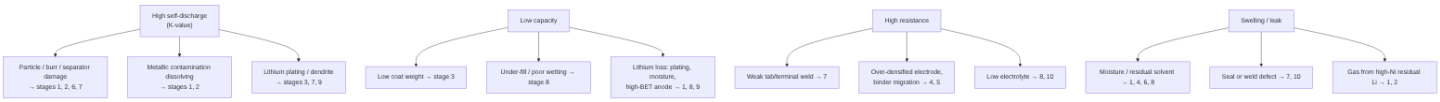
Module 13 — Defect and Troubleshooting Guide, Failure Analysis

Li-ion Cell Manufacturing: an Engineer's Course · Phase C · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: most cell defects end up in one of four symptoms at EOL — **low capacity, high resistance, high self-discharge, or swelling/leak**. Troubleshooting is the art of walking back from the symptom to the stage, using genealogy data first and tear-down second.

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1. Symptom → stage map



2. Defect guide

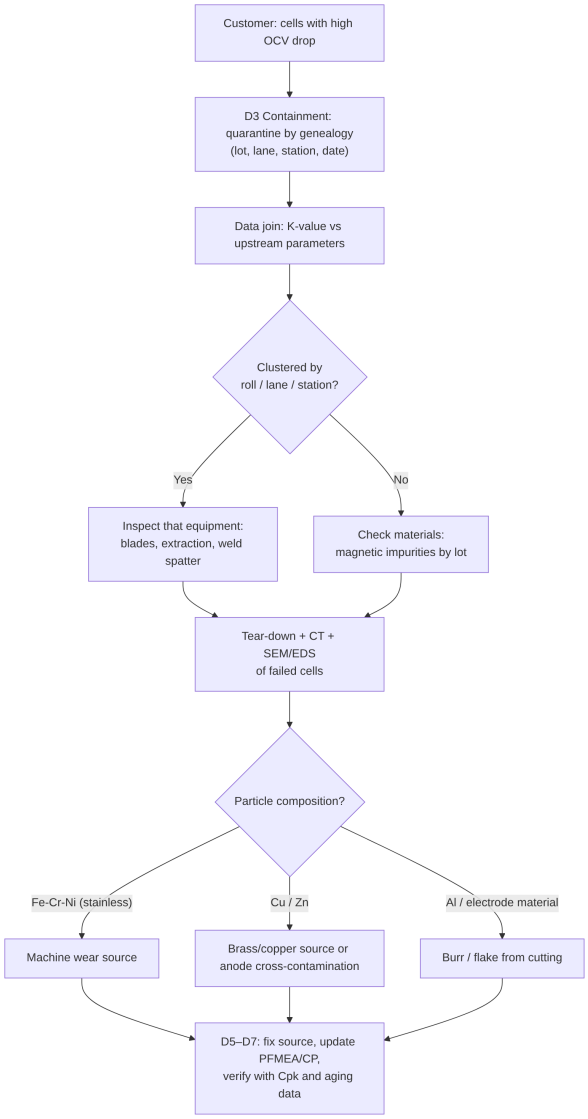
Defect	Root causes	Stage	Detection	Performance impact	Safety impact	Prevention / corrective action
Electrode coating defects (streaks, pinholes, thickness variation)	Agglomerates, bubbles, die lip damage, pump pulsation, foil contamination	2–3	Inline vision, β-gauge, profile	Local capacity mismatch, early fade	Local plating at thin anode spots	Filtration, degassing, lip PM, pulsation damping, auto-skip marked areas
Electrode cracks	Fast drying (capillary stress), thick water-based films, low binder, over-calendering	4–5	Vision, microscopy	Loss of contact, capacity fade	Particle shedding → shorts	Low–high–low drying, binder level, compaction limits
Poor adhesion / delamination	Binder migration, foil oil/oxide, under-drying, calender shear	3–5	Peel test, tape test	Capacity loss, high IR	Loose particles	Drying profile, foil cleanliness spec (dyne), primer layer
Burrs	Worn knives/dies, wrong setup, laser beads	6	Microscopy, vision	—	Separator puncture → internal short	Tool life by metres, burr SPC, laser windows
Misalignment	Web wander, width error, placement error, telescoping	6–7	CCD, X-ray/CT	Capacity loss at edges	Plating at uncovered anode edges, shorts at edges	EPC, vision feedback, tolerance stack-up design
Tab welding defects	Horn/anvil wear, contamination, wrong energy; laser porosity/spatter	7	Weld signature, peel/pull, cross-section, IR	High resistance, heating	Hot spots; spatter particles → shorts	Signature monitoring, PM by count, cleaning
Separator damage	Burrs, particles, tension spikes, gripper damage, heat (hot press, welding)	6–7	Hi-pot, visual, CT	—	Hard short / thermal event	Tension control, clean handling, heat shields
Electrolyte filling defects (under/over-fill, spill, incomplete wetting)	Pump wear, bubbles, nozzle crystallisation, short soak	8	Weighing, formation curves	Fade, high IR, capacity spread	Plating (dry spots); seal failures from spill	Pump PM, 100% weighing, wetting validation, wipe station
Moisture contamination	Wet materials, dew-point excursion, long exposure, bad vacuum drying	1, 6, 8	KF, dew-point logs, gas at formation	Fade, gas, IR rise	HF, gas, venting	Moisture interlocks, exposure clocks, dew-point SPC

Defect	Root causes	Stage	Detection	Performance impact	Safety impact	Prevention / corrective action
Internal short circuits	Metal particles, burrs, separator defects, lithium dendrites	1–9	Hi-pot, OCV drop, temperature at formation	Self-discharge	Thermal runaway risk	Prevention at source (contamination control), screening at hi-pot and aging
Gas generation	Moisture, residual NMP, residual Li (high-Ni), electrolyte decomposition, over-voltage	1, 2, 4, 8, 9	Thickness, pressure, GC of gas	Swelling, contact loss	Venting	Moisture/solvent control, additive tuning, formation recipe
Capacity loss	Low loading, lithium lost to SEI/plating, under-fill, high-BET anode	3, 8, 9	Grading capacity, formation efficiency	Low-grade cells	— (unless plating)	Coat weight control, formation recipe, material specs
High internal resistance	Weak welds, over-densified electrodes, binder-rich surface, low electrolyte, low temperature formation	4, 5, 7, 8	ACIR/DCIR	Low power, heating	Hot spots	Weld control, density targets, fill mass
Leakage	Seal channels, weld pores, damaged pouch, seal-pin weld defects	7, 10	He leak, vacuum decay, weight loss, smell	Electrolyte loss → fade	Flammable electrolyte outside cell, moisture ingress	Seal-area cleanliness, weld monitoring, 100% leak test (prismatic)
Abnormal self-discharge	Micro-shorts (particles, burrs), dissolved metals, shuttle reactions from impurities	1–9	K-value (OCV drop), dynamic limits	Imbalance in packs, reduced range	Possible latent short → thermal event	Contamination control, longer aging for LFP, dynamic limits, µV meters

3. Failure analysis toolbox

Method	Destructive?	What it shows
OCV/ACIR/DCIR, capacity	No	Electrical symptom
dQ/dV and differential voltage analysis	No	Lithium loss vs active-material loss; plating signatures
EIS (impedance spectroscopy)	No	Interface vs bulk resistance; wetting
X-ray / CT	No	Alignment, overhang, particles, tab position, deformation
Thickness / pressure mapping	No	Swelling, gas
Tear-down in glovebox / dry room	Yes	Visual: plating (grey/silver deposits), dry spots, delamination, separator holes
SEM / EDS	Yes	Particle morphology and composition (Fe, Cr, Ni → stainless; Cu; Zn)
ICP-OES/MS	Yes	Metal content in electrolyte or electrodes
GC-MS of gas or electrolyte	Yes	Gas composition, electrolyte decomposition
XPS	Yes	SEI chemistry

4. Troubleshooting flow — self-discharge escape (8D style)



5. Factory example

Pouch cells swell 0.3 mm more than spec at grading, worst in cells made on Mondays. Genealogy: Monday cells used electrode rolls vacuum-dried on Friday and left in the dry-room buffer over the weekend. Dew-point logs show the HVAC setback mode raised dew point to −30 °C on weekends to save energy. **Root cause:** moisture re-absorption during buffer time; HF and gas at formation. **Actions:** no HVAC setback in zones holding product; maximum buffer time interlock in MES; re-dry rule for rolls older than the limit.

6. Interview questions

1. A cell shows low capacity and high resistance together. What are your top three hypotheses and what data do you check first?
2. How do you distinguish lithium inventory loss from active-material loss without tear-down?
3. SEM/EDS finds Fe, Cr, Ni on a separator hole. What does that tell you?
4. Why does plating at the anode edge point to an assembly problem?
5. How do you design containment when the defect shows two weeks after it was made?

7. Quiz

1. Which non-destructive method shows overhang inside a sealed cell?
2. What does a grey/silver deposit on a torn-down anode suggest?
3. Name two causes of gas generation.

4. Which symptom does a weak tab weld cause?
5. Which data link makes containment by root cause possible?

Answers

1. X-ray / CT.
2. Lithium plating.
3. Moisture, residual NMP, residual Li on high-Ni CAM, electrolyte decomposition (any two).
4. High resistance (ACIR/DCIR).
5. Cell-level genealogy (traceability).

8. Hands-on exercise

Build a **troubleshooting matrix** (symptom rows × stage columns) and score how likely each stage is to cause each symptom (0–3). Use it to design a dashboard: for each EOL symptom, which upstream data views should open automatically? Then write a one-page 8D for the Monday swelling example above.

Next: Module 14 — Economics and factory operations

Module 14 — Manufacturing Economics and Factory Operations

Li-ion Cell Manufacturing: an Engineer's Course · Phase D · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: cell cost is mostly **materials**; the factory's job is to waste as little of them as possible (yield) and spread a large **capital** bill over as many kWh as possible (utilisation). In the illustrative India LFP model below, material is ~55–65% of cost, capital ~22–27%, and labour only ~1–2.5%. That is why yield, uptime and capex efficiency — not labour cost — decide whether an Indian plant can compete with China.

The companion spreadsheet `Cell_Cost_Model_1-5-10GWh.xlsx` (same folder) has every input in yellow cells with live formulas.

Part A — Economics

A1. Market anchors (verified)

Fact	Value	Source
Average Li-ion pack price, 2025	$108/kWh (−8% vs 2024)	BNEF, Dec 2025 **[V]**
LFP vs NMC pack average	$81/kWh vs $128/kWh	BNEF **[V]**
Stationary storage packs	$70/kWh (−45%)	BNEF **[V]**
China average pack price	$84/kWh; North America +44%, Europe +56%	BNEF **[V]**
Gigafactory capex	~$80 per kWh of annual capacity (~40 projects); China lowest, US/Japan >2× China	Thunder Said Energy **[V]**
Capex cost in cell price	Amortisation ~8/*kWhplus* 8–12/kWh for a 10% return	Thunder Said **[V]**
Staffing	~70 FTE per GWh/yr; labour ~$6/kWh in the West vs $1–2/kWh in China/emerging markets	Thunder Said **[V]**
Coating + drying share	~11.5% of manufacturing cost, ~46% of energy	EEPower/ORNL **[V]**
Plant energy	~44.6 Wh per Wh capacity (2 GWh NMC333 model), 47 ± 13 Wh/Wh across types	SINTEF **[V]**
Indian reference	Exide's cumulative equity in its cell subsidiary ₹4,902 crore (July 2026) for a 6 GWh phase 1 designed to expand to 12 GWh	Dealroom **[V]** — equity is not identical to capex, but implies ~₹800 crore per GWh order

A2. Cost vocabulary

Term	Meaning
CAPEX	Buildings, dry rooms, utilities, equipment, installation, commissioning
OPEX	Materials, energy, labour, maintenance, consumables, overheads
Conversion cost	Cell cost minus materials — what the factory "adds"
Yield	Good output ÷ input; multiplies through stages (e.g. 12 stages at 99.3% each ≈ 92%)
Utilisation	Actual output ÷ nameplate capacity
Cost per kWh vs per cell	Per cell = per kWh × kWh per cell

A3. Illustrative cost model — LFP prismatic, India greenfield [E]

Assumptions (all [E] unless noted): 280 Ah × 3.2 V = 0.896 kWh/cell; capex 90/75/65*perkWhofannualcapacityfor*1/5/10*GWh*(*anchoredtothe* 80/kWh global average **[V]**); capital charge 20% of capex per year (10-year depreciation + 10% return, consistent with Thunder Said **[V]**); utilisation 80%; base material $42/kWh at 100% yield (BNEF LFP pack $81/kWh × ~75% cell share × ~70% material share); yield 88/92/94%; staffing 150/90/70 FTE per GWh at $10,000 per FTE-year; energy 50/40/35 kWh per kWh at $0.09/kWh; maintenance 3% of capex per year; other overheads $3.0/2.5/2.0 per kWh.

USD per good kWh	1 GWh	5 GWh	10 GWh
Material (÷ yield)	47.7	45.7	44.7
Capital charge	22.5	18.8	16.3
Labour	1.9	1.1	0.9
Energy	4.5	3.6	3.2
Maintenance	3.4	2.8	2.4
Other overhead	3.0	2.5	2.0
Total cell cost	**83.0**	**74.4**	**69.4**
Conversion cost	35.3	28.8	24.7
Cost per cell (0.896 kWh)	**$74.3**	**$66.7**	**$62.2**
Total capex	$90 M	$375 M	$650 M

Reading it: scale cuts ~$14/kWh from 1 to 10 GWh, almost all from capital and yield. Compare with BNEF's China average *pack* price of $84/kWh: an Indian 1 GWh plant on these assumptions would produce *cells* at roughly China's *pack* price — the scale and supply-chain gap in one number.

A4. Sensitivity (5 GWh case) [E]

Change	Total $/kWh	Δ vs base 74.4
Utilisation 60%	82.0	+7.6
Utilisation 90%	71.9	−2.5
Capex −40% (China-like)	65.8	−8.6
Yield 85%	78.2	+3.8
Yield 96%	72.5	−1.9
Material −10%	69.9	−4.5
Material +10%	79.0	+4.6

The biggest levers for an Indian plant are **capex efficiency, utilisation (ramp speed) and materials localisation**; labour is almost irrelevant.

A5. Yield economics

- Scrap value depends on where it happens: electrode scrap before assembly loses materials and some processing; a cell rejected at grading loses everything plus 1–3 weeks of rack time.
- Each yield point at 5 GWh is worth roughly 0.5% of cost (~0.5/*kWh*)*inthismodel—i.e.* 2 M/yr at 4 GWh output **[E]**.
- Ramp-up: new plants typically take 12–24 months to approach mature yields **[T]**; many failed gigafactory projects outside Asia struggled exactly here **[T]**.

Part B — Factory operations

B1. Planning hierarchy

Level	Horizon	Question
Capacity planning	Years	How many lines/racks for the demand curve; phase 2 timing
Production planning (S&OP)	Months	Which cell types, which customers, material call-offs
Scheduling	Days/shifts	Coater campaigns (cathode/anode, recipes), changeovers, formation slot booking
Execution	Real time	MES dispatch, quality holds

Coaters run **campaigns** (long runs of one recipe) because changeovers waste foil and slurry; formation is booked like hotel rooms.

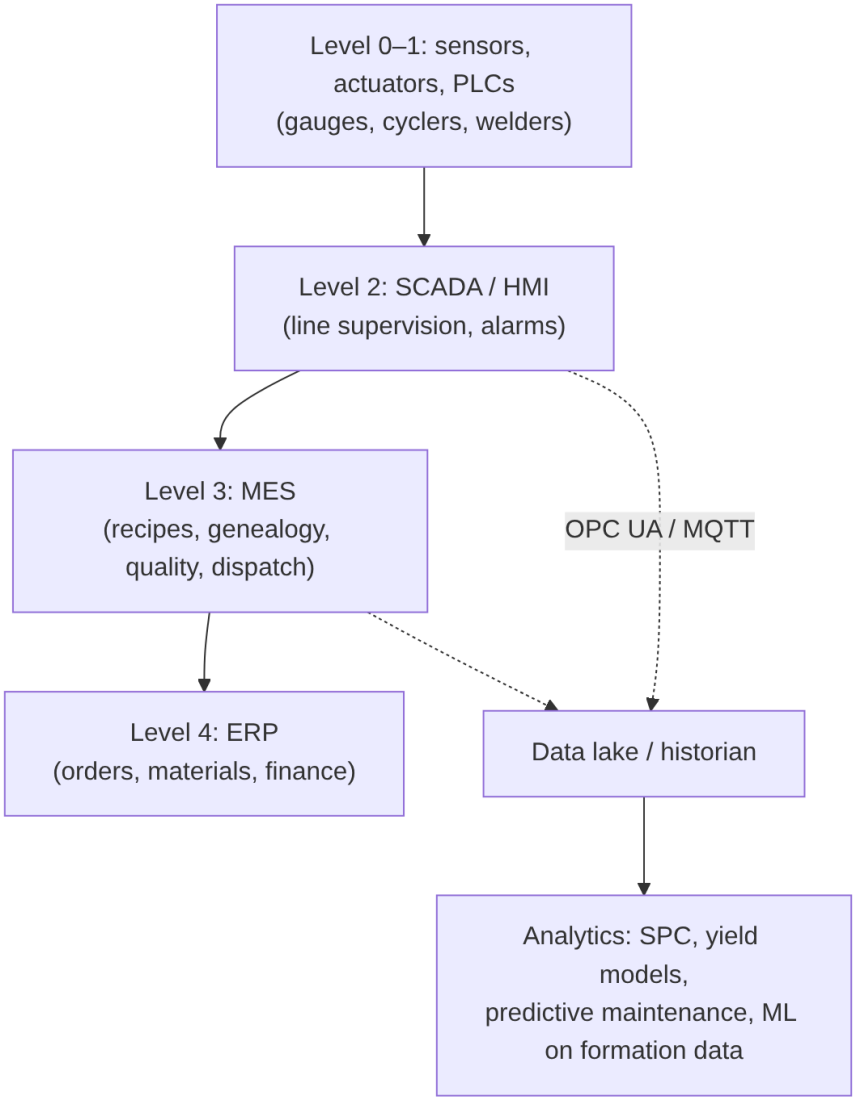
B2. Takt, cycle time, bottleneck, OEE

- **Takt** = available time ÷ required output (Module 10 example: ~4.6 s/cell at 5 GWh).
- **Cycle time** per machine; parallel machines where cycle > takt.
- **Bottleneck**: commonly stacking/winding in assembly and formation capacity in finishing **[T]**; protect it with buffers and priority maintenance.
- **OEE** = Availability × Performance × Quality. Track per coater lane and per assembly machine; use the six big losses as in automotive. Electrode lines lose OEE to splices, roll changes and start-up scrap; assembly to micro-stops and jams; formation to contact faults and empty channels **[T]**.

B3. Yield improvement system

1. Stage-level first-pass yield dashboard, multiplied into overall yield.
2. Daily scrap Pareto by defect code and station.
3. Inline defect maps tied to cell IDs → find which upstream defects really cause EOL failures, stop chasing harmless cosmetic ones.
4. DOE and SPC on special characteristics; Cpk reviews.
5. Golden-batch comparison: parameter fingerprints of the best days.

B4. Digital manufacturing stack



System	Role in a cell plant
MES	Recipe management and locking, electronic batch records, cell-level genealogy (roll → lane → position → cell ID), quality holds, exposure-time clocks, formation slot management
SCADA	Real-time line monitoring, alarms, energy and dry-room monitoring
Equipment data collection	OPC UA / MQTT from every machine; high-frequency gauges and cycler data
Predictive maintenance	Vibration and temperature on calender bearings, coater pumps, fans; weld signatures; laser power trends; cycler channel health
Traceability	Laser-marked codes; roll-position mapping so a vision defect at metre 1,250 of lane 3 is linked to specific cells
Digital twin / simulation	Line balancing, buffer sizing, energy optimisation (e.g. PEM's virtual configuration of cell finishing **[V]** — Electrive)

B5. Roles

Role	Owns
Process engineer	Recipes and windows for a stage (e.g. coating), DOE, process capability, PFMEA content
Equipment engineer	Machine performance, uptime, PM plans, spare parts, OEM interface, upgrades
Manufacturing engineer	Line layout, flow, takt/balancing, new-line industrialisation, cost
Quality engineer	Control plans, SPC, MSA, audits, customer complaints, 8D, PPAP
Production engineer / supervisor	Daily output, shift performance, people, standard work
Data/automation engineer	MES, SCADA, data pipelines, analytics

With your background, the natural entry roles are manufacturing engineering, quality engineering or a costing/industrial-engineering role that combines both — the cost model above is exactly the kind of analysis those teams own.

Interview questions

1. Which cost lever matters most for an Indian cell plant and why?
2. Explain why utilisation affects cost per kWh so strongly.
3. How would you calculate takt time and size formation for a new plant?
4. What data must the MES capture for cell-level traceability?
5. How do you prioritise yield-improvement projects?
6. What is conversion cost, and what is a competitive level?

Quiz

1. BNEF 2025 average LFP pack price?
2. Average gigafactory capex per kWh of annual capacity per Thunder Said?
3. In the illustrative model, what share of 5 GWh cost is material?
4. OEE formula?
5. Which ISA-95 level is MES?

Answers

1. $81/kWh **[V]**.
2. ~$80/kWh **[V]**.
3. ~61% (45.7 / 74.4) **[E]**.
4. Availability × Performance × Quality.
5. Level 3.

Hands-on exercise

Open `Cell_Cost_Model_1-5-10GWh.xlsx` and (a) switch the 5 GWh case to an NMC chemistry with base material $60/kWh and capex $85/kWh **[E]**; (b) find the utilisation at which the 5 GWh plant matches the 10 GWh base cost; (c) build a column chart of the cost stack for the three plant sizes; (d) write a half-page memo on what an Indian plant must do to approach China-level cost.

Next: Module 15 — Global and Indian ecosystem

Module 15 — Global and Indian Cell Manufacturing Ecosystem

Li-ion Cell Manufacturing: an Engineer's Course · Phase D · Evidence labels: [V] verified · [T] typical practice / general knowledge, not re-checked this session · [E] estimate · [P] proprietary

Bottom line: global cell supply is concentrated — CATL and BYD alone installed ~56% of EV battery GWh in 2025. India has moved from announcements to its first commissioned lines in 2026, but operating capacity remains a small fraction of what is announced, and almost all materials and equipment are still imported. **Always separate announced, under-construction, installed, commissioned and producing capacity** — they are different numbers.

Part A — Global market

A1. EV battery installations 2025 (SNE Research via CnEVPost) [V]

Total: **1,187 GWh** (+31.7% vs 901.4 GWh in 2024). Source: CnEVPost, Feb 2026.

Rank	Company	GWh	Share
1	CATL	464.7	39.2%
2	BYD	194.8	16.4%
3	LG Energy Solution	108.8	9.2%
4	CALB	62.8	5.3%
5	Gotion High-tech	53.5	4.5%
6	SK On	44.5	3.7%
7	Panasonic	44.2	3.7%
8	EVE Energy	31.3	2.6%
9	Samsung SDI	28.9	2.4%
10	SVOLT	28.5	2.4%

These are EV installations only; ESS cells (where EVE, CATL, BYD, Hithium and others are large) are counted separately.

A2. Company profiles

Capacity figures are given only where verified; other entries are well-known public facts from general knowledge **[T]** — check the latest annual report before quoting.

Company	Chemistry	Formats	Key manufacturing locations	Capacity	Technology differentiation	Notes
CATL (China)	LFP, NMC, sodium-ion	Prismatic (mainly), some cylindrical	Many sites in China; Germany (Arnstadt), Hungary (Debrecen), Spain JV, Indonesia **[T]**	2025: **772 GWh** production capacity, **321 GWh** under construction, **661 GWh** sales **[V]** (battery-tech.net)	Cell-to-pack, high-volume LFP, fast-charge LFP, sodium-ion commercialisation **[T]**	2025 revenue RMB 423.7 bn, net profit RMB 72.2 bn **[V]**
BYD (China)	LFP (Blade), some NMC	Long prismatic "Blade" (stacked)	China; overseas vehicle plants with expanding cell supply **[T]**	Not verified this session	Vertical integration (cells → vehicles), blade cell-to-pack **[T]**	194.8 GWh EV installs 2025 **[V]**

Company	Chemistry	Formats	Key manufacturing locations	Capacity	Technology differentiation	Notes
LG Energy Solution (Korea)	NMC/NCMA, LFP for ESS	Pouch, cylindrical (incl. 46-series), prismatic in development	Korea, China (Nanjing), Poland (Wrocław), US (Michigan, Arizona, JVs with GM, Honda, Hyundai, Stellantis) **[T]**	Not verified	Pouch NCMA, US localisation, ESS LFP production in Poland **[T]** (Energy-Storage.news)	
CALB (China)	NMC, LFP	Prismatic, cylindrical	China; Portugal project **[T]**	Not verified	High-voltage mid-Ni, "one-stop" prismatic **[T]**	
Gotion (China, VW is a major shareholder)	LFP mainly	Prismatic, cylindrical	China; projects in US, Europe (Germany), Morocco, Slovakia (InoBat JV) **[T]**	Not verified	LFP cost, JV-based expansion **[T]**	Partner to Amara Raja via Gotion-InoBat **[V]** (Autocar Pro)
SK On (Korea)	High-Ni NMC	Pouch (Z-fold stacking)	Korea, US (Georgia; BlueOval SK with Ford), Hungary, China **[T]**	Not verified	High-Ni pouch, Z-fold stacking **[T]**	
Panasonic Energy (Japan)	NCA	Cylindrical (2170, 4680-type)	Japan; US (Nevada with Tesla; Kansas De Soto, announced at 30 GWh **[V-2022 announcement]** (Electrek))	Not verified	High-speed cylindrical manufacturing, high-Ni NCA **[T]**	
EVE Energy (China)	LFP, NMC	Cylindrical (incl. large 46-series), prismatic (ESS), pouch	China; Hungary, Malaysia **[T]**	Not verified	Strong in ESS LFP and small cylindrical **[T]**	
Samsung SDI (Korea)	NCA/NCM	Prismatic, cylindrical	Korea, Hungary, China, US (JVs with Stellantis, GM) **[T]**	Not verified	Prismatic high-Ni, solid-state R&D **[T]**	
AESC (China/Japan)	NMC, LFP	Pouch, prismatic	Japan, UK (Sunderland), France (Douai), Spain, US, China **[T]**	Not verified	Long pouch heritage (Nissan Leaf), European/US localisation **[T]**	

Equipment suppliers (publicly confirmed links are rare). Equipment supplier–cell maker pairings are mostly not disclosed; the verified market picture is: Chinese suppliers held 68% of top-25 equipment sales in 2024 (72% in assembly, 70% in formation/testing), Korean suppliers 29% overall and ~40% in electrode equipment, with PNT second globally in front-end equipment **[V]** (Control Engineering). Lead Intelligent is widely reported as the largest Chinese line supplier **[T]**.

Part B — India

B1. Policy

Policy	Key facts	Source
ACC PLI (Ministry of Heavy Industries)	Launched Oct 2021; ₹18,100 crore outlay; 50 GWh target. Round 1 (Mar 2022) allocated 50 GWh; Hyundai Global Motors withdrew from 20 GWh; round 2 (Sep 2024) awarded 10 GWh. 40 GWh allocated in total	IEEFA, Jan 2026 **[V]**
PLI status	As of Oct 2025: 1.4 GWh (2.8%) commissioned on time, all Ola; no incentives disbursed; 1,118 jobs vs 1.03 M estimated; investment ~₹2,870 crore committed (25.6% of target)	IEEFA **[V]**
PLI conditions	Domestic value addition (25% in 2 years, 60% in 5 years), two-year installation timeline, daily penalty on delays	Vision IAS summary, IEEFA **[V]**
New 10 GWh tender	Bids invited July 2026 (submission 13 Oct, opening 14 Oct 2026), earmarked for grid-scale storage	GreentechLead **[V]**

Policy	Key facts	Source
Cell-component scheme	~₹12,000–13,000 crore proposed for cathode, anode, separator, foil, electrolyte; not yet gazetted	Swarajya **[V]**
Demand drivers	National Electricity Plan: 34 GWh BESS by 2026–27, 236 GWh by 2031–32; VGF support for 43.8 GWh; PM E-DRIVE (₹10,900 crore)	GreentechLead **[V]**

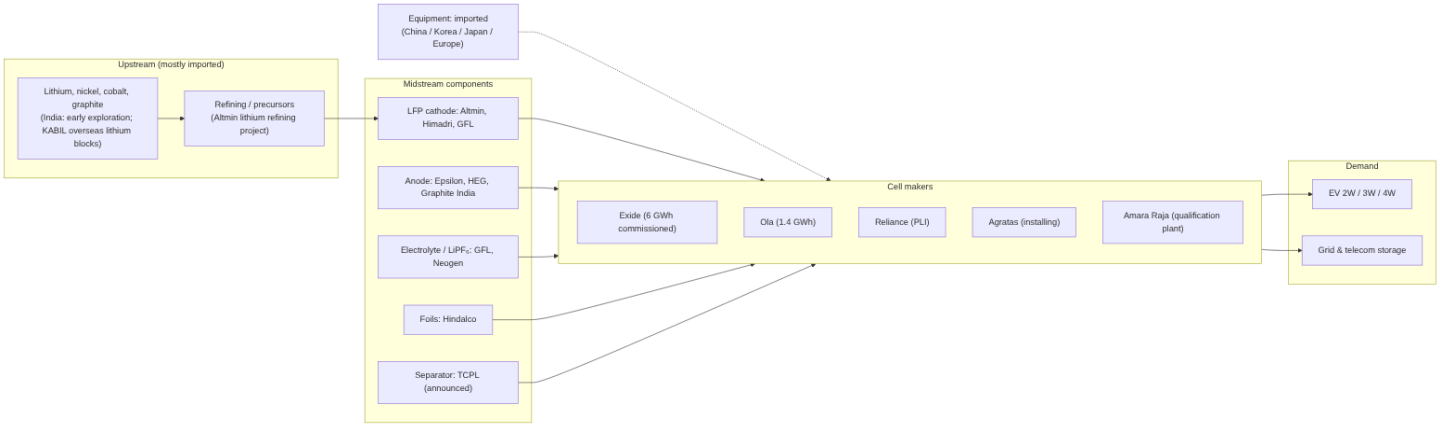
B2. Cell makers — status (October 2026)

Company	Site	Plan	Status as reported	Chemistry / partner	Source
Exide Energy Solutions	Devanahalli, Bengaluru	12 GWh (2 × 6 GWh)	**Phase I 6 GWh commissioned** (reported 28 Sep 2026); revenue targeted from Q3 FY27	LFP prismatic first, NMC later; SVOLT technology licence; non-PLI	Energetica India, Outlook Business
Ola Cell Technologies	Krishnagiri, TN	20 GWh PLI award; limited to 5 GWh until FY2029	1.4 GWh commissioned (Oct 2025)	46-series cylindrical (Ola's "4680 Bharat Cell") **[T]**; PLI	IEEFA
Reliance New Energy	Jamnagar, Gujarat	5 GWh (round 1) + 10 GWh (round 2) PLI	Round-2 indicated on time; round-1 delayed (Oct 2025)	PLI	IEEFA
Agratas (Tata)	Sanand, Gujarat	20 GWh phase 1	First process equipment arrived 16 Sep 2026	—	GreentechLead
Amara Raja	Divitipally, Telangana	16 GWh, ₹9,500 crore	₹500 crore qualification plant (60 MWh) commissioned Jul 2026; 2 GWh commercial plant scheduled 2027	NMC 2170 cylindrical with a Chinese partner; LFP with Gotion-InoBat; non-PLI	GreentechLead, Autocar Pro

B3. Components and materials (India)

Segment	Indian players (status)	Source
LFP cathode	Altmin (Telangana, ₹750 cr plant under construction), Himadri (Odisha, 40,000 t/yr by late 2027), Gujarat Fluorochemicals (small line commissioned)	Swarajya **[V]**
Anode	Epsilon Advanced Materials (qualification plant, ₹145 cr incentive), HEG (₹1,893 cr), Graphite India (₹4,330 cr)	Swarajya **[V]**
Separator	TCPL Packaging announced; no commercial Indian Li-ion separator plant yet	Swarajya **[V]**
Foils	Hindalco: battery Al foil (Odisha); 11,500 t copper foil announced (Gujarat)	Swarajya **[V]**
Electrolyte / $LiPF_6$	Gujarat Fluorochemicals ($LiPF_6$ commercial at Dahej), Neogen Chemicals (electrolyte under licence; $LiPF_6$ with Morita)	Swarajya **[V]**
Equipment	Core process machines imported (China, Korea, Japan, Europe); Chinese commissioning engineers' visa delays slowed PLI plants	IEEFA **[V]**
Critical minerals	Lithium resources identified in J&K and Rajasthan; commercial mining long-dated; NMC CAM supply ~98% Chinese globally per one report	MeraEV, Dealroom

B4. Ecosystem map



B5. Challenges

1. **Import dependence** across minerals, materials and equipment; dependence on imported cells still close to 100% (Oct 2025) **[V]** (IEEFA).
2. **Know-how gap:** none of the PLI winners had large-scale cell experience; Exide and Amara Raja (experienced battery makers) did not qualify **[V]** (IEEFA).
3. **Policy design:** aggressive two-year timelines and high DVA targets **[V]**.
4. **Scale and cost:** see Module 14 — small plants carry high capital cost per kWh.
5. **Testing and certification infrastructure** for cells is limited **[V]** (IEEFA recommends building it).
6. **Talent:** few engineers with dry-room, formation and electrode-process experience **[T]**.

B6. Opportunities

- **Grid storage demand** (236 GWh by 2031–32) gives LFP plants a large domestic market **[V]**.
- **Component localisation** (cathode, anode, foil, electrolyte, cans, laminate film) as incentives arrive **[V]**.
- **Equipment and spares localisation**: tooling, slitting knives, dies, fixtures, trays, dry-room HVAC, intralogistics — areas where India's automotive supplier base can enter **[E]**.
- **Services:** commissioning, maintenance, testing labs, recycling (scrap from ramp-ups is valuable).
- **Careers:** process, quality, equipment and costing engineers with automotive discipline are scarce and needed now **[E]**.

Interview questions

1. What is the difference between announced, installed, commissioned and producing capacity? Give an Indian example of each.
2. Why did ACC PLI under-deliver, and what would you change?
3. Which Indian component segment do you expect to localise first, and why?
4. Compare CATL's and LGES's technology strategies.
5. Why does LFP dominate Indian plans?

Quiz

1. CATL's 2025 EV battery market share (SNE)?
2. Total global EV battery installations in 2025?
3. How much ACC PLI capacity was commissioned on time by Oct 2025?
4. Which Indian cell maker commissioned 6 GWh in Sep 2026?
5. Which Indian company makes $LiPF_6$ commercially?

Answers

1. 39.2% **[V]**.
2. 1,187 GWh **[V]**.

3. 1.4 GWh **[V]**.
4. Exide Energy Solutions **[V]**.
5. Gujarat Fluorochemicals **[V]**.

Hands-on exercise

Build a one-page **capacity tracker** for Indian cell plants with columns: company, site, chemistry/format, announced GWh, under construction, installed, commissioned, producing, source and date. Fill it from this module, then search for one update per company and record what changed. This is the discipline the brief asked for: never mix stages of capacity.

Next: Module 16 — Capstone: interview guide, glossary, resources

Module 16 — Capstone: Interview Guide, Glossary and Resources

Li-ion Cell Manufacturing: an Engineer's Course · Phase D · Evidence labels: [V] verified · [T] typical practice · [E] estimate · [P] proprietary

Bottom line: you are ready when you can (1) walk the line from powder to pallet and explain *why* each control exists, (2) translate every battery quality problem into the APQP/PFMEA/SPC language you already speak, and (3) put a cost on a yield, uptime or capex decision.

Part A — Capstone project

Design, on paper, a 5 GWh/yr LFP prismatic cell line for India. Use your exercises from Modules 1–15.

Deliverable	Content	From module
1. Cell design sheet	280 Ah LFP: loadings, densities, N/P, overhangs, fill mass	3, 4, 5, 7
2. Process flow diagram	All 12 stages with inspection points	0
3. Equipment list	Machine counts by stage, OEM shortlist	10
4. Line balance	Takt, cycle times, buffers, formation channels, aging positions	6, 8, 10
5. Layout	Zones, dew points, flows, fire compartments, expansion	11
6. PFMEA + control plan	Top 30 failure modes, special characteristics, reaction plans	12
7. Defect playbook	Symptom → stage → data → action	13
8. Cost model	Capex, $/kWh, sensitivities, break-even utilisation	14
9. Supply plan	Indian vs imported materials and equipment, risks	1, 15
10. Ramp-up plan	FAT/SAT, run-at-rate, yield learning curve, staffing	10, 14

Present it as a 15-slide deck or a 10-page document — the format a plant director or interviewer would expect.

Part B — Technical interview preparation guide

B1. How interviews for cell-plant roles usually run [T]

1. **Fundamentals:** cell components, chemistries, SEI, why moisture matters.
2. **Process depth:** one or two stages in detail (the one you claim expertise in).
3. **Quality:** PFMEA, SPC, Cpk, 8D, MSA with battery examples.
4. **Problem-solving case:** "self-discharge rejects doubled — what do you do?"
5. **Commissioning/ramp-up:** FAT/SAT, run-at-rate, OEE.
6. **Cost/business:** where cost comes from; capex/yield trade-offs.
7. **Behavioural:** cross-functional work with OEM technicians (often Chinese/Korean), shift leadership, safety culture.

B2. Question bank with answer outlines

#	Question	What a strong answer covers
1	Walk me through the cell process.	Three blocks; 12 stages; why dry room; formation as bottleneck; where defects are created vs detected
2	Why PVDF/NMP for cathode, water for anode?	Electrochemical stability at potentials; NMP dissolves PVDF; cost and toxicity; water-based cathode efforts
3	What controls coat weight?	Pump-metered flow, web speed, width; β/X-ray gauge closed loop; N/P safety link

#	Question	What a strong answer covers
4	Explain binder migration.	Drying physics; low–high–low; adhesion loss; how to detect (peel)
5	Porosity vs density?	Formula; energy vs rate; Bruggeman; wetting time
6	Burr specification?	Separator thickness; tool life by metres; inspection; SC classification
7	Winding vs stacking?	Speed, space use, stress, tabs; format fit
8	Why hi-pot before filling?	Detect shorts cheaply; voltage/limits; false-pass risk
9	Fill mass setting?	Pore volume + reserve; g/Ah; weighing MSA; Cpk
10	What is SEI and why slow formation?	EC reduction; lithium loss; plating risk; ORNL evidence on fast formation trade-offs
11	K-value testing?	OCV drift; fixed vs dynamic limits; LFP flat curve; temperature compensation
12	Self-discharge escape case	Containment by genealogy; data join; tear-down SEM/EDS; source fix; PFMEA update
13	Dry room design?	Dew point; desiccant wheel; pressure cascade; moisture loads; energy
14	Special characteristics of a cell?	Coat weight/N/P, burr, overhang, moisture, fill mass, K-value, welds
15	OEE on a coater vs a stacker?	Splices/roll changes vs micro-stops; quality losses as marked sections
16	Biggest cost levers?	Material/yield, capex/utilisation; labour minor; numbers from Module 14
17	Formation sizing?	Channels = rate × hours ÷ availability; energy recovery
18	How do you run a material change?	PCN, PPAP-style revalidation: compaction curve, slurry behaviour, formation efficiency, life tests
19	India's ecosystem?	PLI status, Exide commissioning, imports, component scheme; announced vs operating capacity
20	Why do you, from automotive, fit here?	APQP/PFMEA/SPC rigour, commissioning, costing; humility on electrochemistry; examples

B3. Your story (suggested framing)

- **Bridge:** "Eight years in automotive product development and costing — should-cost, BOM analysis, PPAP/APQP, ECN ownership — plus paint-shop quality where environment control and defect Paretos rule. A cell plant combines the two: a paint-shop-like environment problem and a cost problem dominated by materials and yield."
- **Proof:** bring the cost model, the PFMEA and the capacity tracker from this course.

Part C — Glossary

Term	Meaning
ACIR	AC internal resistance, usually measured at 1 kHz
Active material	Material that stores lithium (cathode CAM, anode AAM)
Aging	Holding cells to stabilise SEI and screen self-discharge
Agglomerate	Cluster of particles not broken apart in mixing
Areal capacity	mAh per cm² of electrode
Areal loading (coat weight)	mg of dry coating per cm²
AS/RS	Automated storage and retrieval system
β-gauge	Beta-ray gauge measuring coat weight
BET	Specific surface area by gas adsorption
Binder	Polymer holding particles together and to foil (PVDF, CMC/SBR)
Binder migration	Binder carried to the surface during fast drying
Blade cell	BYD's long thin prismatic LFP cell
Burr	Raised metal edge from cutting
Calendering	Compressing electrodes between rolls
CAM	Cathode active material
CEI	Cathode–electrolyte interphase
CID	Current interrupt device in cylindrical caps

Term	Meaning
CMC	Carboxymethyl cellulose (anode thickener/dispersant)
Coating window	Stable operating region of a slot die
Comma coater	Knife-over-roll coater
Conversion cost	Cell cost minus materials
CTP	Cell-to-pack design
DCIR	DC internal resistance from a current pulse
Degassing	Removing formation gas before final seal
Dew point	Temperature at which air moisture condenses; dry-room metric
dQ/dV	Differential capacity analysis
Dry electrode	Solvent-free electrode process
Dry room	Room with very low dew point
DVA	Domestic value addition (PLI condition)
EC / EMC / DMC	Ethylene, ethyl methyl, dimethyl carbonate (solvents)
EIS	Electrochemical impedance spectroscopy
EOL	End-of-line test
EPC	Edge position control (web guiding)
FAT / SAT	Factory / site acceptance test
FEC / VC	Fluoroethylene / vinylene carbonate (SEI additives)
First-cycle efficiency	Discharge ÷ charge capacity on first cycle
Flotation dryer	Oven where the web floats on air jets
Formation	First controlled charge that forms SEI
Gelation	Slurry turning to gel (PVDF + base/moisture)
Genealogy	Full traceability of a cell's materials and process data
Gigafactory	GWh-scale cell plant
Grading	Sorting cells by capacity, IR, OCV
Grind gauge (Hegman)	Measures largest particles in slurry
Gurley	Separator air permeability
HAZ	Heat-affected zone of laser cut
He leak test	Helium mass-spectrometer leak detection
Hi-pot	High-voltage insulation test
Hot calendering	Calendering with heated rolls
ICP-OES/MS	Elemental analysis method
Intermittent coating	Coating with gaps along the web
Jelly roll	Wound electrode assembly
K-value	OCV drop rate during aging (mV/day)
Karl Fischer	Moisture titration
Lamination stacking	Laminating electrode–separator units before stacking
LFP / NMC / NCA / LTO	Cathode/anode chemistries
$LiPF_6$	Electrolyte salt; hydrolyses to HF
Lithium plating	Metallic lithium deposited on anode
Mandrel	Core on which electrodes are wound
MES	Manufacturing execution system
Micro-short	Small internal short causing self-discharge
N/P ratio	Anode-to-cathode capacity ratio
NMP	N-methyl-2-pyrrolidone solvent
Notching	Cutting tab shapes into electrode margins

Term	Meaning
OCV	Open-circuit voltage
OEE	Overall equipment effectiveness
Overhang	Anode beyond cathode / separator beyond anode
PCN	Product/process change notification
PLI (ACC)	Production-linked incentive for advanced chemistry cells
Porosity	Void fraction of electrode
Pot life	Time a slurry stays usable
Pouch	Laminated-foil cell housing
Prismatic	Rigid rectangular can cell
PVDF	Polyvinylidene fluoride binder
Residual lithium	$LiOH/Li_2CO_3$ on high-Ni cathode surfaces
SBR	Styrene-butadiene rubber binder
Seal pin	Pin welded into prismatic fill port
SEI	Solid electrolyte interphase on the anode
Separator	Porous film between electrodes
Shutdown	Separator pore closure on heating
Slitting	Cutting rolls into lanes
Slot die	Pre-metered coating head
Solids content	Mass fraction of solids in slurry
Springback	Thickness recovery after calendering
Tabless	Full-edge current collection in cylindrical cells
Takt time	Available time ÷ required output
Tap charge	Small initial charge after filling
Telescoping	Axial shift of layers in a jelly roll
Tortuosity	Path-length factor for ion transport
Ultrasonic welding	Solid-state welding of foil stacks
UN 38.3	Transport test standard for lithium cells
Wetting	Electrolyte penetration into pores
Z-fold	Stacking with zig-zag continuous separator

Part D — Recommended resources

D1. Books [T — well-known titles; check editions]

Title	Why
Korthauer (ed.), *Lithium-Ion Batteries: Basics and Applications*, Springer, 2018	Includes production chapters by German industry and academia
Kirby Beard (ed.), *Linden's Handbook of Batteries*, 5th ed., McGraw-Hill, 2019	Reference for chemistries and cell design
Yoshio, Brodd, Kozawa (eds.), *Lithium-Ion Batteries: Science and Technologies*, Springer, 2009	Classic materials and cell technology
Newman & Balsara, *Electrochemical Systems*, Wiley	Theory behind porous electrodes (optional, deep)

D2. Papers and guides

Reference	Use
PEM RWTH & VDMA, *Production Process of a Lithium-Ion Battery Cell*, 5th ed. (free) — link	Whole process
Kwade et al., *Nature Energy* 2018 — doi	Process–structure–property

Reference	Use
Liu et al., *iScience* 2021 — doi	Cost, energy, throughput per step
Wood et al., *Joule* 2019 — doi	Formation
Wood, ORNL NAATBatt 2019 slides — pdf	Formation and wetting data
Zhang et al., electrode drying review, *Adv. Energy Mater.* 2022 — preprint	Drying
A Flexible Model for Benchmarking the Energy Usage of Automotive Lithium-Ion Battery Cell Manufacturing, *Batteries* 2021 (SINTEF) — record	Plant energy
Degen et al., *Nature Energy* 2023, energy consumption of current and future cell production **[T — not opened]**	Energy
Duffner et al., *Nature Energy* 2021, post-Li-ion compatibility with Li-ion lines **[T — not opened]**	Sodium-ion comparison
Hawley & Li, *Journal of Energy Storage* 2019, electrode manufacturing review **[T — not opened]**	Electrode processes
IEEFA & JMK, ACC PLI assessment, Jan 2026 — report	India policy
BNEF Battery Price Survey 2025 — press release	Prices

D3. Videos, courses and communities [T — search and verify availability]

Resource	Use
The Limiting Factor (YouTube)	Deep dives into cell design and manufacturing (cylindrical, tabless, dry electrode)
PEM RWTH Aachen and Fraunhofer FFB channels/webinars	Production research talks
The Battery Show India / Europe conference talks	Industry practice, equipment makers
Volta Foundation, *The Battery Report* (annual, free)	Industry overview
Faraday Institution reports	Manufacturing and supply-chain studies
NPTEL (IIT) courses on electrochemical energy storage	Indian-context fundamentals
Equipment OEM application notes (Dürr, Bühler, Hirano, Manz, Trumpf)	Machine-level practice

Final self-assessment

- [ ] I can draw the 12-stage flow from memory with environments and inspection points.
- [ ] I can derive porosity, N/P, takt, formation channels and cost per kWh by hand.
- [ ] I can write a PFMEA line and a control-plan line for any stage.
- [ ] I can explain three ways a metallic particle reaches a cell and how each is stopped.
- [ ] I can state India's operating vs announced cell capacity with sources and dates.
- [ ] I have the capstone pack ready to show in an interview.

Back to start: Module 0 — Roadmap and line overview